



Data driven classification of X-ray sources

by

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Abstract

Modern X-ray telescopes like *Chandra* and *XMM-Newton* have detected hundreds of thousands of unresolved X-ray sources in the sky. Current methods to analyze the X-ray spectra of these sources are inefficient; detailed X-ray spectroscopy to understand the X-ray emission is too time-consuming, and using hardness ratios to approximate source properties has only been efficient at discriminating broad spectral features, e.g. absorption and whether spectra are "hard" or "soft" (whether the spectra extend to high energy, or not).

In this thesis, we utilize this power of CCDs to differentiate between line-dominated and continuum spectra. First, we trained a sequential neural network to separate the spectra of active galactic nuclei (AGN) and young stars through supervised learning. Our model could classify the spectra with an accuracy of 92%. We also developed X-ray colors tailored to identify the Fe emission lines to select candidate millisecond pulsars (MSPs) in the galactic bulge and test if they can explain the observed galactic center excess in γ -rays. We finally developed a semi-supervised classification model on the 4XMM-DR13 data to distinguish AGN, stars, CVs and stellar mass NS/BH systems.

Preface

This thesis is an original work by Pavan R. Hebbar under the supervision of Prof. Craig O. Heinke. All data used are publicly available.

I have published the contents of Chapter 3 in Hebbar and Heinke (2023). I extracted the *Chandra* spectra and trained the model. Co-author Craig O. Heinke assisted me in interpreting the statistical significance of the results and their theoretical implications.

I have submitted the contents of Chapter 4 to AAS and received comments from the referee. I simulated artificial X-ray spectra of neutron stars (NS) and cataclysmic variables (CVs), selected X-ray colors that could separate the continuum sources from Fe-K line-dominated CVs, and checked if the candidate MSP population could explain the γ -ray excess towards the galactic centre. Dr. Jiaqi Zhao used hardness ratios to select *XMM-Newton* sources in the Galactic Bulge. Prof. Craig Heinke assisted us in selecting appropriate distribution of NS and CV parameters for the simulation and interpreting the results.

In Chapter 5, I extracted the 4XMM-DR14, and developed the algorithms for the machine learning algorithms. Prof. Craig Heinke helped me understand the physical relevance of the machine learning results.

“Science is built up with facts, as a house is with stones. But a collection of facts is no more a science than a heap of stones is a house.”

Subrahmanyan Chandrashekar



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Contents

1	The X-ray Universe	2
1.1	Types of X-ray sources	3
1.1.1	Chromospherically active stars	3
1.1.2	Active Galactic Nuclei	4
1.1.3	Cataclysmic variables	5
1.1.4	Non-interacting neutron stars	6
1.1.5	X-ray binaries	7
1.2	Serendipitous X-ray source catalogues	8
1.2.1	<i>Chandra</i> Source Catalog	8
1.2.2	<i>XMM-Newton</i> Serendipitous Source Catalogue	9
1.3	On the road to this thesis	9
2	An overview of machine learning concepts	11
2.1	Foundations of Machine learning	12
2.1.1	Data and feature representation	13
2.1.2	Training, Validation, and Testing	14
2.1.3	Loss Functions and Regularization	15
2.1.4	Optimization	16
2.2	Supervised training	17
2.2.1	Logistic Regression	17
2.2.2	Artificial Neural Networks	18
2.2.3	Convolutional Neural Networks	19

2.3	Unsupervised training	19
2.3.1	Dimensionality reduction	19
2.3.2	Autoencoders	21
2.4	Semi-supervised learning	22
2.4.1	Autoencoders with classification loss	23
2.4.2	Pseudo-labels	23
3	Supervised classification of stars in Orion nebula from AGN in CDFS	25
3.1	Introduction	25
3.1.1	Artificial neural networks	29
3.2	Data Acquisition and Analysis	30
3.2.1	Setting up the ANN	37
3.3	Results	43
3.3.1	Variation of performance with properties of sources.	47
3.3.2	Performance at low counts	50
3.3.3	Weights applied by ANN on energy bins	51
3.3.4	Applying trained ANN model to observed data	51
3.3.5	Using ML to detect extragalactic sources in COUP	52
3.3.6	Changing response of Chandra	52
3.4	Discussions	58
3.4.1	Incorrect classification of Sources	58
3.4.2	Applications	59
3.4.3	Current and Future missions	61
3.5	Conclusion	62
4	Identifying candidate X-ray binaries and millisecond pulsars in Galactic Bulge using Fe-K line diagnostics	64
4.1	Introduction	64
4.2	Data Acquisition and Analysis	67

4.2.1	XMM-Newton sources	68
4.2.2	Chandra sources	69
4.2.3	Simulated source selection	71
4.2.4	X-ray color analysis.	73
4.3	Results	74
4.4	Discussion	78
4.4.1	X-ray sources with counterparts	78
4.4.2	X-ray variability	79
4.4.3	Background AGN	80
4.4.4	Probability of CVs among candidate quiescent sources . .	82
4.4.5	Comparison with Hailey et al. (2018)	84
4.4.6	Implications for the γ -ray GCE	86
4.5	Summary & Conclusion	88
5	Semi-supervised classification of XMM-Newton spectra	98
5.1	Introduction	98
5.1.1	Convolutional Neural Networks	100
5.1.2	Autoencoders	101
5.1.3	Semi-supervised learning	101
5.2	Data Acquisition and analysis	103
5.2.1	Extraction of <i>XMM-Newton</i> Spectra	103
5.2.2	Generating a set of labeled spectra	105
5.2.3	Semi-supervised learning of the X-ray spectra	107
5.3	Results	110
5.4	Discussion	117
5.4.1	Comparison with hardness ratios	117
5.4.2	Visualizing the CNN activations	118
5.4.3	Separation between AGN and stars in latent space	120
5.4.4	Extending to 4-class classification	120

5.5	Conclusion	123
6	Summary and conclusions	126
A	Separating COUP and CDFS spectra	130
A.1	Typical spectra of simulated stars and AGN	130
A.2	TensorFlow model of the ANN	131
A.3	Choosing number of nodes, L_1 value, and the size of training set	132
B	Identifying candidate MSPs and X-ray binaries in the Galactic Bulge	135
B.1	Calculating absorption column for Galactic Bulge sources. . . .	135
B.2	Estimating the number of cataclysmic variables in our sample. .	136

List of Tables

3.1	Performance of our machine learning model on different datasets	43
4.1	Descriptive version of the “Spectral fitting of <i>XMM-Newton</i> sources” table	90
4.2	Descriptive version of the “Spectral fitting of <i>Chandra</i> sources” table	92
4.3	Descriptive version of the “ <i>XMM-Newton</i> candidates without lines” table	95
4.4	Descriptive version of the “ <i>Chandra</i> candidates without lines” table	96
5.1	Performance comparison between pseudo-label and autoencoder models for <i>XMM-Newton</i> data under different cases.	112

List of Figures

3.1	Example spectra of an isolated neutron star (4U 0142+61), active star (COUP 053445.2-052504), SNR (Cassiopeia A), and AGN (Centaurus A). We also show the approximate positions of Fe-L, Ne-K, Mg-K, Si-K, S-K, and Fe-K emission lines. The spectra of active stars and SNRs are dominated by these lines while that of AGN and NSs are largely continuous. AGN can sometimes show an Fe-K line.	28
3.2	Distribution of the parameters used to simulate the spectra of stars for training our ANN. Since the Orion nebula is close to the galactic plane, N_H is high. Most stars have $kT < 2$ keV. The few sources with $kT \approx 14$ keV are likely to be heavily absorbed stars where emission lines could not be fit (and a higher temperature plasma could fit the relatively hard spectra). We see some correlation between kT_1 and kT_2 , and a strong correlation between EM_1 and EM_2 . Since we match the fluxes of stars and AGN, specifying the emission measure values only contributes to maintaining the ratio of emission from the two plasmas.	33
3.3	Properties of AGN sample used for training and testing our ANN.	35

3.4	Normalized (divided by total counts) mean spectra of the observed COUP stars (solid blue) and CDFS AGN (solid purple). For the purpose of plotting, we only show the mean spectra calculated from sources with net counts more than 100. From the figure, we see that the spectra of stars show a prominent Ne-K/Fe-L emission line at ~ 1 keV along with slight hints of Mg-K (~ 1.3 keV), Si-K (~ 1.8 keV), and S-K (~ 2.3 keV), while the AGN show no such lines. Since the CDFS AGN have lower flux than the COUP stars in general, we notice that the spectra of AGN have more noise (both background and Poisson) than the stars.	38
-----	--	----



3.5	Histogram of the net counts of the observed COUP stars (solid blue) and CDFS AGN (solid purple). We have normalized the histograms such that the area under both histograms is unity (to avoid distortion due to more COUP sources than CDFS AGN). We notice that the COUP stars are typically ~ 10 times brighter than the CDFS AGN.	39
-----	--	----

3.6	Normalized histogram of the ratio of background-to-net counts for the observed COUP stars and CDFS AGN. Since CDFS AGN are mostly faint, most of them have a very high background contribution.	40
-----	---	----

3.7	Architecture of the ANN model. The input layer consists of 527 nodes corresponding to the number of channels in the <i>Chandra</i> ACIS-S spectra in the energy range 0.3–8.0 keV. We use one hidden layer with 10 nodes which are fully connected to the input layer. The hidden layer uses a ReLU activation and an L_1 regularization with $\lambda = 0.001$. These hidden nodes are then used to calculate the values in the two output nodes that use a softmax activation function and signify the probability of the source being an active star or an AGN.	42
3.8	Normalized mean of the artificially generated AGN and star spectra simulated without including the background. These spectra are similar to that in Fig. 3.4, but less noisy since they are the mean of 100,000 spectra. With less noise, the emission features in the spectra of stars are more clear. The emission-line-like features in the AGN spectra are due to a combination of the intrinsic absorption columns, redshifts, and the <i>Chandra</i> response. Red-shifted Fe-K emission lines cause the appearance of bumps in the mean AGN spectra in the 2–4 keV energy range.	44
3.9	Distribution of total counts, in the 0.3–8.0 keV energy range, in the simulated sample of AGN and stars generated without including the background. Despite equating the 0.5-10 keV flux of simulated stars and AGN, stars have higher X-ray photon counts since they have softer spectra than AGN.	45
3.10	Distribution of background-to-net count ratio in the simulated AGN and star spectra when the observed background is included. The plot only represents spectra with net counts greater than 100 in the 0.3–8.0 keV energy range. Many of these spectra have a high contribution from the background as these sources are fainter than typical sources detected by <i>Chandra</i>	46

3.11	Distribution of net counts of star and AGN spectra, simulated with a reduction of the background contribution by a factor of 10.	47
3.12	Distribution of the background-to-net count ratio in the simulated AGN and star spectra simulated with the reduced background. The y-axis is normalized such that the figure shows a density function. This distribution is more typical of the X-ray sources detected by <i>Chandra</i> .	48
3.13	Distribution of probabilities of the source being AGN, as calculated by our trained model. Notice that most of the stars and AGN have been correctly identified with a certainty of more than 95%. In this article we have used a cutoff of 50% to differentiate stars and AGN.	49
3.13	Variation in performance of our ANN model for different properties of the sources. In each case the histogram represents the number of spectra in the given binning of the property. The y-axis on the right corresponds to the values of confusion matrix (1-Accuracy, 1-Recall/TNR etc.).	55
3.14	Change in the accuracy with net counts for simulated spectra with net counts > 1.0 . From the figure, we see that our algorithms does not perform well on sources with 1–10 net counts (accuracy marginally better than 50%). The accuracy is $\gtrsim 70\%$ for sources with net counts $\gtrsim 20$ and $\gtrsim 80\%$ for net counts $\gtrsim 70$.	55
3.15	Weights to each energy channel in the first layer of a trained ANN model. We notice that the model assigns larger weights (magnitude only) to the spectra around the position of the Ne, Fe, Mg, and Si lines. The different curves correspond to the weights of ten nodes in our ANN.	56
3.16	Performance of the trained ML model with net counts, and with background contribution, on the observed spectra.	57

3.17	Mean normalized spectra of correctly identified stars and AGN, compared to those that are not classified properly. Notice that the stars that have been identified as AGN do not show strong emission lines, are heavily absorbed, and are harder than the stars that have been correctly identified.	58
4.1	<i>XMM-Newton</i> EPIC (<i>left</i>) and <i>Chandra</i> ACIS (<i>right</i>) observations around the Galactic center. We combine the EPIC-PN and EPIC MOS exposures, using a weight of 0.4 for MOS data to account for the differing effective areas. We also overlay contours of N_H , and positions of all the sources we consider. Sources detected in EPIC-PN but not EPIC-MOS (blue upper triangle), detected in EPIC-MOS but not EPIC-PN (downward blue triangle), and detected in both EPIC-PN and MOS (blue diamond) are marked separately. <i>Chandra</i> ACIS sources are shown in blue circles. We only consider <i>XMM Newton</i> sources with $N_H \geq 5 \times 10^{22} \text{ cm}^{-2}$ and <i>Chandra</i> sources within $30'$ of the Galactic Center.	67
4.2	Mean normalized spectra of simulated MSPs and observed sources with EPIC-PN (<i>left</i>), EPIC-MOS (<i>center</i>), and ACIS (<i>right</i>) detectors. Each spectrum is divided by its net counts in the 2–10 keV energy range to get its normalized spectrum. The mean of the normalized spectra for observed sources shows an excess in 6–7 keV corresponding to the fluorescent Fe (6.4 keV) and Fe-K (6.7 keV & 7.0 keV) lines. This emission is from CVs embedded in our sample. The observed spectra also differ from simulated spectra in terms of hardness (possibly due to a more complex N_H & L_X distribution) and apparent absorption features (due to over-subtraction of the background).	72

4.3	Distribution of the net and background counts in the EPIC-PN, EPIC-MOS, and ACIS sources (observed sources and simulated MSPs). We only show simulated sources whose net counts are greater than 3 times the square root of their background counts. In general, these sources show a high background due to the diffuse emission close to the Galactic Center. Our simulated sources cover the entire range of net counts and background contributions spanned by the observed sources.	73
-----	--	----



4.4	Distribution of simulated MSPs and observed sources in the X-ray color vs. net counts plane. We calculate the X-ray color as the ratio of the net counts in the 6.2–7.2 keV region, to the sum of the net counts in the 5.8–6.2 and 7.2–7.6 keV regions, so larger colors indicate Fe line emission. We also show the 2σ bounds on the color calculated empirically from our simulated sample of MSPs. The observed sources show a greater scatter in the X-ray colors, particularly to higher X-ray color, compared to the simulated MSPs, due to the presence of line-emitting CVs among the observed sources. Our final list of candidate continuum sources, that appear to lack lines both by X-ray color and in spectral fitting, are marked as stars.	75
-----	--	----

4.5	Sky positions of candidate continuum sources detected in <i>XMM-Newton</i> (blue circles) and <i>Chandra</i> (red circles) observations. Among the XMM detections, sources only identified through EPIC-pn (EPIC-MOS) analysis are shown as upward (downward) triangles, and sources detected in both EPIC-pn and EPIC-MOS are marked as diamonds. <i>Left</i> , a view of the full region. <i>Right</i> , a zoomed view of the Galactic center region. The background maps are the combined exposure maps of <i>XMM-Newton</i> EPIC-pn, EPIC-MOS, and <i>Chandra</i> ACIS observations (EPIC-MOS and ACIS exposure maps were weighted by 0.4 compared to EPIC-pn exposure maps). <i>Chandra</i> detections dominate the central few arcminutes, while <i>XMM-Newton</i> detections dominate further away.	77
4.6	Limiting flux for AGN in our analysis for EPIC-PN (<i>left</i>), EPIC-MOS (<i>center</i>) & ACIS-I (<i>right</i>) detectors. The limiting flux indicates the minimum flux required for the AGN to be detected with ≥ 250 counts, after suffering Galactic absorption, given the total exposure time towards that region. The flux is calculated using N_H values and combined exposures of all observations used in our analysis. We use a grid size of $4.5'$.	80
4.7	Comparing the Fe colors of simulated CVs with that of observed sources. Using twice the EWs of Fe lines of IPs in Xu et al. (2016) fits our observed sources better. We use these simulated CVs to estimate the probability of CVs among our candidate continuum sources.	83

4.8	Fe excess (<i>left</i>) and spectra (<i>right</i>) of Hailey et al. (2018) sources detected in CSC v2.0. Only 2CXO J174539.4-290041 and 2CXO J174540.3-290049 lie within the 2σ bounds of MSP-like Fe colors. We also show the spectra of these sources, which reveal Fe lines in the other three; 2CXO J174540.4-290036, 2CXO J174539.4-290046, 2CXO J174539.2-290049.	85
4.9	Limiting unabsorbed luminosity for GB sources to have ≥ 250 counts in our analysis for EPIC-PN (<i>left</i>), EPIC-MOS (<i>center</i>) & ACIS-I (<i>right</i>) detectors. The flux is calculated using N_H values and combined exposures of all observations used in our analysis. We use a grid size of $4.5'$	86
5.1	Distribution of net counts between 0.5–10.0 keV in our selected PN (<i>left</i>) and MOS (<i>right</i>) spectra. We also show the distribution of net counts for different kinds of X-ray sources in our labelled set. We only use spectra with ≥ 100 counts for our analysis. . . .	106
5.2	Distribution of background-to-net count ratio for our selected PN (<i>left</i>) and MOS (<i>right</i>) spectra (in the 0.5–10.0 keV range). We also show these distributions for different kinds of X-ray sources in our labelled set.	106
5.3	Mean normalized PN spectra of the different classes of X-ray sources in our labelled set. The spectrum of stars shows a peak around ~ 0.8 keV. This is due principally to the combined effects of Fe-L/Ne-K emission lines. The mean spectra of AGN, CVs and NS/BH are largely featureless.	107
5.4	Mean normalized MOS spectra of different classes of X-ray sources in our labelled set. These spectra show similar features as the mean normalized PN spectra.	108

5.5	Performance of our models vs. net counts in the spectra. The top-left sub-figure shows the performance of PN spectra using the pseudolabeling algorithms, the top-right sub-figure shows the performance of MOS spectra with the pseudolabeling algorithm, the bottom-left sub-figure shows the performance of the autoencoder model for the PN spectra, and the bottom-right sub-figure shows the performance of the autoencoder model on the MOS spectra. In general, we see that the model classifies the spectra with an accuracy, recall and precision $\geq 90\%$ on spectra with net counts $\gtrsim 300$	113
-----	--	-----

5.6	Performance of our models with background-to-net-count ratio in the spectra. The top-left sub-figure shows the performance of PN spectra using the pseudolabeling algorithm, the top-right sub-figure shows the performance of MOS spectra with the pseudolabeling algorithm, the bottom-left sub-figure shows the performance of the autoencoder model for the PN spectra, and the bottom-right sub-figure shows the performance of the autoencoder model on the MOS spectra. The performance of the models decreases with increasing background.	114
-----	--	-----

5.7	Mean normalized spectra of correctly vs. incorrectly classified sources. The top-left sub-figure shows the mean PN spectra using the pseudolabeling algorithm, the top-right sub-figure shows the mean MOS spectra with the pseudolabeling algorithm, the bottom-left sub-figure shows the mean PN spectra for the autoencoder model, and the bottom-right sub-figure shows the mean MOS spectra using the autoencoder model. Stars mis-identified as AGN do not tend to show the strong peak in the 0.8–1.0 keV range. The increased noise in the incorrectly classified spectra is from the lower photon counts in these spectra, and the small number of incorrectly classified sources in the test set ($\mathcal{O}(10)$).	115
5.8	Performance of our trained ML models on PN and MOS spectra with < 100 counts. The top-left sub-figure plots the PN spectra classified using the pseudolabelling algorithm, the top-right sub-figure plots the MOS spectra classified with the pseudolabelling algorithm, the bottom-left sub-figure shows the classifications on PN spectra using the autoencoder model, and the bottom-right sub-figure shows classifications on the MOS spectra using the autoencoder model. The performance of our models declines steeply for the spectra with low photon counts.	116
5.9	Mean normalized spectra of correctly vs. incorrectly classified sources. The top-left sub-figure shows the mean PN spectra using the pseudolabeling algorithm, the top-right sub-figure shows the mean MOS spectra with the pseudolabeling algorithm, the bottom-left sub-figure shows the mean PN spectra for the autoencoder model, and the bottom-right sub-figure shows the mean MOS spectra using the autoencoder model. The spectra show similar features as the mean normalized spectra of stars and AGN in the labeled set.	117

5.10	Hardness ratios of 1000 predicted AGN and stars, calculated from the PN spectra. We show the predictions of the pseudo-label model on the left, and those of the autoencoder on the right. We see that both models are able to predict stars over a wide range of intrinsic hardnesses and absorption column densities.	118
5.11	Grad-CAM heatmap for an example AGN and a star. Depending on the spectra, different features can drive the classification of the source. We note that the left plot shows specific searches by the algorithm for line features at 0.6 (O), 0.8 (Fe), 1.0 (Ne), 1.3 (Mg) and 1.8 (Si) keV, typical of stars.	119
5.12	Unsupervised UMAP projection of the latent variables encoded using the trained autoencoder model. We see that the labeled AGN and stars are separated as distinct clusters in this manifold. The figures on the left show the distributions of the latent variables for the PN spectra, and those on the right for the MOS spectra. We see that the labeled AGN and stars are mostly separated in these projections.	121
5.13	Example confusion matrices from 4-class pseudo-labeling (left) and autoencoder (right) methods on the PN spectra. While the algorithms are able to identify AGN and stars, the ‘NS/BH’ and ‘CV’ sources are not classified accurately.	122
5.14	UMAP projection of the latent space in the 4-class autoencoder. AGN and stars show good separation. While we see grouping of some of the ‘NS/BH’ sources, the ‘NS/BH’ and the ‘CV’ classes do not show clear clustering of X-ray spectra as seen for the AGN and stars.	122

A.1	Normalized model spectra of stars for 90-percentile intervals in N_H and kT . The spectra have been normalized such that the sum of counts in all energy channels is one. These spectra also do not account for the Poisson error due to low counts. For low N_H , we see a dominant Fe-L/Ne-K emission line, which disappears for higher N_H (we only see Si and S lines for $N_H = 4.8 \times 10^{22}$). Similarly at lower kT , we clearly see Fe-L, Ne-K, Mg-K and Si-K lines, but these aren't dominant at higher temperatures.	131
A.2	Normalized spectra of AGN for 90-percentile intervals in N_H , and Γ . We fix the redshift to 0.7 (most probable value in the distribution of z). The features in the spectra are mainly due to the effective area of <i>Chandra</i> ACIS and absorption edges. In this figure, we only show AGN of type <code>tbabs*ztbabs*pegpwr1w</code> . Some AGN also have an additional soft component and an Fe-K line. 4% of the AGN are better represented by a <code>pexrav</code> model.	131
A.3	(<i>Left</i>) Mean accuracy on the observed spectra and the (<i>Right</i>) standard deviation in accuracy on the simulated spectra for different numbers of nodes and the L_1 regularization values, λ s. We notice that while the accuracy seems to increase with the number of nodes and decreasing λ , the standard deviation also increases, making these models less robust when the data has low signal-to-noise ratios.	133
A.4	Best fit weights for the ANN model with 20 nodes in the hidden layer and L_1 regularization value, $\lambda = 0.0001$. We see that these weights are more variable than those in Fig. 3.15.	134
A.5	Change in classification accuracy with the number of spectra used for training. The accuracy seems to saturate at $\sim 90\%$ at $\sim 10^5$ training points.	134

B.1	Comparing the absorption column density from spectral fitting ($N_{H,spec}$) to that calculated from dust extinction maps ($N_{H,dust}$).	136
B.2	<i>Left:</i> Probabilities that observed sources have no detected Fe lines $p(NL)$; <i>Center:</i> probabilities that simulated CVs do not show sufficient evidence of Fe lines i.e. $p(NL CV)$; <i>Right:</i> probability that sources with no detected Fe lines could be CVs. All results shown here are for sources that have Fe colors consistent with that of MSPs. We also show the histograms of the observed sources, simulated CVs, and candidate continuum sources (with MSP-like colors). Given that none of our simulated CVs with ≥ 3000 counts had colors consistent with simulated MSPs, we assume that all sources with ≥ 3000 counts and showing no lines on spectral analysis are continuum sources.	138

Chapter 1

The X-ray Universe

We have come a long way since the first X-ray survey by the *Uhuru* satellite, which had a limiting sensitivity of $\sim 10^{-11}$ ergs s $^{-1}$ cm $^{-2}$ (Giacconi et al. 1971; Forman et al. 1978; see Santangelo et al. 2022 for a detailed history of X-ray instruments). The use of Charge-Coupled Devices (CCDs) in X-ray detectors (see Lumb et al. 1991; Edgar 2011 for reviews) allowed the measurement of direction, time of arrival and energy of X-ray photons simultaneously, and revolutionized the field of X-ray astronomy. Modern X-ray telescopes such as *Chandra* (Weisskopf 2003) and *XMM-Newton* (Jansen et al. 2001) have much better sensitivity and angular resolutions, allowing them to detect and study faint sources against the background; greater temporal resolution to analyze X-ray variability; and sufficient spectral resolution to probe the emission processes of X-ray sources.

These advances in X-ray astronomy have allowed us to identify several kinds of X-ray sources, such as normal stars, isolated neutron stars (NSs), accreting NSs and black holes (BHs) in X-ray binaries (XRBs), active galactic nuclei (AGN), accreting white dwarfs in cataclysmic variables (CVs), supernova remnants, etc. These sources can have temperatures $\gtrsim 10^6$ K, extreme densities (up to $\gtrsim 10^{14}$ g cm $^{-3}$), and strong gravitational and magnetic fields, allowing us to probe the behaviour of matter in these extreme conditions (refer to

Pavlov et al. 2002; Kaspi et al. 2006; Remillard and McClintock 2006; Lattimer and Prakash 2007; Padovani et al. 2017 for detailed reviews). However, our understanding of these sources is limited by the low photon count rates from most X-ray sources, making it difficult to identify the type of X-ray source. Efficient ways to categorize X-ray sources in large catalogs would enable us to study the populations of these sources across the Galaxy and the universe, and understand the structure and evolution of stars and galaxies.

1.1 Types of X-ray sources

Several astronomical sources emit X-rays through thermal processes (e.g. bremsstrahlung & black body radiation) and non-thermal mechanisms (e.g. synchrotron radiation & inverse Compton scattering). In this section, I shall give a brief overview of the important classes of X-ray emitters that are relevant to this thesis. We shall particularly focus on the X-ray spectra of these sources in the 0.2–10.0 keV energy range.

1.1.1 Chromospherically active stars

X-rays are emitted by most stars, especially stars with high magnetic activity resulting in bright coronae, through bremsstrahlung (free-free) radiation and emission lines (Vaiana et al. 1981; Schmitt 1997; Drake et al. 2000). This X-ray emission can be further boosted by increased rotation rates for stars in close, tidally-locked binary systems (e.g. Walter and Bowyer 1981, the increased rotation rates result in higher magnetic activity). Given that the X-rays are emitted from hot, low-density plasma, the X-ray spectra of these stars are characterized by strong emission lines. Notable emission features in their X-ray spectra include the Fe-L lines ($\sim 0.8 - 1.0$ keV), Ne-K lines (~ 1.0 keV), Mg-K lines (~ 1.3 keV), and Si-K lines (~ 1.8 keV). The Ne-K and Fe-K lines are particularly strong resulting in a distinctive peak around $\sim 0.8 - 1.0$ keV in

their spectra. Given that the X-ray emission is from hot, low density coronae, the X-ray spectra of these sources can be fit using a thermal plasma model like `apec` (Smith et al. 2001) with plasma temperature $kT \lesssim 1$ keV (Getman et al. 2005b). In some cases, an additional hotter plasma component, with $kT > 1$ keV, is required to explain the higher energy X-rays.

Studying X-ray emission from these active stars is crucial for understanding stellar activity and the physical processes occurring in stellar coronae (Güdel 2004). Examining the X-ray spectra allows us to probe the effect of magnetic fields on hot plasma, plasma heating mechanisms, and energy transport in the atmosphere of these stars. Studying the variability and spectrum of X-ray emission offers insights into stellar evolution, particularly in terms of how stellar winds, rotation, and magnetic fields interact in these dynamic environments.

1.1.2 Active Galactic Nuclei

Most galaxies host massive black holes at their centres. Active galactic nuclei (AGN) refer to those central BHs that actively accrete matter from their surroundings and emit strongly across the electromagnetic spectrum (check Padovani et al. 2017, for detailed review). The X-ray emission from AGN is usually dominated by inverse Compton scattering of low-energy photons from the accretion disk, by the hot optically thin corona surrounding the BH. Synchrotron emission from ultra-relativistic electrons can be important for blazars and radio-loud AGN with strong jets, and during transient events. Given the non-thermal nature of the X-ray emission, their spectra can usually be fit by a power-law continuum with $\Gamma \sim 1.7 - 2.0$. The processing of high-energy X-rays in the accretion disk and the surrounding gas could lead to additional features, like a soft energy excess ≤ 2 keV, and a prominent Fe-K line at 6.4 keV.

The study of X-ray emission from AGN provides crucial insights into the physical properties of the supermassive black holes, their accretion disks, and

the surrounding environments. Analyzing the X-ray emission allows us to study the accretion flows and how they power the outflows and relativistic jets from the central BH. These outflows and jets affect the local and global environment of the host galaxy and the intergalactic medium, influence the star formation rate, and thus play an important role in the large-scale evolution of galaxies (Fabian 2012).

1.1.3 Cataclysmic variables

Cataclysmic variables (CVs) are close white-dwarf (WD) binary systems with a WD accreting matter from its low-mass companion star. These WDs may have weak magnetic fields, allowing the formation of an accretion disk that reaches down to the WD surface, or magnetic, causing matter to be funnelled onto the magnetic poles via either a partial accretion disk ("intermediate polars", with $B \sim 10^6 - 10^7$ G), or no accretion disk ("polars", with $B \gtrsim 10^7$ G). X-rays are emitted primarily from the shock-heated accretion stream near the WD surface. This nature of X-ray emission leads to hot (few tens of keV) thermal bremsstrahlung spectra with strong ionized Fe-K emission lines at 6.7 and 7.0 keV, sometimes accompanied by super-soft black-body like emission with $kT \sim 0.02 - 0.1$ keV. The presence of cooler matter in the accretion disk can also lead to a fluorescent 6.4 keV Fe-K emission feature (e.g. Fabian et al. 2000).

CVs are important laboratories to understand accretion-related physics. The relative simplicity of non-relativistic accretion in CVs allows us to learn the accretion geometry, shock structure and boundary layer effects before applying them to understand more complicated accretion-driven compact objects like X-ray binaries and AGN (e.g. Mukai 2017). Some CVs are progenitors of Type Ia supernova and hence have important applications in understanding the large scale structure and evolution of the universe. CVs, especially hard intermediate polars, contribute to a major chunk of the unresolved X-ray emission in the Galactic ridge, and understanding their population would allow us to search for

other kinds of X-ray sources, like millisecond pulsars (see next section), in the Galaxy.

1.1.4 Non-interacting neutron stars

Neutron stars (NSs) are the remnants of massive stars that exploded in supernovae, leaving behind a compact object with a mass typically between 1.4 and $2.0 M_{\odot}$. Isolated (non-accreting) neutron stars include pulsars with energetic pulsar wind nebulae, magnetars with extremely strong magnetic fields ($\geq 10^{14}$ G), young central compact objects found in supernova remnants, and rapidly rotating low-magnetic field neutron stars with hot magnetic poles (Kaspi et al. 2006). These conditions result in thermal black-body like emission from the NS surface/atmosphere (Pavlov et al. 2002), and non-thermal emission from the NS magnetosphere and/or the pulsar wind nebulae surrounding the NS (Slane 2017). Binary millisecond pulsars like black widows and redbacks can also host an intra-binary shock that can emit bright high-energy X-rays (Roberts 2013). The soft thermal emission can usually be modelled by a black-body model or a neutron star atmosphere model and the hard, non-thermal emission can be modelled by a power-law continuum with $\Gamma \sim 1 - 2$ (Zavlin et al. 2002). The X-ray spectra of some bright NS may also show features such as cyclotron absorption lines, which provide crucial information about the strength of the magnetic field near the star's surface (Bignami et al. 2003).

Studying isolated NS spectra is crucial for understanding the fundamental physics of dense matter. Determining the surface temperature and magnetic field structure of NSs through X-ray analysis allows us to study neutron star cooling properties and magnetic field decay without the interference of accretion from a companion (e.g. Heinke 2025). This provides valuable insights into constraining the neutron star's interior physics. Comparing the properties of different populations of neutron stars like magnetars, rotation powered pulsars, and central compact objects allows us to understand their formation mechanisms

and thus probe supernovae explosions. Understanding the properties of the magnetosphere and pulsar wind nebulae provides us critical insights into the behaviour of matter under extreme physical conditions.

1.1.5 X-ray binaries

X-ray binaries (XRBs) are interacting binary systems in which a central neutron star (NS) or black hole (BH) accretes matter from a companion donor star. They are broadly classified into low-mass X-ray binaries (LMXBs) and high-mass X-ray binaries (HMXBs) based on the mass of the companion, and typically the mass-transfer method and NS magnetic field (Charles and Coe 2006). LMXBs have late-type donors with mass $\lesssim 1 M_{\odot}$ that generally transfer material through Roche-lobe overflow (matter at the closer edge of the companion star feels an equal gravitational tug from the compact object, so a stream of matter flows toward the compact object), while in HMXBs, the compact object accretes matter ejected by an early-type, high-mass star ($M \gtrsim 10 M_{\odot}$) through stellar winds, and/or excretion disks (matter ejected at the equator by fast-spinning Be stars). Soft X-rays are emitted by matter in the accretion disk, while inverse-Compton scattering of these photons by electrons in the hot ionized corona around the central NS or BH produces the hard X-ray continuum (Done et al. 2007). Given the complex nature of the emission, the spectra of XRBs are characterized by a combination of soft thermal disk emission and a hard power-law tail with $\Gamma \sim 1.7\text{--}2.0$, often accompanied by Fe-K emission features around 6.4 keV.

XRBs provide crucial insights into the nature of compact objects and the physics of accretion. The spectral and variability properties of transient LMXBs that exhibit X-ray outbursts allow us to study accretion disk instabilities, the formation of relativistic jets, and transitions between hard and soft spectral states. In NS XRBs, X-ray spectra and light curves constrain the mass and radius of the neutron star, providing valuable probes of the neutron star equation of

state. In BH XRBs, studying the inflow of matter toward the central black hole helps us understand the behaviour of matter in strong gravitational potentials and the coupling between accretion flows and jet formation (Miller-Jones et al. 2012).

1.2 Serendipitous X-ray source catalogues

The previous section discussed the importance of studying X-ray emission from various astrophysical questions to understand fundamental questions in physics and astronomy. However, the X-ray sky is vast, and the field of view of X-ray telescopes like *Chandra* and *XMM-Newton* is only a few tens of arcminutes. Thus, collecting data from X-ray sources across the universe can be challenging.

Dedicated observations of X-ray sources through X-ray telescopes allow us to answer important science questions. These observations also detect several other X-ray sources in the field of view. Serendipitous source catalogs of these X-ray detections generated through standard pipelines provide important clues in studying X-ray source populations, understanding the cosmic X-ray background, monitoring time-variable sources, and revealing previously unknown objects (e.g. Kim et al. 2007). In this work, we shall mainly discuss two such catalogs from the X-ray telescopes *Chandra* and *XMM-Newton*.

1.2.1 *Chandra* Source Catalog

The *Chandra* Source Catalog (CSC) is the comprehensive catalog of X-ray sources detected by the *Chandra X-ray observatory* (Evans et al. 2010). The latest version (2.1) provides the properties of 407,806 unique point and extended sources, with more than 1.3 million detections from ACIS and/or HRC-I observations released to the public before the end of 2021 (Evans et al. 2024). The total sky coverage is ~ 730 square degrees.

The latest version of CSC includes astrometric, photometric, spectral and

variability properties along with information on the significance of each detection. It also provides access to high level data products like X-ray light curves, spectra, exposure maps, etc. Thus, the catalog serves as a great tool for population studies, as well as for detailed analysis of individual sources.

1.2.2 *XMM-Newton* Serendipitous Source Catalogue

The *XMM-Newton* Serendipitous Source Catalogue is a comprehensive list of all *XMM-Newton* source detections provided by the *XMM-Newton* Survey Science Centre on behalf of ESA (Webb et al. 2020). The latest release, 4XMM-DR14, presents the properties of 1,035,832 detections of 692,109 unique sources detected from the 13864 *XMM-Newton* EPIC observations taken before the end of 2023. The total sky coverage with ≥ 1 kilo-second exposure is 1383 square degrees. The Survey Science Centre also provides the stacked catalogue, 4XMM-DR14s, that is compiled from 1751 groups of overlapping observations.

The 4XMM Newton Catalogue provides an extensive list of parameters including the positions, source extents, fluxes across 5 narrow bands and 3 broad bands, hardness ratios across the different bands, and intra- and inter-observation variability of each source. It also provides us access to high-level data products such as spectra and light curves from each of the EPIC-PN cameras. *XMM-Newton* EPIC cameras have moderate angular, spectral, and time resolution, with a wider field of view and more effective area than *Chandra*, making it ideal to study different kinds of X-ray sources across the universe.

1.3 On the road to this thesis

While the CSCv2.1 and 4XMM-DR14 catalogs provide properties and standardized data products for hundreds of thousands of X-ray sources, most of these are serendipitously detected and have not been well-studied. X-ray spectra of these sources could help us understand the X-ray emission and the type

of X-ray source. But a detailed analysis of the X-ray spectra would be too time-consuming for all the detected X-ray sources. Hardness ratios could allow us to estimate source properties, but the X-ray emission lines of faint sources can sometimes be confused with their continuum, leading to incorrect classification of X-ray sources. Therefore, we need more efficient classification techniques that can distinguish the spectra of different X-ray classes.

In this thesis, we explore methods to distinguish between the X-ray spectra of different X-ray sources using machine learning (ML) and tailored X-ray colours. We argue that identifying strong X-ray emission lines in moderate-resolution X-ray CCD spectra, and thus separating line-dominated X-ray sources from continuum ones, can allow us to classify X-ray sources. We shall briefly introduce machine learning concepts in chapter 2. In chapter 3, we test if supervised machine learning through neural networks can distinguish between the *Chandra* spectra of AGN and stars. Through the results of this analysis we determine the conditions under which ML algorithms work the best and get their limitations. In 4, we use tailored X-ray colors to identify a candidate MSP population in the Galactic Bulge, and check if a population of MSPs can explain the observed γ -ray excess around the galactic centre. We finally attempt to classify the *XMM-Newton* spectra in the 4XMM-DR14 catalogue through semi-supervised learning in chapter 5. We conclude our observations and look into future prospects in chapter 6

Chapter 2

An overview of machine learning concepts

The enormous amounts of data collected from high-throughput astrophysical observatories like the Gaia mission (Gaia Collaboration et al. 2023), the Zwicky Transient facility (Bellm et al. 2019), CHIME (CHIME Collaboration et al. 2022) and eROSITA (Predehl et al. 2021) have transformed the field of astrophysics. Modern telescopes routinely detect millions of sources spanning a wide range of flux, morphology, variability and spectral shapes. Extracting meaningful physical insights from such heterogeneous high-dimensional datasets requires identifying complex patterns and non-linear relationships, and detecting possible anomalies. In recent years, machine learning (ML) has been shown to be a powerful tool for automated classification, regression and clustering of astrophysical sources with high accuracy and scalability for the vast amount of data collected in astronomy (Ball and Brunner 2010; Fluke and Jacobs 2020; Ivezić et al. 2020).

Classical statistical and parametric model fitting approaches, though powerful, often rely on simplifying assumptions such as linearity and typically operate in a low-dimensional feature space. Therefore, they become inadequate when dealing with complex, non-linear relationships, noisy measurements, or

incomplete data — problems often found in large astronomical catalogs. Several astrophysical problems like classification of X-ray sources through their multi-wavelength properties, photometric red shift estimation, and transient detection involve **multi-dimensional, non-linear parameter dependencies that are difficult to model analytically**. In contrast, ML algorithms are extremely capable of learning intricate relationships directly from the data.

The large datasets in astronomy, **minimal ethical concerns** and the general **culture of sharing observations** make astronomy an ideal ground to use ML. ML has been successfully applied in a broad range of astrophysical research, such as estimation of galaxy redshift from photometry (Salvato et al. 2019), classifying galaxies based on their morphology (Dieleman et al. 2015), identifying transient events (Bloom et al. 2012; Goode et al. 2025), detecting exoplanets (Shallue and Vanderburg 2018), and classifying X-ray sources (Tranin et al. 2022) based on multi-wavelength properties. With future projects like Vera C. Rubin Observatory’s Legacy Survey of Space and Time and the ngVLA, ML will play a crucial role in **understanding the data**.

2.1 Foundations of Machine learning

ML methods aim to extract patterns and predictive relationships from data without **explicit analytical methods and rule based approaches**. At its core, ML seeks to learn a mapping between an input vector x (of observables, e.g. X-ray fluxes, hardness ratio, brightness, etc.) and an output y (derived properties or source class). This mapping is defined by the model $f(x, \theta)$, where θ are model parameters that are optimized from the input data. The performance of the model depends not only on the choice of the mapping algorithm, **but also on how the data is represented in the loss function used for learning, and the optimization strategy involved**. In this section, we briefly explain the different elements of **an ML workflow**.

2.1.1 Data and feature representation

ML methods require data to be expressed in numerical forms that can be used during optimization of the model. The input space will then consist of N input vectors, where each input $\mathbf{x}_i \in \mathbb{R}^d$ can be expressed as

$$\mathbf{x}_i = [x_{i1}, x_{i2}, \dots, x_{id}]^T, \quad (2.1)$$

where each value x_{ij} is called a feature value of the input. Thus the input space can be expressed as $\mathcal{X} \in \mathbb{R}^{n \times d}$, where the columns are different feature vectors. Depending on the nature of the data the features may correspond to scalar values, structured arrays or derived parameters from pre-processing the data, dimensionality reduction techniques, etc.

Choosing features is an important step in constructing an ML workflow. A well-chosen feature space ensures that each feature provides new information about the data, removes irrelevant or dependent properties and enhances the separability of the data. Traditional ML involved manual feature selection based on domain expertise and dimensionality reduction techniques such as principal component analysis (PCA). However, with deep learning, the initial layers replace manual feature selection with representation learning, where the network discovers the most discriminative features directly from the data, leading to superior performance.

In several cases, feature vectors are standardized using z-score standardization,

$$\tilde{x}_{ij} = (x_{ij} - \mu_j) / \sigma_j, \quad (2.2)$$

where μ_j and σ_j are the mean and standard deviation of the feature vectors. This makes the magnitude of all vectors similar and ensures that all the features can contribute to the learning process.

2.1.2 Training, Validation, and Testing

An ML task seeks to learn a mapping $f_\theta : X \rightarrow Y$ from input space X to Y , parametrized by θ . The aim of training is to optimize the model parameters given the training data. In general, we aim to train the model such that it not only performs well on the seen training data, but also on new, unseen, data.

To ensure generalization of the model, the available data is divided into:

- **Training set:** Used to update model parameters. That is, given a training dataset, $\{\mathbf{x}_i, y_i\}_{i=1}^N$, we aim to maximize the likelihood $L(y_i, f_\theta(\mathbf{x}_i))$.
- **Validation set:** Used to monitor performance during the training and tune hyperparameters (fixed during optimization process) to improve the performance.
- **Test set:** Used only during the final evaluation, i.e. not seen by the model during training or selection of hyperparameters.

Common evaluation metrics to check the performance of the model are:

- **Accuracy:** Fraction of correct predictions

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (2.3)$$

- **Recall:** Fraction of true positives properly identified in the dataset

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (2.4)$$

- **True Negative Rate (TNR):** Fraction of true negatives properly identified in the dataset

$$\text{TNR} = \frac{\text{TN}}{\text{TN} + \text{FP}} \quad (2.5)$$

- **Precision:** Fraction of true positives among the data predicted by our model to be positive.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (2.6)$$

- **Negative Predictive Value (NPV):** Fraction of true negatives among predicted negatives.

$$\text{NPV} = \frac{\text{TN}}{\text{TN} + \text{FN}} \quad (2.7)$$

TP, TN, FP, and FN refer to true positives, true negatives, false positives and false negatives in our dataset. If a model performs very well on the training data, but not on the validation and test sets, then the model is said to overfit the training data. However, if the model is too simple to capture the underlying structure of the data, we say it is under-fit.

2.1.3 Loss Functions and Regularization

The loss function quantifies the discrepancy between the predicted output and the true labels. For most classification problems, we use the *cross-entropy loss*

$$L_{CE} = -\frac{1}{N} \sum_{i=1}^N \sum_{k=1}^K y_{ik} \ln p_{ik} \quad (2.8)$$

where N is the number of data points, K is the number of classes, y_{ik} is the one-hot encoded label, and p_{ik} is the model's predicted probability for class k . In one-hot encoding, the output labels $y_i \in \mathbb{R}^{K \times \mathbb{C}}$, such that $y_{ik} = 1$ if $\mathbf{x}_i$ belongs to class k , else $y_{ik} = 0$. E.g., if a data point belongs to the first class in a 4-class classification model, the corresponding label would be $[1, 0, 0, 0]$.

To prevent overfitting, we can regularize the model by adding penalty terms to the loss function. A common sign of overfitting is the flattening of the loss gradients, that results in high fluctuations and magnitude of the model parameters. Therefore, common regularization techniques add penalty terms

based on the magnitude of model parameters. These include L_1 regularization,

$$L_1 = \lambda_{L1} \sum_j |\theta_j| \quad (2.9)$$

and L_2 regularization,

$$L_2 = \lambda_{L2} \sum_j \theta_j^2 \quad (2.10)$$

2.1.4 Optimization

After designing the loss function, we iteratively update the model parameters to minimize the loss function. One of the simplest form of this update includes:

$$\theta_{t+1} = \theta_t - \eta \nabla_{\theta} L(\theta_t), \quad (2.11)$$

where η is the learning rate controlling the step size. However, computing gradients over the entire input space can be computationally expensive. To overcome this, stochastic gradient descent (SGD) and its variants are commonly used. In SGD, we divide the input data into mini-batches and update the model after calculating gradients over each mini-batch. We say that the model has trained over one epoch when we have cycled through all the mini-batches in the input data. SGD also introduces noise in the optimization process, allowing the model to escape local minima. Several methods have been proposed to improve the stability and convergence of the SGD methods. The most common among them, Adam (Kingma and Ba 2014), computes an exponentially weighted average of the past gradients (g_t) and their squares. For each update, we calculate:

$$\begin{aligned} m_t &= \beta_1 m_{t-1} + (1 - \beta_1) g_t, \\ v_t &= \beta_2 v_{t-1} + (1 - \beta_2) g_t^2, \\ \theta_{t+1} &= \theta_t - \eta m_T / (\sqrt{v_t} + \varepsilon) \end{aligned} \quad (2.12)$$

Typical values of $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\varepsilon = 10^{-8}$.

Machine learning methods can be broadly classified into supervised, unsupervised and semi-supervised training based on the availability of the output labels. In supervised training, all the input vectors $\mathbf{x}_i$ have known output labels, y_i , and the aim is to find a function to predict y_i , from $\mathbf{x}_i$. In unsupervised learning, none of the input data have output labels, and the algorithm tries to find interesting structure and clusters within the data. When labels are provided for a subset of the input data, we call it semi-supervised training. In these methods, we use the unlabelled data to find important features of the input data, and then use the labelled data to find a mapping between the input data and output labels.

2.2 Supervised training

Supervised training is the most commonly used ML technique, where the model is trained using a labeled dataset containing input-output pairs $(\mathbf{x}_i, y_i)$. During training, the parameters are optimized to minimize the discrepancy between the predicted and true outputs, as defined by a suitable loss function. Once trained, the model should be able to infer labels for previously unseen data enabling predictive classification or regression tasks. In this section, we shall detail the supervised training methods used in this thesis.

2.2.1 Logistic Regression

Logistic regression (LR) is one of the simplest linear learning models for binary classification. It assumes a Bernoulli problem where the probability of an instance belonging to class $y = 1$ is given by a sigmoid transformation, $\sigma(z)$, of a linear combination of input features, i.e.,

$$P(y = 1|\mathbf{x}) = \sigma(\mathbf{w}^T \mathbf{x} + b) = \frac{1}{1 + e^{-(\mathbf{w}^T \mathbf{x} + b)}} \quad (2.13)$$

where $\mathbf{w}$ are the model weights and b is the bias. The sigmoid function ensures that the output probability is in the range $(0, 1)$. Thus, the decision boundary between the classes corresponds to $P(y = 1|\mathbf{x}) = 1$ i.e. $\mathbf{w}^T \mathbf{x} + b = 0$ (hence a linear classifier).

For multi-class classification (with K classes), LR can be generalized into softmax regression:

$$P(y = k|\mathbf{x}) = \frac{e^{(\mathbf{w}_k^T \mathbf{x} + b_k)}}{\sum_{j=1}^K e^{(\mathbf{w}_j^T \mathbf{x} + b_j)}} \quad (2.14)$$

Note that each class, k , will have its own set of weights $\mathbf{w}_k$. Such models can be trained using the cross-entropy loss (described by Eq. 2.8).

2.2.2 Artificial Neural Networks

Several real world problems cannot be solved by linear classifiers. Artificial Neural Networks (ANNs) extend the logistic regression to detect complex decision boundaries by introducing one or more hidden layers of neurons that apply non-linear transformations on the input data. A typical feed-forward neural network consists of an input layer, one or more hidden layers, and an output layer. The layers are connected by weights $\mathbf{W}$ and biases $\mathbf{b}$, that are learned during training. Mathematically, each hidden layer can be expressed as:

$$\mathbf{h}^{(l)} = g(\mathbf{W}^{(l)} \mathbf{h}^{(l-1)} + \mathbf{b}^{(l)}), \quad (2.15)$$

where $\mathbf{h}^{(l)}$ is the output of the l^{th} layer and $g(\cdot)$ is the chosen non-linear activation function. Common activation functions include the sigmoid function, $1/(1+e^{-z})$, Rectified Linear Unit, $\text{ReLU max}(0, z)$, the hyperbolic tangent, $\tanh(z)$, and softmax in the final layer for output probabilities. The ANN model can then be optimized by the cross-entropy loss.

2.2.3 Convolutional Neural Networks

Convolutional neural networks (CNNs) are a type of neural networks designed to exploit the spatial, sequential, or temporal structure of the data. Unlike ANNs where each feature is treated independently, CNNs look for local connectivity in the dataset through convolution kernels that slide across the input to detect patterns. The convolution operation is given by:

$$(\mathbf{I} * \mathbf{K})(i) = \sum_m \sum_n I(i - m)K(m) \quad (2.16)$$

where $\mathbf{I}$ is the input (here we assume 1D input, but it can also be a 2D image/map), and $\mathbf{K}$ is the kernel that extracts local features. Each kernel produces a feature map that highlights the locations where the feature is located in the input. We then apply the activation functions and pooling layers that reduce the resolution of the feature map, while preserving important information. One of the common pooling functions, max-pooling, outputs the maximum value over a region of specified size.

Finally the extracted features can be flattened and passed into one or more fully connected layers to calculate the output.

2.3 Unsupervised training

Unsupervised learning aims to uncover the underlying structure and patterns in the data, and learn representations in a lower dimension, without using labelled examples. Unsupervised algorithms aim to infer the distribution of the input data and find relationships across multiple dimensions.

2.3.1 Dimensionality reduction

Machine learning on high-dimensional datasets suffers from the curse of dimensionality — as the volume of the input space increases and the input data

distribution becomes sparse, quantities such as Euclidean distance become less meaningful since all the data points are further away from each other, and we need more complex models with a large number of parameters, increasing the chance of overfitting. Thus, it is important to remove redundant or correlated features and project the data into a lower-dimensional space that captures the most important structures while minimizing the information lost. This allows for more efficient training, noise reduction, and visualization of the data.

Principal Component Analysis

Principal component analysis (PCA) is a linear transformation technique to identify the directions (principal components) of maximum variance in the data. Mathematically, given the input $\mathbf{X} \in \mathbb{R}^{N \times d}$ (i.e N input vectors with d dimensions each), PCA learns an orthogonal projection that maximizes the variance in the projected data. PCA first performs the eigenvalue decomposition on the covariance matrix, $\mathbf{C}$,

$$\begin{aligned}\mathbf{C} &= \frac{1}{N-1}(\mathbf{X} - \bar{\mathbf{X}})^T(\mathbf{X} - \bar{\mathbf{X}}), \\ \mathbf{C}\mathbf{w}_k &= \lambda_k\mathbf{w}_k,\end{aligned}\tag{2.17}$$

where $\mathbf{w}_k$ is the k^{th} eigenvector, and λ_k is the corresponding eigenvalue representing the variance along each eigenvector. The data is then projected onto the first m eigenvectors (principal components),

$$\mathbf{Z} = \mathbf{X}\mathbf{W}_m,\tag{2.18}$$

where $\mathbf{W}_m = [\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_m]$.

This reduced space can then be used for visualization or as input features for downstream ML methods. However, PCA can only capture linear relationships among features, and completely ignore components with smaller eigenvalues. Thus, datasets with non-linear relationships, or separation not along the direc-

tion of highest variance, might lose crucial information from this dimensionality reduction.

Uniform Manifold Approximation and Projection

The Uniform Manifold Approximation and Projection (UMAP) aims to learn a low-dimensional embedding that preserves both local and global structure of the data. UMAP first constructs a weighted k-nearest neighbor graph (kNN) to capture local structure. It assigns a membership strength to model the similarity between data points i and j :

$$\mu_{ij} = \exp \left(-\frac{d(i, j) - \rho_i}{\sigma_i} \right), \quad (2.19)$$

where $d(i, j)$ is the distance between points i & j , ρ_i is the distance to the nearest neighbour, and σ_i controls smoothness. The algorithm then optimizes a low-dimensional embedding that minimizes the cross-entropy between the high-dimensional fuzzy graph and its low-dimensional counterpart. This cross-entropy is given by,

$$\mathcal{L}_{\text{UMAP}} = \sum_{i \neq j} \left[\mu_{ij} \ln \frac{\mu_{ij}}{\nu_{ij}} + (1 - \mu_{ij}) \ln \frac{1 - \mu_{ij}}{1 - \nu_{ij}} \right], \quad (2.20)$$

where ν_{ij} represents the similarity/membership strength in the lower dimensions.

Thus UMAP ensures that points that are close in the original space remain close after the dimensionality reduction.

2.3.2 Autoencoders

Autoencoders are neural networks designed for unsupervised representation learning. They aim to learn a compact encoding of the input data by training the network to reconstruct its own input after passing through a bottleneck latent space with lower dimensions than the input data.

The autoencoder consists of two parts: an encoder $f_{\theta}(\mathbf{x})$ that maps input

$\mathbf{x}$ to a lower-dimensional latent representation $\mathbf{z}$, and a decoder $g_\phi(\mathbf{z})$ that reconstructs the original input from the latent vector $\mathbf{z}$, i.e. $\mathbf{z} = f_\theta(\mathbf{x})$ and $\hat{\mathbf{x}} = g_\phi(\mathbf{z})$. The model is then trained by minimizing a reconstruction loss function.

The mean square error is one of the most commonly used reconstruction loss functions.

By constraining the dimensionality of the latent space $\mathbf{z}$, the autoencoder is forced to learn the most informative features of the data. Autoencoders can have deep networks for encoders and decoders and additional loss terms to impose additional properties of the model parameters, latent space, and reconstructed outputs, making them extremely flexible for a wide range of problems. When combined with a classification objective, autoencoders can be used for semi-supervised learning, leveraging both labelled and unlabelled data.

2.4 Semi-supervised learning

In astrophysics and several other real-world scientific applications, the availability of the labelled data is limited by the difficulty or computational cost, while unlabelled data is abundant. Semi-supervised learning (SSL) addresses this challenge by using both the labelled and unlabelled set to improve the generalization of the models. SSL aims to exploit the structures and patterns in the unlabelled data to guide the learning on the labelled set.

Suppose the dataset consists of a labelled set $\mathcal{D}_L = \{(\mathbf{x}_i, y_i)\}$ and unlabelled set $\mathcal{D}_U = \{\mathbf{x}_j\}$. The objective of SSL is to minimize the total loss function,

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{sup}}(\mathcal{D}_L) + \alpha \mathcal{L}_{\text{unsup}}(\mathcal{D}_U), \quad (2.21)$$

where $\mathcal{L}_{\text{sup}}$ is the supervised training loss that includes both the input and the target output, e.g. cross-entropy, and $\mathcal{L}_{\text{unsup}}$ is the unsupervised loss that checks for the consistency or meaningful outputs of the model on the unlabelled input vectors, e.g. reconstruction loss. The hyperparameter α balances the two losses.

2.4.1 Autoencoders with classification loss

Adding a classification objective to the autoencoders can ensure that the latent space can ensure separability between the classes while also being able to reconstruct the input. In this framework, along with the encoder $f_\theta(\mathbf{x})$ and decoder $g_\phi(\mathbf{z})$, the model also has a classifier $h_\psi(\mathbf{z})$ that predicts the output classes.

Thus the total autoencoder loss is given by

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{recon}}(\mathbf{x}, g_\phi(f_\theta(\mathbf{x}))) + \alpha \mathcal{L}_{\text{class}}(y, h_\psi(f_\theta(\mathbf{x}))), \quad (2.22)$$

where the reconstruction loss, $\mathcal{L}_{\text{recon}}$, is calculated for all the input data but the classification loss, $\mathcal{L}_{\text{class}}$, is only computed for the labelled inputs.

2.4.2 Pseudo-labels

Another method of SSL is to extend the classification model trained on the labelled data to the unlabelled set (Lee 2013). The idea is to use the model's own predictions on the unlabelled data as temporary “pseudo-labels”, thereby adding to the labelled dataset. During training, the unlabelled samples that the model classifies with a high confidence are assigned these pseudo-labels and included in the supervised loss computation. These pseudo-labels are re-calculated after every weight update.

During training, we first predict class probabilities $p(y|\mathbf{x}_j)$. The pseudo-label can then be defined as $\hat{y}_{j_j} = \mathbf{argmax}_c p(y = c|\mathbf{x}_j)$ for the unlabelled inputs $\mathbf{x}_j \in \mathcal{D}_u$. The loss functions can then be modified as :

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{sup}}(\mathcal{D}_L) + \alpha \sum_{\mathbf{x}_j \in \mathcal{D}_U} \mathbf{1}(\mathbf{max}(p(y|\mathbf{x}_j)) > \tau) \mathcal{L}_{\text{sup}}(\mathbf{x}_j, \hat{y}_j), \quad (2.23)$$

where $\mathbf{1}$ is the indicator function, i.e., 1 if the condition is satisfied and 0 if it is false, and τ is the confidence threshold. By scheduling α and τ during



the training process so that the unlabelled set doesn't contribute much to the training in the initial stages but more during the final epochs, we can reduce the risk of propagating errors from incorrect pseudo-labels while also regularizing the model to prevent overfitting on the labelled set.

Pseudo-label method is based on the cluster-assumption that the nearby points in feature space share the same label, and that the decision boundaries should lie in a low density region. Thus by self-training iteratively on its own predictions, the model tries to stabilize the decision boundaries by avoiding high density regions.

Chapter 3

Supervised classification of stars in Orion nebula from AGN in CDFS

3.1 Introduction

The X-ray sky consists of a variety of extremely hot objects ($kT \sim 1$ keV). These sources include chromospherically active stars (e.g., Güdel [2004](#); Preibisch and Feigelson [2005](#)), supernova remnants (SNRs; e.g., Vink [2012](#)), isolated neutron stars (NSs; e.g., Kaspi et al. [2006](#); Pavlov et al. [2002](#)), X-ray binaries (XRBs; e.g., Remillard and McClintock [2006](#); Campana et al. [1998](#)), and active galactic nuclei (AGN; e.g., Netzer [2015](#); Padovani et al. [2017](#)). These different kinds of X-ray sources emit radiation through distinct physical processes — active stars and SNRs emit X-rays predominantly from thermal bremsstrahlung and line radiation, young isolated NSs show thermal blackbody-like emission, NSs with strong magnetic fields can also produce synchrotron radiation, and X-ray binaries and AGN cool via inverse Compton scattering (refer Bradt [2014](#), for detailed review of the emission processes).

X-ray sources, especially compact objects, host exotic environments with

extremely hot temperatures, high densities and immensely strong gravitational and magnetic fields (see Remillard and McClintock 2006; Lattimer and Prakash 2007; Turner and Miller 2009, for detailed reviews). The surfaces of young NSs can have temperatures $T \sim 10^6$ K, the centres of NSs can reach densities $\rho \sim 10^{14} - 10^{15}$ g/cm⁻³, magnetars can host magnetic fields of $10^{14} - 10^{15}$ G (Olausen and Kaspi 2014), and black holes (BHs) exert gravitational fields that test the limits of modern physics. Thus, studying these compact objects can help us understand important physics such as nuclear forces, interaction of matter with strong magnetic fields, etc. X-ray emission is universal from these sources due to their temperatures and the high-energy processes involved in emitting radiation, thus making X-ray surveys ideal to search for and study compact objects. However, our knowledge of many of these sources has been constrained by the small numbers of identified objects.

Over the last decades, we have launched several powerful X-ray telescopes to observe these high-energy X-ray sources, including the *Chandra X-ray observatory* (CXO or *Chandra*, for short), the *XMM-Newton* telescope, *eROSITA* instrument etc. These instruments record the position, time of arrival and energy of each X-ray photon detected. These telescopes have detected hundreds of thousands of X-ray sources in the sky — e.g. the *Chandra* Source Catalog (CSC) has $\sim 300,000$ unique sources (Evans et al. 2010), the XMM-Newton Serendipitous Source Catalog has detected $\sim 600,000$ X-ray sources (Webb et al. 2020), and the eROSITA Final Equatorial Depth Survey (eFEDS) survey has detected $\sim 30,000$ sources (Brunner et al. 2022), with several million X-ray sources expected in the final eROSITA survey (Predehl et al. 2021). However, most of these sources haven’t been studied in detail and we do not know the type of the X-ray source. Detailed individualized X-ray spectroscopy to model the X-ray emission of individual X-ray sources, understand their properties, and detect compact objects can be time-consuming for these large catalogs. Automated spectroscopic catalogs (e.g. Corral et al. 2015) can be useful, though

these also have limits (the selection of models, time and storage required for the fitting). Hardness ratios, that compare the number of X-ray photons in the soft X-ray band (say, 0.5–2 keV) and the hard X-ray band (say, 2–6 keV), can also be used to estimate the properties of the X-ray source (Yokogawa et al. 2000; Prestwich et al. 2003; Brassington et al. 2008), and quantile analysis provides an alternative with substantial benefits (Hong et al. 2004). However, hardness ratios do not have strong discriminatory power for faint (and sometimes even moderately bright) sources, which can lead to inaccurate classification of X-ray sources (e.g. Hebbar et al. 2019). Thus, we need efficient ways to identify the X-ray sources detected in large X-ray surveys.

CCD detectors that are used in most X-ray telescopes have an energy resolution of $\Delta E \sim 0.1\text{--}0.2$ keV (Fano 1947; Strüder et al. 2001; Predehl et al. 2021). Such moderate-resolution X-ray spectra are capable of detecting and resolving emission and absorption lines from elements such as neon (Ne), magnesium (Mg), silicon (Si), and iron (Fe). The coronae of active stars emit significant X-ray energy through the Ne-K & Fe-L lines. (Both Ne & Fe emit at ~ 1 keV. The dominant line among them depends on conditions in the stellar coronae. The comparison between the relative strengths of these lines is not important for the moderate resolution spectra used here.) The spectra of SNRs also have bright emission lines from Mg, Si, S, etc. Most AGN have a continuum dominated inverse Compton power-law spectrum from the hot corona. Some AGN also have additional soft energy components and fluorescent emission lines from the surrounding colder gas. But these line features are usually much fainter than the continuum component, except for the Fe-K line at ~ 6.4 keV (which can be redshifted) (Matt et al. 1997; García et al. 2013). NSs typically have continuum X-ray spectra with no emission lines. Typical spectra of these sources are shown in Fig. 3.1. Thus, an ability to identify these emission lines and distinguish between continuum- and line-dominated X-ray spectra will allow us to understand the nature of the X-ray source.

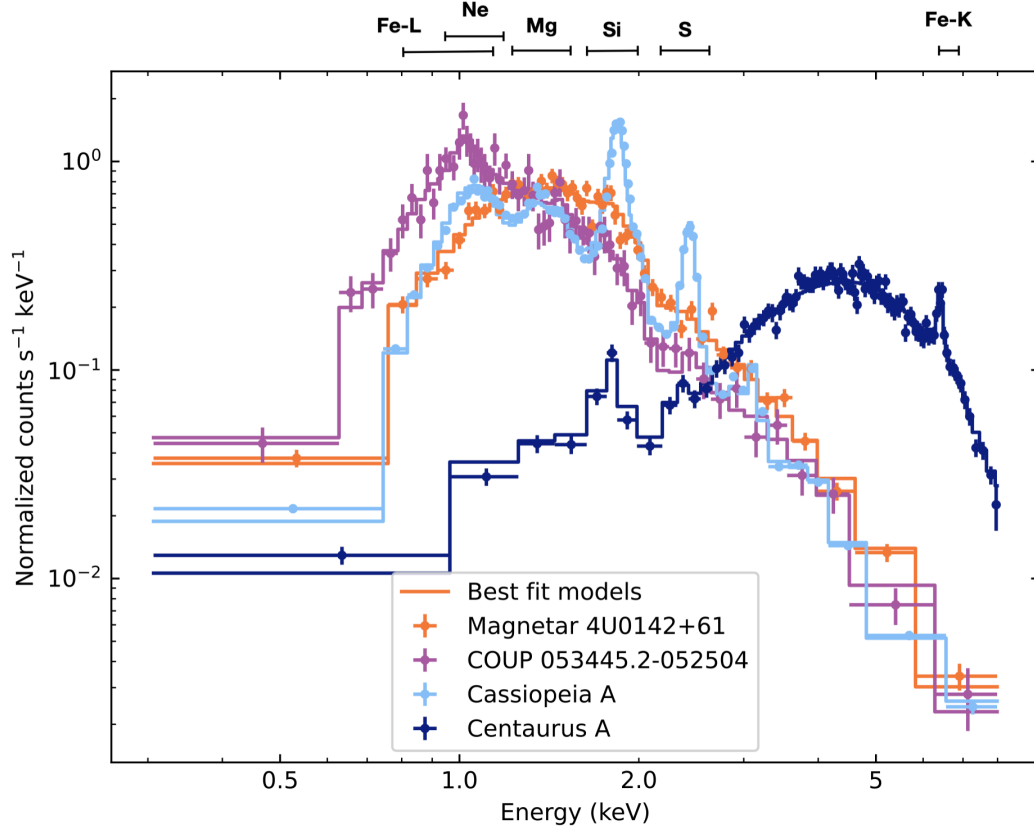


Figure 3.1: Example spectra of an isolated neutron star (4U 0142+61), active star (COUP 053445.2-052504), SNR (Cassiopeia A), and AGN (Centaurus A). We also show the approximate positions of Fe-L, Ne-K, Mg-K, Si-K, S-K, and Fe-K emission lines. The spectra of active stars and SNRs are dominated by these lines while that of AGN and NSs are largely continuous. AGN can sometimes show an Fe-K line.

The large datasets used in astronomy, minimal privacy and ethical concerns in sharing data, etc. make astronomy research a great potential application for machine learning (ML). Machine learning algorithms have been used for calculating photometric redshifts (e.g., Carrasco Kind and Brunner 2013), classification of galaxies and identification of AGN using optical observations (e.g., Cavioti et al. 2014; Rozo et al. 2016; Chattopadhyay et al. 2019, etc.), classification of supernovae (Möller and de Boissière 2020) and modelling their X-ray spectra (Parker et al. 2022; Matzeu et al. 2022), identification of exoplanets (Shallue and Vanderburg 2018), etc. Recently Yang et al. (2021), Schneider

et al. (2021), and Tranin et al. (2022) have used ML algorithms to identify the multiwavelength counterparts of X-ray sources in CSC v2.0, eFEDS, and XMM-Newton data, respectively, using properties such as angular separation, X-ray flux, X-ray hardness ratios, Gaia magnitude and colors, etc., and classify the X-ray sources. While such methods work efficiently for sources far from the Galactic plane, it can be very time-consuming to find multiwavelength counterparts of X-ray sources in highly absorbed or crowded regions such as the Galactic bulge and globular clusters. We also anticipate that combining information from the X-ray spectra themselves along with multiwavelength information will provide more accurate source identification.

3.1.1 Artificial neural networks

Artificial neural networks (ANNs) are supervised learning algorithms that use hidden layers to learn non-linear representations of the data. In a classical feed-forward ANN, a weight matrix is applied on the input data that transforms the dimensionality of input. Then, we apply non-linear activation functions on this transformed input to calculate the values of nodes in the first hidden layer. This process can be repeated on the first hidden layer to calculate the next layer, and so on, and finally, the output layer. Typical activation functions include Rectified Linear Unit (ReLU) and its variants, sigmoid function, softmax function, etc. The choice of activation functions depends on the problem and nature of output needed (check Bishop and Nasrabadi (2007) for detailed reference). In particular, using the sigmoid and the softmax functions that give values between 0 and 1 on the output layer gives a probabilistic interpretation for binary and multi-class classification.

As non-linear classifiers, the ANNs can learn complex relationships in the data and thus lead to better accuracy of classification results. The probabilistic nature of the output allows us to change the threshold based on our requirement. ANNs can also be modified for semi-supervised training through auto-encoders,

and use physical expressions for better interpretation of the results and the training process. Thus ANNs can be adapted to a wide variety of problems. However, ANNs need a large dataset for training and are prone to over-fitting. Thus, it is important that we use proper regularization to avoid spurious weights and test the trained model with a separate dataset that is not used for training (a test set).

In this paper, we aim to develop ANN algorithms that distinguish different kinds of X-ray sources based on their moderate-resolution X-ray spectra from CCDs. We test if ANNs can differentiate young stars from AGN, in simulated and real Chandra X-ray Observatory data. We describe our methods of data generation and the setup of our analysis in §3.2. We show the results of our classification and investigate the robustness of our ANN model in §3.3. We discuss the implications of our results and its broader application in §3.4. We summarize the results of our analysis and discuss future prospects in §3.5.

3.2 Data Acquisition and Analysis

We use ANNs as a supervised training algorithm to test their feasibility. Training and testing a complex ANN requires a large dataset of labelled sources. Thus, we rely on XSPEC `fakeit` simulations of the spectra of stars and AGN for the training. The *Chandra* Orion Ultradeep Project (COUP, Getman et al. 2005b) and the *Chandra* Deep Field South (CDFS, Giacconi et al. 2002) surveys are ideal for extracting the properties of stars and AGN for the simulated spectra. Most of the sources in the COUP survey of a star forming region are young stars, while the majority of the sources detected in the CDFS are AGN. All the COUP observations were conducted in January 2003, thus removing the effects of the changing soft energy response of *Chandra* ACIS. The CDFS observations were taken during the years 2000, 2007 and 2010. We only consider the observations in year 2000 for our analysis as they are the ones used in the CDFS AGN

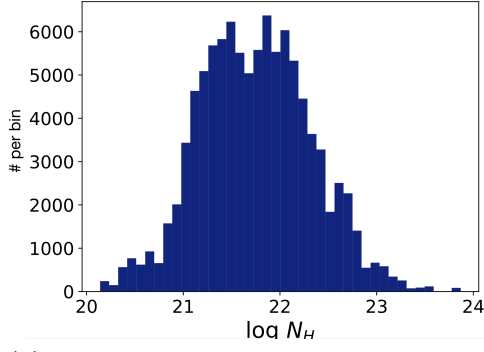
spectral properties catalog (Tozzi et al. 2006). This also ensures a better soft energy response for our AGN data. With a few hundred sources detected in each of the COUP and CDFS surveys, they provide us samples of observed spectra to test our results.

We use the X-ray sources in the *Chandra* Orion Ultradeep Project (COUP, Getman et al. 2005b) survey for our ensemble of young stars. The COUP survey used the ACIS-I instrument of *Chandra* for an exposure time of 838 ks and detected 1616 X-ray point sources, that are mostly active young stars, and modeled them using one or two-component thermal plasma spectra. We utilize their catalog of objects identified as stars, and do not consider sources that are marked uncertain. The COUP catalog specifies the best-fitting parameter values of these models for each COUP spectrum. We use the distribution of these properties to extract the general properties of X-rays from young stars. Since we are interested in getting the distribution of the properties of the COUP spectra, we also omit sources with marginal (null hypothesis property from χ^2 is between 0.05 and 0.005) and poor (null hypothesis probability from χ^2 is less than 0.005) fits. These poor fits could be due to contamination from surrounding sources, high absorption leading to poor fitting of lines, spectra more complicated than that of thermal-equilibrium plasma, etc. Using the membership information of the COUP catalog provided by Getman et al. (2005d), we only select sources within the Orion Nebula Cluster. This gives us a set of 1045 sources. We also remove sources with flags for: deviations in the emission lines (presence of narrow spectral features not explained by the model, probably due to different elemental abundance; 62 such sources), soft or hard excess (mostly due to poor subtraction of a non-uniform background around weak sources; 214 such sources), confusion from nearby sources (54 such sources), two components with different absorption column densities (6 such sources) or a poor fit (poor χ^2 statistics and/or based on visual examination; 89 such sources). Note that these are only removed for the creation of the distribution of spectral properties, not

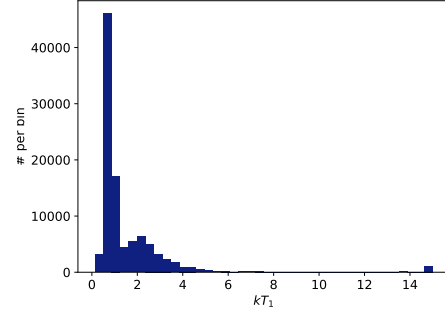
for testing our trained ML model.

This filtering leaves us with 679 sources. Of these 679 sources, 406 sources were modeled with single plasma models and 273 sources were fit with two-temperature plasma models. We use the distributions of N_H (absorption column densities), kT_1 , kT_2 (temperatures of plasma [stellar atmosphere in this case]), and the emission measures of the components of these COUP X-ray spectra models to simulate spectra of 100,000 young stars in our sample. We fix the abundance values to 0.3 times the solar values in accordance with Getman et al. (2005b). We show the distribution of parameters used to simulate our active star spectra in Figs. 3.2. From Figs. 3.2a & 3.2b, we notice that 90% of the stars have $\log N_H$ (in cm^{-2}) $\in (20.82, 22.68)$ and $kT_1 \in (0.43, 3.3)$ keV. We use the `tbabs*apec` and `tbabs*(apec+apec)` models in XSPEC v12.11.1 and the `akeit` command to simulate spectra with an exposure time of 1 megasecond (Ms). (The `tbabs` component is used to model the absorption with Wilms et al. (2000a) abundances and the `apec` models X-ray emission from plasma in thermal equilibrium). Tsujimoto et al. (2005) detected a 6.4 keV Fe fluorescence line in seven of the COUP stars. However, this fluorescent emission is only present in young stellar objects with significant disks, and is weak (usually <150 eV, and weaker than the nearby 6.7 keV Fe line). Therefore, we do not add this line to our simulated stellar spectra.

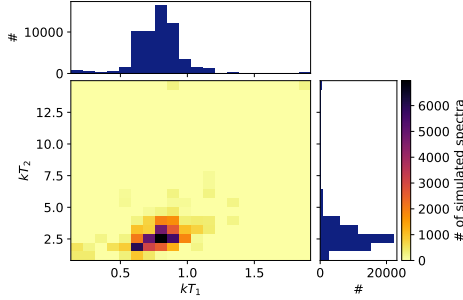
The properties of the AGN X-ray spectra are selected from the *Chandra* Deep Field South AGN spectral properties catalog (CDFSAGNCXO, Tozzi et al. 2006). This catalog lists the properties of 321 high redshift AGN in the 1 Ms *Chandra* Deep Field South (CDFS) survey. Most of these sources were fit with an absorbed power-law model (Compton thin or ‘C-thin’ model). For 8 sources, the fit included an additional soft power-law component (same slope as the absorbed power-law but no absorption; Soft component or ‘Soft-C’ model), and 14 AGN were fit with a reflection-dominated model (Compton thick or the ‘C-thick’ model). Tozzi et al. (2006) fixed the Galactic absorption (from our



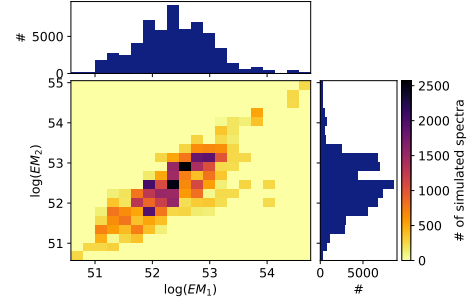
(a) Distribution of absorption column densities, N_H .



(b) Distribution of temperatures, kT_1 .



(c) 2D histogram of kT_1 , and kT_2 used to simulate stars with two-temperature plasma models (kT_1 refers to colder plasma).



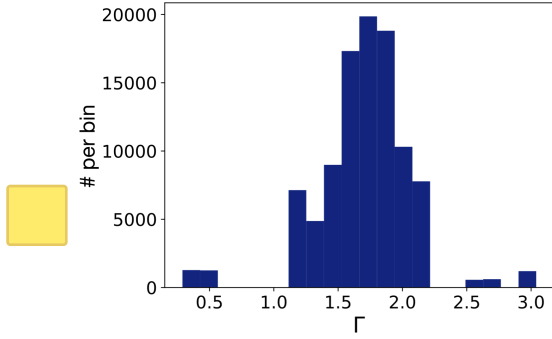
(d) 2D histogram of emission measures used to simulate stars with two-temperature plasma models.

Figure 3.2: Distribution of the parameters used to simulate the spectra of stars for training our ANN. Since the Orion nebula is close to the galactic plane, N_H is high. Most stars have $kT < 2$ keV. The few sources with $kT \approx 14$ keV are likely to be heavily absorbed stars where emission lines could not be fit (and a higher temperature plasma could fit the relatively hard spectra). We see some correlation between kT_1 and kT_2 , and a strong correlation between EM_1 and EM_2 . Since we match the fluxes of stars and AGN, specifying the emission measure values only contributes to maintaining the ratio of emission from the two plasmas.

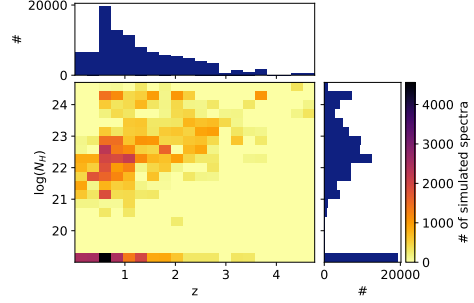
Galaxy) to $N_{H,Gal} = 8 \times 10^{19} \text{ cm}^{-2}$. Accordingly, we use a C-thin model for 93%, C-thick model for 4%, and Soft-C model for 3% of the 100,000 artificial AGN spectra that we generated. For models where the intrinsic absorption of the host galaxy could not be constrained [i.e. where Tozzi et al. (2006) reported intrinsic absorption column density $N_{H,int} = 0$], we use a value of 10^{19}

cm^{-2} . Note that effects from Galactic absorption and the response of *Chandra* dominate the soft spectra in these AGN.

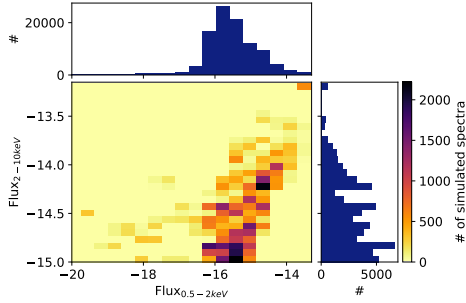
Fe line emission has been detected in 34% of well-detected CDFSAGNCXO AGN with spectroscopic redshifts, with equivalent widths between 100–3000 eV (Liu et al. 2017). Since we desire our ML algorithms to account for the possibility of Fe-K emission in the AGN spectra, we introduce an Fe-K line in 50% of our simulated spectra of AGN. This may or may not be representative of the AGN population in the CDFS sample or across the universe (it is unclear, since our modelled equivalent width distribution reaches to small values that may not be detected in e.g. the CDFS sample), but this ensures that AGN with and without lines are represented in our sample. **Our goal is that our model can identify AGN with or without the Fe-K line.** The equivalent width of the Fe-K emission line was assumed to follow a uniform distribution between 100–3000 eV. The position of the Fe-K line is calculated based on the chosen redshift for each AGN, and a rest-frame energy of 6.4 keV. We show the properties of our AGN sample in Figs. 3.3. We notice that 90% of the sources have $\log N_H$ (in cm^{-2}) $\in (19.05, 24.2)$. Among sources where N_H could be properly constrained, the range is $(21.37, 24.2)$, $z \in (0.01, 2.56)$ and $\Gamma \in (1.2, 2.2)$. We use the XSPEC models `cflux*tbabs*ztbabs*pegpwlw`, `tbabs*pegpwlw` + `cflux*ztbabs*pegpwlw`, and `cflux*tbabs*pexrav` for the C-thin, Soft-C and C-thick AGN respectively (we use `cflux` on the absorbed flux, as the CDFSAGNCXO catalog only reports the absorbed flux values) to generate an artificial AGN spectra sample with an exposure time of 1 Ms. The `tbabs` and `ztbabs` are used to model the absorption from the Galaxy (fixed to $8 \times 10^{-19} \text{cm}^{-2}$ in the direction of the CDFS sample) and the redshifted intrinsic absorption in the host galaxy. The `pegpwlw` component models the power-law emission from the AGN. We add an additional `tbabs*pegpwlw` component to model the soft component in Soft-C AGN. In these AGN, we use the same power-law indices for the two components, in accordance with the spectral fitting in Tozzi



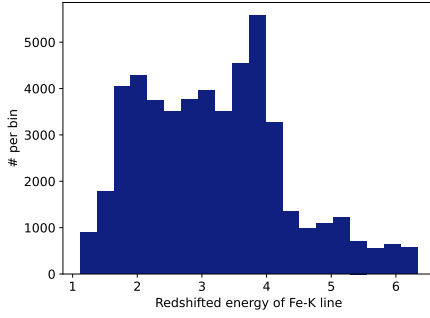
(a) Distribution of power-law indices for AGN distribution. Most spectra have $\Gamma \sim 1.8$.



(b) 2D histogram of absorption column density, N_H and redshift, z . For sources where Tozzi et al. (2006) could not constrain the N_H , we use a fixed $N_H = 10^{19} \text{ cm}^{-2}$. AGN with higher redshift usually have higher absorption.



(c) 2D histogram of hard (2–10 keV) and soft (0.5–2.0 keV) flux used for simulating our AGN spectra. The two fluxes generally show a positive correlation except at low values of soft flux, which could be due to high absorption in these sources.



(d) Positions of the Fe-K emission line in the AGN spectra after accounting for the redshift. This figure only considers the AGN spectra in which an Fe-K emission was added (50% of all AGN spectra). We assume a uniform distribution of equivalent widths from 100–3000 eV for these spectra.

Figure 3.3: Properties of AGN sample used for training and testing our ANN.

et al. (2006). We use `pexrav` model to generate the reflected AGN spectra in C-thick AGN. We add an additional `tbabs*ztbabs*gaussian` component for AGN with an Fe-K line.

We cross-match the X-ray positions of the COUP and CDFSAGNCXO sources with the CSC v2.0 catalog using a search radius of $3.0''$, and download

the spectra from the COUP and CDFS observations. We are able to retrieve the spectra of 1373 stars out of the 1616 COUP sources, and 296 AGN out of the 321 CDFS sources. (We fail to extract all sources because the CSC pipeline masks the corners of the ACIS detectors, where the background is very high, to minimize the detection of erroneous sources.) The CDFS observations were taken in four epochs – 1 Ms between 1999-2000, 1 Ms in 2007, 2 Ms in 2010, and 3 Ms in 2014. Since the soft energy response of *Chandra* ACIS has been degrading, we only use the 1999-2000 observations in this paper. These are also the observations used by Tozzi et al. (2006) to study the AGN properties. In order to generate the artificial spectra of our sample, we use the response matrix and effective area files of the cross-matched COUP and CDFS sources. We show the mean spectra of these sources in Fig. 3.4. We only used sources with net counts greater than 100 in Fig. 3.4 to reduce the noise. We also show the distribution of the net counts and the background contribution to these sources in Figs. 3.5 & 3.6.

We note a few caveats of using the COUP and CDFS surveys. Both COUP and CDFS surveys have exposure time ~ 1 Ms, much longer than the exposure times of typical *Chandra* observations. Thus, the net counts of sources in COUP are higher than those of typical stars, and the faint AGN in CDFS have larger background-to-net count ratios than most AGN detected. From Figs 3.5, we notice that the COUP stars are ~ 10 times brighter than CDFS AGN, implying that the spectra of our sample stars will have higher signal-to-noise ratio than our sample AGN. Therefore, we use the flux distribution from the CDFS AGN in the 0.5–10 keV regime to simulate the spectra of stars so that both AGN and stellar spectra have similar signal-to-noise ratios. We do this to check if our ML method is inherently better at picking AGN/stars. However, this doesn't imply that all our sources have the same noise. The ~ 2 orders of magnitude range in the flux of CDFS AGN ensures that we can check our performance with varying net-counts and background contribution.

We attempt to analyze how the background can affect the performance of our methods by simulating three kinds of spectra from AGN and stars — without background, with the observed background, and with a reduced background rate (i.e., we increased the flux of the sources by a factor of 10 and reduced the exposure to 100 ks). We consider a reduced background because the targets we selected are unrepresentative of typical Chandra observations (as noted above), so the reduced background analysis will be more helpful to estimate potential future performance. We also notice the COUP sources are ~ 10 times brighter than the AGN in the CDFS survey. Therefore, we use the fluxes from the CDFS AGN in the 0.5–10 keV regime to simulate the spectra of stars so that both AGN and stellar spectra have similar signal-to-noise ratios.

3.2.1 Setting up the ANN

We incorporate a **sequential** artificial neural network (ANN) model using TensorFlow (Martín Abadi et al. 2015) for our primary analysis. In this model, we add one hidden layer with ten nodes (we also tried other configurations, but found no significant increase in the accuracy with a higher number of nodes/layers) and a Rectified Linear Unit (ReLU) activation function (we show the neural network architecture in Fig. 3.7). The output layer has sigmoid activation ensuring the probabilistic interpretation of the output values. As the problem of classifying the X-ray spectra is **essentially a Bernoulli problem** (i.e. an X-ray spectrum could be from an AGN with probability p and star with probability $1-p$), we use a **binary cross-entropy loss for training**. The **binary cross-entropy loss is defined by**,

$$H_p(D) = -\frac{1}{N} \sum_{i=1}^N y_i \log p(y_i) + (1 - y_i) \log(1 - p(y_i)), \quad (3.1)$$

where $H_p(D)$ signifies loss calculated over dataset D , N is the number of input data, $y_i \in \{0, 1\}$ (we chose 0 for stars and 1 for AGN), and $p(y_i)$ is the probability

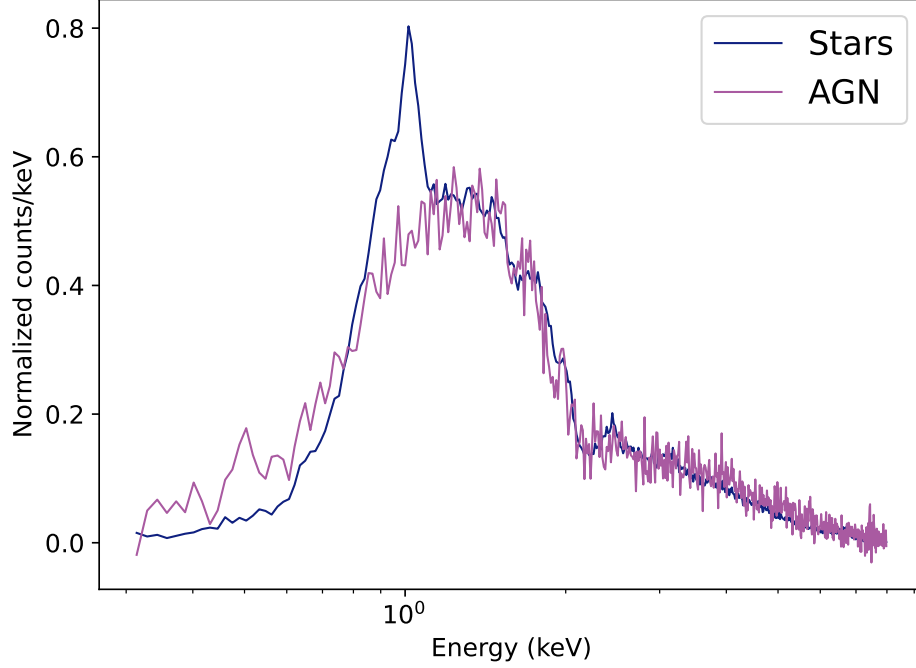


Figure 3.4: Normalized (divided by total counts) mean spectra of the observed COUP stars (solid blue) and CDFS AGN (solid purple). For the purpose of plotting, we only show the mean spectra calculated from sources with net counts more than 100. From the figure, we see that the spectra of stars show a prominent Ne-K/Fe-L emission line at ~ 1 keV along with slight hints of Mg-K (~ 1.3 keV), Si-K (~ 1.8 keV), and S-K (~ 2.3 keV), while the AGN show no such lines. Since the CDFS AGN have lower flux than the COUP stars in general, we notice that the spectra of AGN have more noise (both background and Poisson) than the stars.

that $y_i = 1$. We use the Adam optimization algorithm (Kingma and Ba 2014) for training the model since it results in less computation time and faster convergence. We fix the L_1 regularization value, $\lambda = 0.001$, to avoid over-fitting of the data (we test for various values). We consider the 0.3–8.0 keV energy interval while training our ANN classification model, to limit the background contribution (not to be confused with the 0.5–10 keV flux values reported in Tozzi et al. (2006) and used in our simulations). In order to limit the Poisson noise, we only use simulated spectra with greater than 100 net counts in the 0.3–10 keV energy range for training (this net count criteria is not for simulating

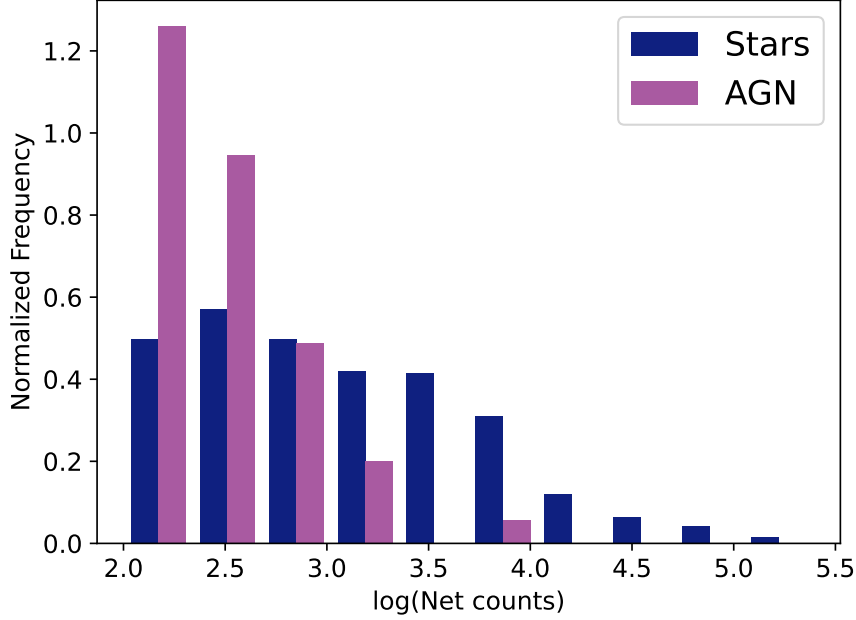


Figure 3.5: Histogram of the net counts of the observed COUP stars (solid blue) and CDFS AGN (solid purple). We have normalized the histograms such that the area under both histograms is unity (to avoid distortion due to more COUP sources than CDFS AGN). We notice that the COUP stars are typically ~ 10 times brighter than the CDFS AGN.

sources but only for training and testing ANN). We use 80% of the spectra for training the model and use the remaining 20% as our test set. We perform 10-fold cross-validation while training the model, to ensure that we do not overfit the data. We then apply the trained ANN model to classify spectra of the test set and evaluate our performance. We perform 20 such iterations to study the behaviour of our ANN model.

For several ML applications, the input is standardized by subtracting each feature by the mean of that feature across all classes, and dividing the difference by the standard deviation of that feature. In general, this allows for better fitting of the weights. We also perform ANN classification with standardized input, but the improvement in classification accuracy is $\lesssim 1\%$. In astrophysics, having unequal number of sources in each source class is common, which can

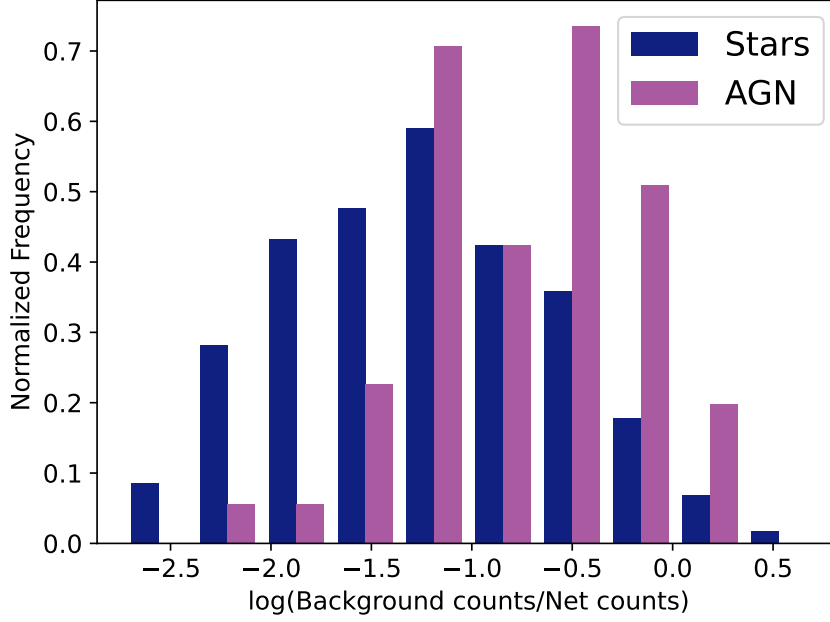


Figure 3.6: Normalized histogram of the ratio of background-to-net counts for the observed COUP stars and CDFS AGN. Since CDFS AGN are mostly faint, most of them have a very high background contribution.

lead to the mean being biased to one particular class. Therefore, we choose to present results that do not use standardized inputs.

We analyze the results of our classification using the following metrics (for the purpose of this paper, we consider AGN as positives and stars as negatives):

- **Accuracy:** The total fraction of correct predictions across the entire set.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (3.2)$$

- **Recall:** The ratio of true positives to the total number of positives in the dataset. In our case, this signifies the fraction of AGN identified correctly.

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (3.3)$$

- **True Negative Rate (TNR):** The ratio of true negatives to the total number of negatives in the dataset, i.e. the fraction of stars identified correctly.

$$\text{TNR} = \frac{\text{TN}}{\text{TN} + \text{FP}} \quad (3.4)$$

- **Precision:** The ratio of true positives to the number of samples that have been classified as positives. In our case, it is the fraction of true AGN among the spectra classified as AGN.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (3.5)$$

- **Negative Predictive Value (NPV):** The ratio of true negatives to the number of samples that have been classified as negative, i.e. the fraction of true stars among the spectra that have been identified as stars.

$$\text{NPV} = \frac{\text{TN}}{\text{TN} + \text{FN}} \quad (3.6)$$

In the above equations TP, TN, FP, and FN stand for the true positives (AGN identified correctly), true negatives (stars identified correctly), false positives (stars identified as AGN), and false negatives (AGN identified as stars), respectively.

We also evaluate the performance of the trained ML model on the observed data. This set consists of all COUP stars (including sources that have been flagged for deviations in emission lines, poorly or marginally fit spectra, soft/hard excess, etc; only extragalactic sources are removed) and CDFS AGN that have more than 100 counts. Since there are only 108 such CDFS AGN as compared to 679 COUP stars, we randomly select 108 COUP stars so that both classes are equally sampled. Note that we select a new random sample of COUP stars with each of the 20 iterations to avoid any hidden biases.

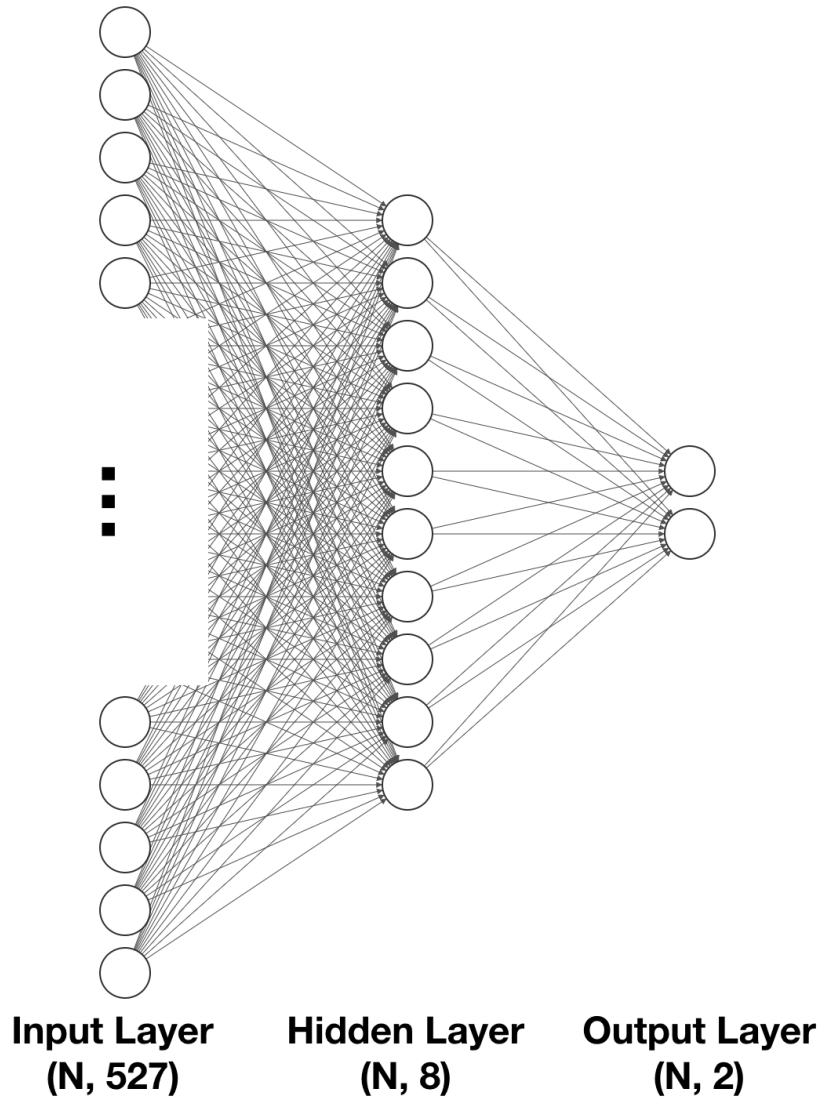


Figure 3.7: Architecture of the ANN model. The input layer consists of 527 nodes corresponding to the number of channels in the *Chandra* ACIS-S spectra in the energy range 0.3–8.0 keV. We use one hidden layer with 10 nodes which are fully connected to the input layer. The hidden layer uses a ReLU activation and an L_1 regularization with $\lambda = 0.001$. These hidden nodes are then used to calculate the values in the two output nodes that use a softmax activation function and signify the probability of the source being an active star or an AGN.

3.3 Results

We first analyze the spectra of AGN and stars without including the background. We show the mean spectra of stars and AGN in Fig. 3.8. From the figure, we notice that AGN in general have harder spectra than their stellar counterparts (note that some AGN in our sample also have soft spectra as seen from Fig. 3.3a). In our case this is further amplified due to their higher (intrinsic) N_H and the presence of redshifted Fe-K lines in the CDFS AGN spectra. We also notice that the spectra of stars have dominant Ne, Fe, Mg, Si, and S lines. The feature at ~ 2 keV in the mean AGN spectrum is due to a combination of N_H , z , position of the Fe-K line, and the response of the *Chandra* ACIS detector, and is not a real emission line. We show the distribution of the net counts in the 0.3 – 8.0 keV energy range in Fig. 3.9. We see that the stars have higher count rates than AGN on average by a factor of ~ 5 , even though their fluxes in the 0.5–10 keV interval are similar. This is because of the softer spectra of stars in comparison to AGN (i.e. we match the X-ray fluxes of stars and AGN in the 0.5–10 keV range while simulating spectra, but consider 0.3–8.0 keV for the classification and analysis.).

Applying our ANN model to this dataset gives us an overall accuracy $\sim 89\%$, recall $\sim 83\%$, TNR $\sim 92\%$, precision of $\sim 86\%$, and NPV of $\sim 91\%$. However, when this trained model is used to classify the set of observed spectra, we get an accuracy of only $\sim 81\%$. The decreased accuracy is due to the poor performance of the classification algorithms at low net counts, where the contribution of

Table 3.1: Performance of our machine learning model on different datasets

Type of Dataset	Testing data	Accuracy	Recall	TNR	Precision	NPV
No background	Simulated spectra	$(88.9 \pm 0.9)\%$	$(83 \pm 5)\%$	$(92 \pm 3)\%$	$(86 \pm 3)\%$	$(91 \pm 2)\%$
	Observed spectra	$(81 \pm 3)\%$	$(67 \pm 7)\%$	$(93 \pm 5)\%$	$(92 \pm 6)\%$	$(74 \pm 4)\%$
Observed background	Simulated spectra	$(88.7 \pm 0.6)\%$	$(86 \pm 3)\%$	$(91 \pm 2)\%$	$(87 \pm 2)\%$	$(90 \pm 2)\%$
	Observed spectra	$(89 \pm 3)\%$	$(84 \pm 5)\%$	$(95 \pm 2)\%$	$(94 \pm 3)\%$	$(85 \pm 4)\%$
Reduced background	Simulated spectra	$(91.7 \pm 0.8)\%$	$(91 \pm 2)\%$	$(92 \pm 2)\%$	$(91 \pm 2)\%$	$(93.0 \pm 2)\%$
	Observed spectra	$(91 \pm 2)\%$	$(90 \pm 3)\%$	$(92 \pm 4)\%$	$(91 \pm 3)\%$	$(91 \pm 3)\%$
Reduced background, Net Counts > 1	Simulated spectra	$(86 \pm 1)\%$	$(86 \pm 3)\%$	$(85 \pm 3)\%$	$(85 \pm 2)\%$	$(86 \pm 2)\%$
	Observed spectra	$(82 \pm 3)\%$	$(77 \pm 4)\%$	$(87 \pm 4)\%$	$(86 \pm 4)\%$	$(80 \pm 3)\%$

Note: The error-bars indicated above correspond to one standard deviation

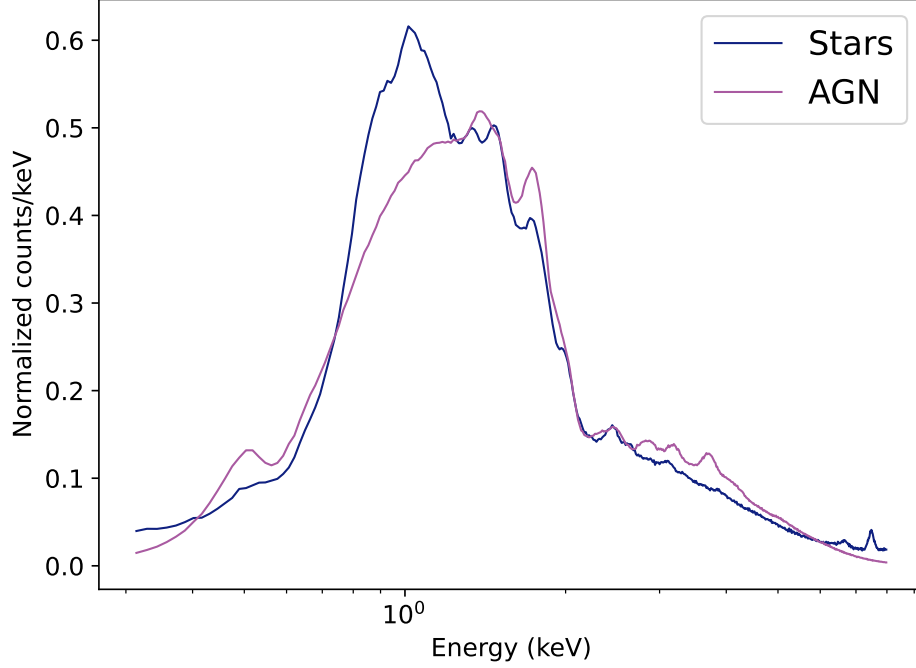


Figure 3.8: Normalized mean of the artificially generated AGN and star spectra simulated without including the background. These spectra are similar to that in Fig. 3.4, but less noisy since they are the mean of 100,000 spectra. With less noise, the emission features in the spectra of stars are more clear. The emission-line-like features in the AGN spectra are due to a combination of the intrinsic absorption columns, redshifts, and the *Chandra* response. Red-shifted Fe-K emission lines cause the appearance of bumps in the mean AGN spectra in the 2–4 keV energy range.

background cannot be ignored.

Next, we explore the effect of background on our simulations. For this purpose, we simulate the spectra of AGN and stars using similar background levels as the COUP and CDFS sources. Fig. 3.10 shows the histogram of the ratio of background counts to net counts for the simulated sources in the 0.3–8.0 keV energy range. (The background counts have been scaled such that they correspond to the source extraction region). We notice that the stars and AGN in the COUP and CDFS catalogs have a high background, in general, with many sources having background counts > 0.1 times their net counts. Our classification accuracy is $\sim 89\%$ on these simulated spectra. When we apply

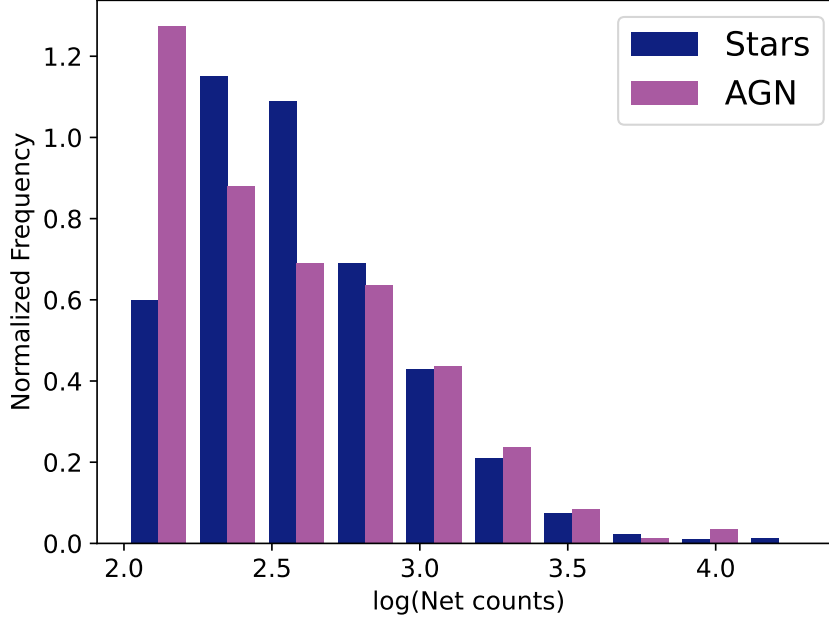


Figure 3.9: Distribution of total counts, in the 0.3–8.0 keV energy range, in the simulated sample of AGN and stars generated without including the background. Despite equating the 0.5–10 keV flux of simulated stars and AGN, stars have higher X-ray photon counts since they have softer spectra than AGN.

the trained model to the observed spectra of CDFS and COUP sources, we get a similar classification accuracy of $\sim 89\%$. This performance is better than the ANN model trained on simulated spectra without background.

The high background in our simulated flux is due to the low fluxes of the CDFS AGN (as well as the off-axis positions of a substantial portion of the sources). In general, most X-ray sources detected by *Chandra* have a lower background contribution than these AGN. For instance, a typical *Chandra* observation is of order 30 ks, rather than the 1 Ms CDFS, reducing the typical background by a factor of 33. However, more *Chandra* sources with >100 counts can be found in longer observations; we use 100 ks observations for our standard as a compromise. To consider the performance of our classification algorithms for typical stars and AGN, we thus consider simulated spectra of stars and AGN

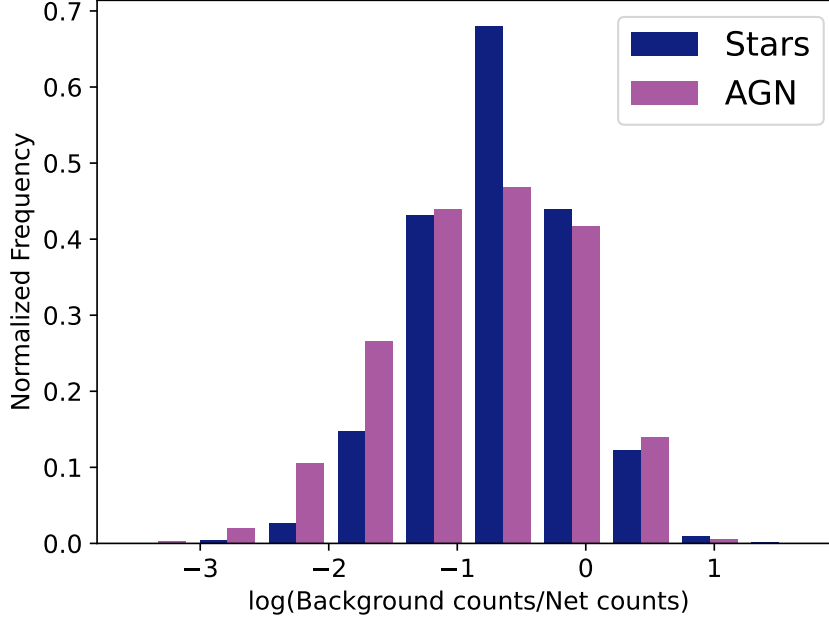


Figure 3.10: Distribution of background-to-net count ratio in the simulated AGN and star spectra when the observed background is included. The plot only represents spectra with net counts greater than 100 in the 0.3–8.0 keV energy range. Many of these spectra have a high contribution from the background as these sources are fainter than typical sources detected by *Chandra*.

where the background contribution has been reduced by a factor of 10.

We show the distribution of the net counts and the background-to-net count ratio in Figs. 3.11 & 3.12 respectively. These values of background are more typical of X-ray sources detected by *Chandra*. Applying our ANN model to these spectra gives an overall accuracy $\sim 92\%$, recall of $\sim$ precision $\sim 91\%$, and TNR and NPV $\sim 92.5\%$. Applying the trained model to observed spectra gives an accuracy of $\sim 91\%$. One of the reasons why the model trained on reduced background performs better on the observed data, as compared to the model trained on the observed background, could be that the lower noise from the background contribution helps better constrain the values of the best fit model. We summarize the results of our classification on the simulated and the training data in Table 3.1.

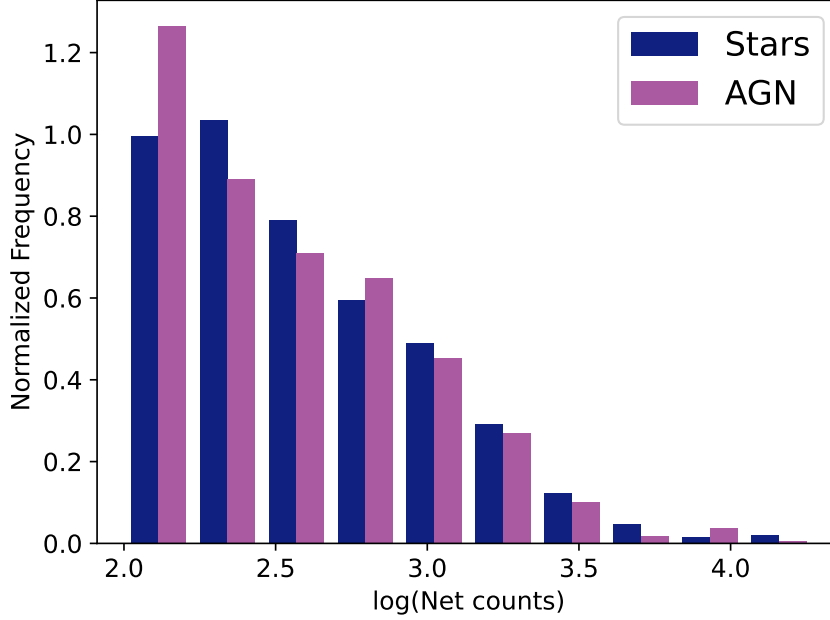


Figure 3.11: Distribution of net counts of star and AGN spectra, simulated with a reduction of the background contribution by a factor of 10.

We also look at the certainty with which our model identifies the AGN vs. stars. Fig. 3.13 shows the distribution of probabilities of the input source being an AGN as estimated by our model. From the figure, we note that our model is able to distinguish most AGN and stars with very high confidence. We used a cutoff of 50% to classify sources as AGN/stars in this paper. For situations where we need a purer sample of AGN, we can use a higher cutoff, or we can use a lower cutoff if we want to lower the number of AGN that could be identified as stars.

3.3.1 Variation of performance with properties of sources.

Since the model trained on the reduced background spectra, which represent typical *Chandra* sources, gives us the best results for both simulated and observed spectra, we use these to analyze if our classification model preferentially selects stars or AGN of certain properties. We show the performance metrics with

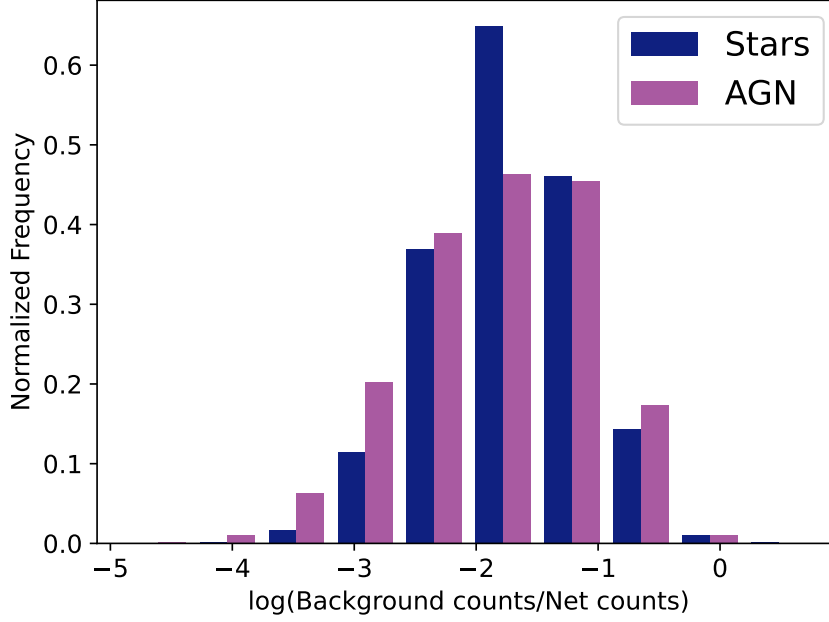


Figure 3.12: Distribution of the background-to-net count ratio in the simulated AGN and star spectra simulated with the reduced background. The y-axis is normalized such that the figure shows a density function. This distribution is more typical of the X-ray sources detected by *Chandra*.

respect to different properties of stars and AGN in Fig. 3.13.

We first analyze the performance with respect to the net counts and the background contribution (background-to-net count ratio). The performance improves as net counts increase. As the net counts increase, the Poisson noise decreases, and hence the emission lines can be more easily identified. Similarly, as the background contribution decreases, the background noise can be distinguished from the true emission lines leading to better differentiation of stars and AGN. In general, we get good results (better than 90%) for sources with net counts greater than 200 and/or background-to-net count ratio smaller than 0.05.

We then study the performance with respect to the absorption column densities used for the AGN and star spectra. We see that the true negative rate (i.e., the fraction of stars detected correctly) decreases steeply for $N_H \gtrsim 10^{22}$

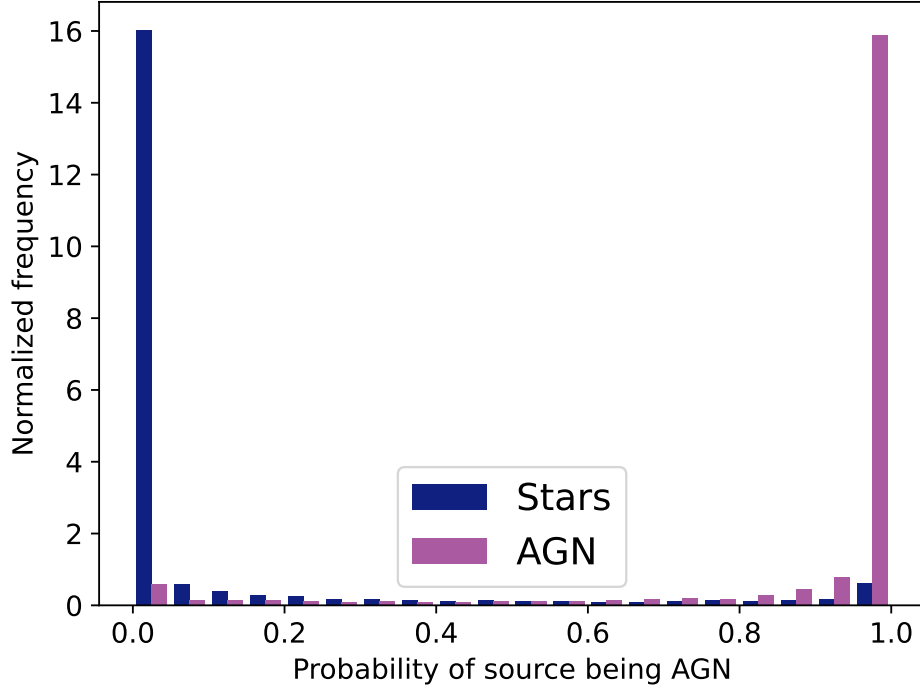


Figure 3.13: Distribution of probabilities of the source being AGN, as calculated by our trained model. Notice that most of the stars and AGN have been correctly identified with a certainty of more than 95%. In this article we have used a cutoff of 50% to differentiate stars and AGN.

cm^{-2} . This is because high absorption column densities block the soft X-rays which are more important to the detection of the Ne, Fe-L, Mg and Si lines that identify the spectra as stars. We expect such high N_H only for stars very close to the Galactic plane and those in dense nebulae. The detection of AGN suffers only slightly with the increasing N_H . This is because as N_H increases, the count rate at softer energies decreases, thus increasing the Poisson noise, which could be confused as lines. At $N_H \gtrsim 10^{24}$, the performance improves because of the reprocessed X-ray emission from the Compton thick model we use in this range.

The detection of stars also decreases with the increase in the plasma temperature used to model X-ray emission. For high temperatures ($kT > 2$ keV),

the continuum emission dominates while lines fade, making it harder to detect the line emission. However, we expect that most of the stars have $kT < 2$ keV (e.g. Güdel (2004)). Higher temperatures are mostly due to improper fitting of highly absorbed sources.

The detection of AGN varies with the hardness of the X-ray emission. Our algorithm seems to detect harder AGN at a higher rate. This is expected since most CDFS AGN have $\Gamma = 1.75 \pm 0.02$. We notice that the performance is better on Compton thick AGN, and AGN with an additional soft component, as compared to Compton thin AGN. The detection fraction of AGN increases slightly with the redshift of AGN. The decrease in recall at $z > 4$ may be due to very few data points in these bins.

We notice that our model detects AGN with an Fe-K line more efficiently than the ones without Fe-K emission. This can be deduced through the increasing recall as the equivalent width of the Fe-K emission line increases. The change in recall with the position of the Fe-K emission line seems to be consistent within the error-bars.

3.3.2 Performance at low counts

We also test the performance of the ML algorithms on sources with reduced background and net counts less than 100. For this purpose, we select all sources with net counts > 1 in our simulated and observed dataset to train and test our performance. The overall accuracy is 86% on the simulated sample and 82% on the observed spectra. The recall, TNR, precision and NPV have similar values as the accuracy. We list the values of the performance metrics for the observed and simulated data in Table 3.1. We also show the variation in accuracy with net counts in Fig. 3.14. The accuracy is very poor ($\lesssim 50\%$) for net counts $\lesssim 8$. The accuracy is $\sim 60\%$, $\sim 74\%$, and $\sim 83\%$ for net counts $\sim 8 - 21$, $\sim 21 - 55$, and $\sim 55 - 146$, respectively. For these low count sources, including the information from their positions in the sky, detection of multiwavelength

counterparts and their properties along with the classification probabilities from X-ray spectra alone will help in improving the overall classification accuracy.

3.3.3 Weights applied by ANN on energy bins

Since we use a simple ANN model with only one neural network layer, the weights assigned to the input layer can be interpreted as the relative importance of the energy channel to classifying the source as a star or an AGN. We show the weights assigned to each energy channel in Fig. 3.15. From this figure, we see that the X-ray spectra in the energy range $\sim 0.8\text{--}1.4$ keV are strongly selected, indicating the preference given to the Ne, Fe, Mg, and Si lines. The bumps in the spectra at energy bins $\gtrsim 2$ keV are probably due to the model searching for the redshifted Fe-K line. We notice that training the ANN model using only the spectra without an Fe-K line reduces the weights in this regime. Thus, our model indeed tries to identify emission lines in the spectra to classify the source, and does not use the hardness ratio of the spectra alone.

3.3.4 Applying trained ANN model to observed data

From the values in Table 3.1, we see that while the ANN models trained on the data simulated with the observed background or with the reduced background yield only slightly poorer performance on the observed data as compared to that on the simulated spectra, the model trained on spectra without background performs very poorly on the observed spectra. Comparing the performance of these models with respect to the net counts and the background contribution of the observed spectra, shown in Fig. 3.16, gives us more insight into this behavior.

We see that the performance of the ANN model trained on the spectra without background contribution degrades steeply at low net counts ($\lesssim 1000$) or when the ratio of background-to-net counts is high ($\gtrsim 0.02$). This is because

the contribution of background is high in this regime. This background noise can be confused with emission lines, leading to many AGN being classified as stars. Looking at the performance of the ML models trained with background (observed and/or reduced background) with respect to the background-to-net count ratio also allows us to understand the slightly poorer overall accuracy on the observed data. Comparing Fig. 3.16d and Fig. 3.14d, we see that the observed AGN have much higher background than the simulated AGN. Both plots show that the accuracy is greater than 90% for background counts $\lesssim 5\%$ of the net counts from the source.

3.3.5 Using ML to detect extragalactic sources in COUP

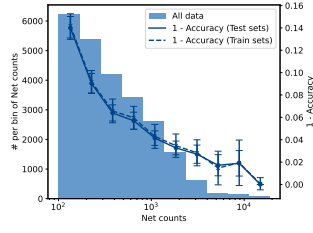
Using the ANN model trained on the data simulated with the reduced background, we try to identify the extragalactic sources detected in the COUP catalog. Among the sources marked as extragalactic in Getman et al. (2005d), 63 sources have net counts greater than 100. Our ANN model is able to identify ~ 40 of these 63 extragalactic sources. This corresponds to a recall of $\sim 63\%$ which is much lower than the recall on the CDFS spectra. This poorer performance is because most AGN in the COUP catalog are heavily absorbed and have a high background (60% of extragalactic sources have background-to-net count ratio $\gtrsim 0.3$, and 84% of extragalactic sources have background-to-net count ratio $\gtrsim 0.1$).

3.3.6 Changing response of Chandra

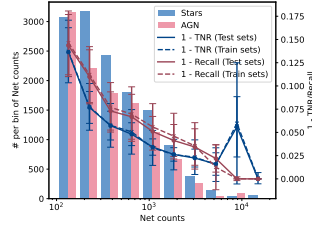
The response function of *Chandra* has been changing over the years, as the effective area at soft energies degrades. Thus the same star could have different emission line strengths (especially < 2 keV) across the years. We generate 10,000 artificial spectra of stars and AGN each, with the ACIS-I aimpoint responses from the *Chandra* Cycle 25 call for proposals to check if our model could be

applied if similar *Chandra* observations were taken in 2023. We find that our model (trained on the COUP responses) can can classify the simulated stars having >100 net counts with a accuracy of $(84 \pm 3)\%$, retrieve $(77 \pm 7)\%$ of AGN (recall), and identify $(90 \pm 3)\%$ of the stars (TNR) i.e. the performance on identifying stars is only slightly decreased, but that for AGN decreases a lot. We will explore the training of ML models with the changing response of *Chandra* in our future works.

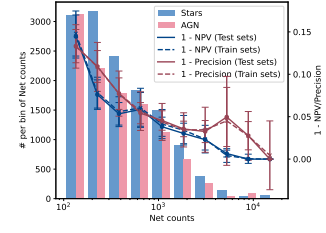




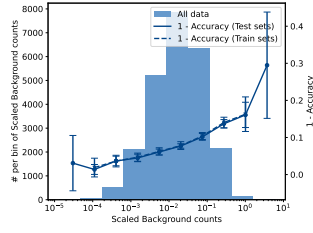
(a) Change in accuracy with net counts. The accuracy increases with net counts, due to the decrease in the Poisson noise, and is $>90\%$ for net counts >300 .



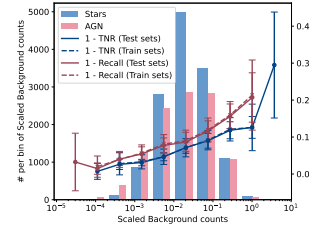
(b) Change in recall (fraction of AGN retrieved) and TNR (fraction of stars retrieved) with net counts in the spectra. These show similar trends as the accuracy.



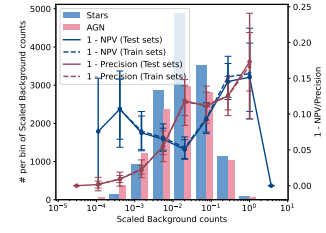
(c) Change in precision (fraction of true AGN among sources classified as AGN) and NPV (fraction of true stars among sources classified as stars) with net counts. Both show similar trends as the accuracy.



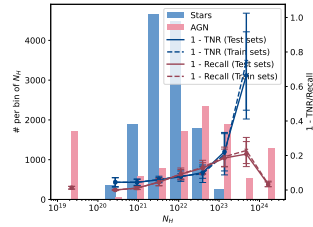
(d) Change in accuracy with the ratio of background-to-net counts. The accuracy decreases with increasing background contribution. Accuracy $> 90\%$, for background-to-net count ratio < 0.08 .



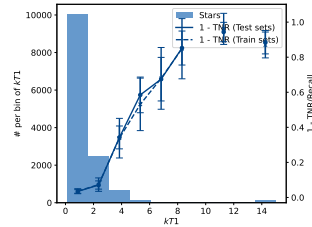
(e) Change in recall and TNR with the changing background-to-net count ratio. These show similar trends as the accuracy.



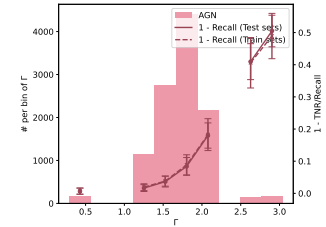
(f) Change in precision and NPV with the background-to-net count ratio. As background increases, more stars can be confused as AGN.



(g) Change in recall and TNR with absorption column density. High absorption ($N_H > 10^{22} \text{ cm}^{-2}$) leads to very low flux $< 2 \text{ keV}$, leading to difficulties in the detection of Fe-L, Mg and Si lines, and thus to identifying stars.



(h) Change in fraction of stars detected with the changing temperature of the thermal plasma used. At high temperatures, the atoms are fully ionized and the spectra are continuum dominated, making it harder to identify the spectra as stellar.



(i) Change in fraction of true AGN detected with the power-law index of AGN. Our model tends to select AGN with harder spectra. This is expected since most of the AGN have $\Gamma < 2$.

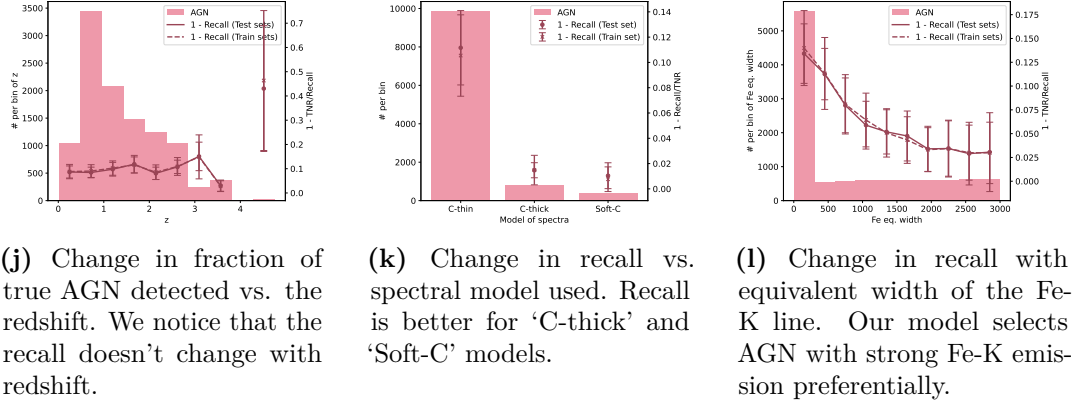


Figure 3.13: Variation in performance of our ANN model for different properties of the sources. In each case the histogram represents the number of spectra in the given binning of the property. The y-axis on the right corresponds to the values of confusion matrix (1-Accuracy, 1-Recall/TNR etc.).

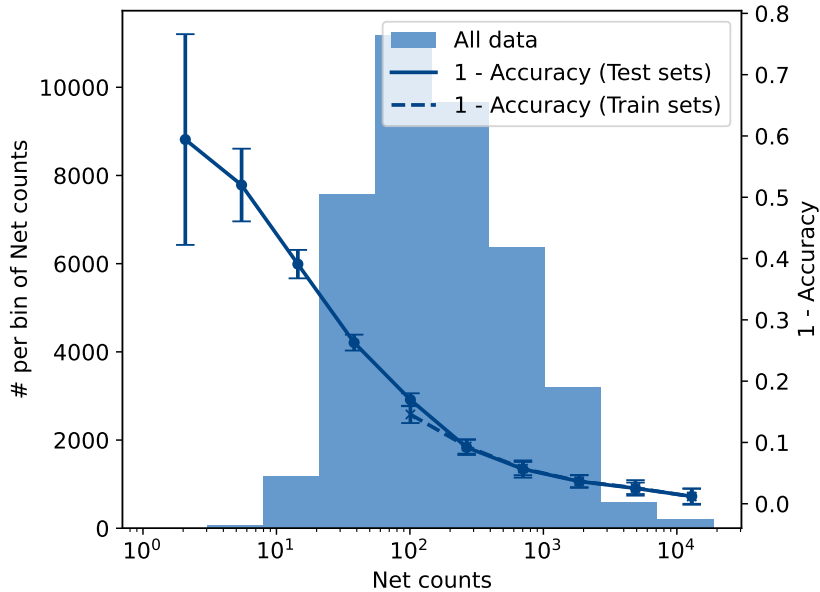


Figure 3.14: Change in the accuracy with net counts for simulated spectra with net counts > 1.0 . From the figure, we see that our algorithms does not perform well on sources with 1–10 net counts (accuracy marginally better than 50%). The accuracy is $\gtrsim 70\%$ for sources with net counts $\gtrsim 20$ and $\gtrsim 80\%$ for net counts $\gtrsim 70$.

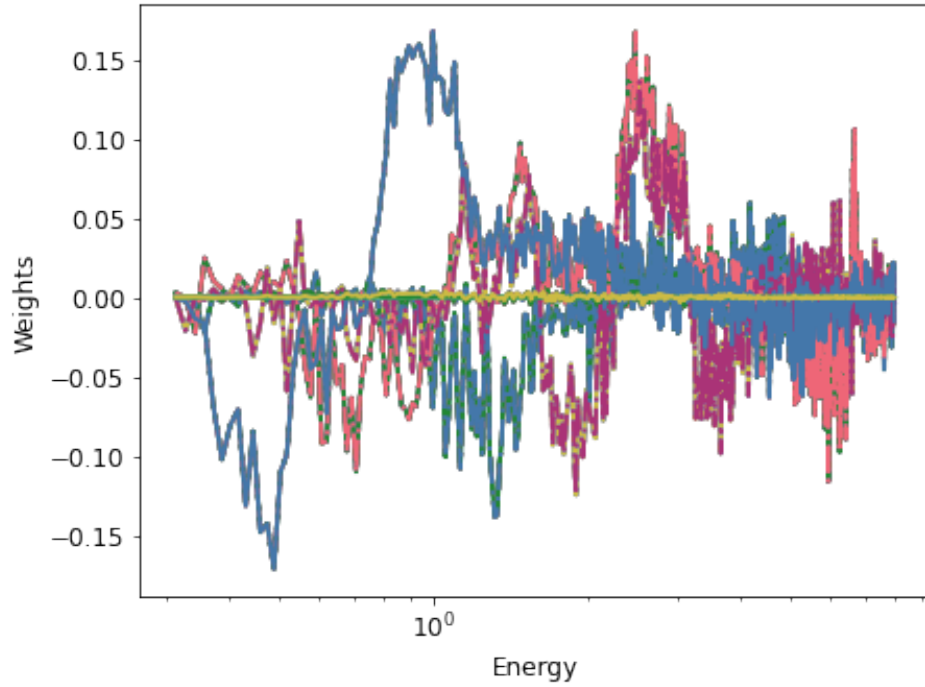
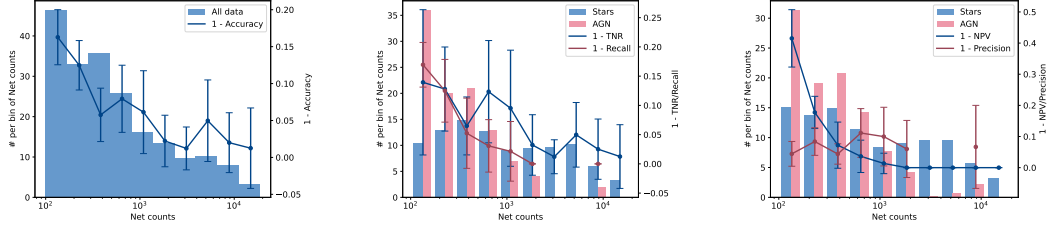


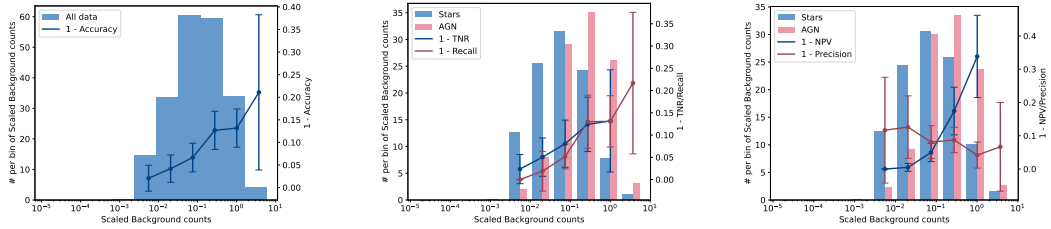
Figure 3.15: Weights to each energy channel in the first layer of a trained ANN model. We notice that the model assigns larger weights (magnitude only) to the spectra around the position of the Ne, Fe, Mg, and Si lines. The different curves correspond to the weights of ten nodes in our ANN.



(a) Change in accuracy with the net counts in the spectra. The accuracy increases with net counts, due to the decrease in Poisson noise and in general, a smaller background contribution.

(b) Change in recall (fraction of AGN retrieved) and TNR (fraction of stars retrieved) with the net counts in the spectra. These show similar trends as the accuracy.

(c) Change in precision (fraction of true AGN among sources classified as AGN) and NPV (fraction of true stars among sources classified as stars) with the net counts. Both show similar trends as the accuracy.



(d) Change in accuracy with the ratio of background-to-net counts. The accuracy decreases with the increase in the background contribution.

(e) Change in recall and TNR with the background-to-net count ratio. These show similar trends as the accuracy.

(f) Change in precision and NPV with the background-to-net count ratio. As background increases, more AGN can be confused as stars.

Figure 3.16: Performance of the trained ML model with net counts, and with background contribution, on the observed spectra.

3.4 Discussions

3.4.1 Incorrect classification of Sources

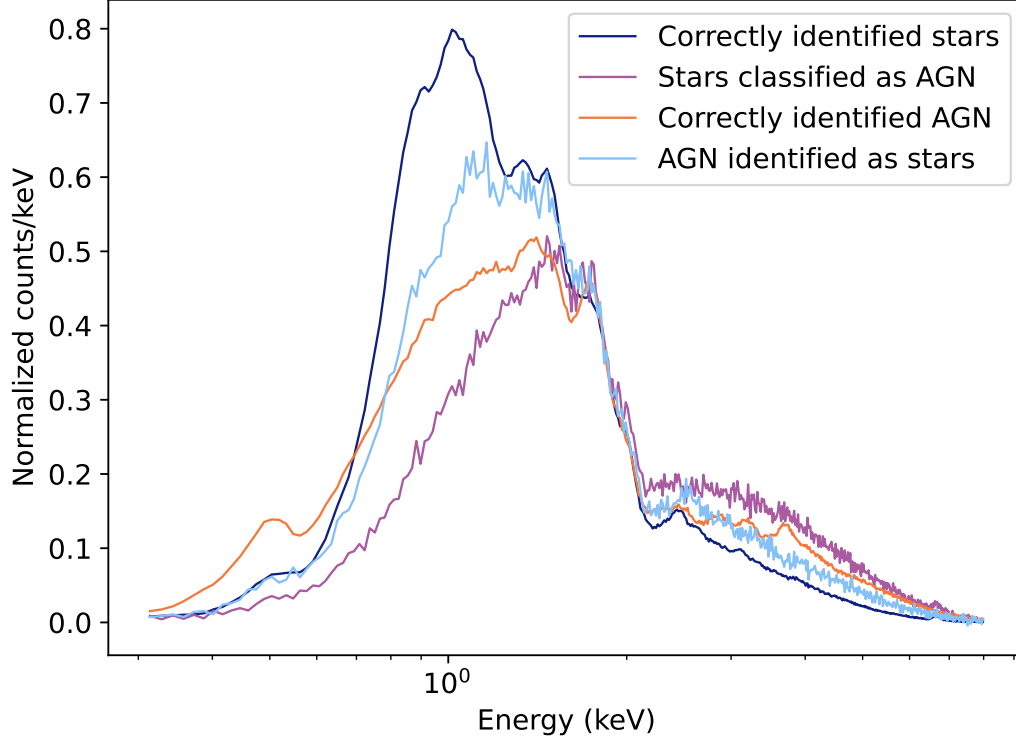


Figure 3.17: Mean normalized spectra of correctly identified stars and AGN, compared to those that are not classified properly. Notice that the stars that have been identified as AGN do not show strong emission lines, are heavily absorbed, and are harder than the stars that have been correctly identified.

From our results, we notice that the trained ANN model uses a combination of emission-line strengths (interpreted from Figs. 3.14g, 3.14h & 3.15) and hardness of the spectra (Fig. 3.14i). Fig. 3.17 shows the mean spectra of stars and AGN that have been correctly/incorrectly identified. From the figure, we notice that incorrectly classified stars show stronger absorption, show fainter emission lines, and/or are harder than correctly identified stars, i.e., their spectra are not line-dominated. AGN that have been identified as stars seem to show higher absorption and/or are softer than correctly identified AGN.

We further explore the mis-classification of softer AGN. The poor recall of AGN for $\Gamma > 2.0$ is because these AGN are affected more by increasing N_H , i.e. for $N_H \lesssim 10^{21} \text{ cm}^{-2}$, all AGN have similar recall ($\sim 98\text{--}99\%$), but at $N_H \sim 10^{22} \text{ cm}^{-2}$, the recall for AGN with $\Gamma > 2.0$ decreases to $\sim 72\%$ while the recall for hard AGN with $\Gamma < 2.0$ is $\sim 96\%$. This can be explained as classification of AGN with hardness similar to stars will mainly be based on identification of soft X-ray emission lines, which can be confusing at high absorption. Our sample of soft AGN with $\Gamma > 2.0$ need a net count $\gtrsim 3000$ photons, or a strong Fe-line with equivalent width $\gtrsim 1500 \text{ eV}$, to be detected with a recall of $> 90\%$. The improvement in the detection of AGN with the increasing fraction of continuum soft-emission also explains why our performance is better on AGN with soft component and Compton thick AGN.

3.4.2 Applications

Based on our results, our algorithm works very well on sources with sufficient X-ray photons to enable the detection of Fe-L, Mg and Si lines against the background and Poisson noise. Based on the distribution of weights (Fig. 3.12) and the decreased accuracy of classification on stars with $kT > 2 \text{ keV}$, we see that the trained ANN picks out the energies corresponding to emission lines and identifies line-dominated X-ray spectra. Thus, ANNs could provide an efficient and practical approach to classify serendipitous X-ray sources.

X-ray emission from most isolated NSs in our Galaxy can be fit with a power-law model with photon index $\Gamma \approx 1\text{--}2$, a blackbody of temperature $0.05\text{--}0.3 \text{ keV}$, or a combination of these, and do not show any lines (Pavlov et al. 2002). Those which are dominated by the power-law will then be similar to AGN, without the Fe-K lines. Based on Fig. 3.13l, we expect that our model should be able to identify such NSs from chromospherically active stars in our Galaxy with a good recall (80–90% if they follow a similar distribution of net counts, column density, and background contribution). Our results indicate

that the additional presence of softer components (e.g. a blackbody) does not affect our performance. Many neutron star X-ray spectra are dominated by a blackbody-like thermal component. Discriminating between blackbody-like spectra and soft thermal plasma spectra (e.g. APEC models with temperatures < 2 keV) will be even easier, as these low-temperature plasma spectra are even more dominated by lines than the harder spectra.

X-ray binaries have spectra similar to those of AGN in the energy range of 0.1-10 keV. Thus our results can also be applied to identifying X-ray binaries. However, distinguishing X-ray binaries from AGN based on their spectra alone would be more challenging, and we would need to use the information from location of the source (X-ray binaries are primarily found along the Galactic plane and globular clusters), X-ray variability and multiwavelength observations to properly differentiate X-ray binaries and AGN. The presence of a fluorescent Fe line will also be important for distinguishing AGN at significant redshift from XRBs. Our results show that ANNs can detect the Fe-line irrespective of the redshift.

Another application of this method would be in identifying millisecond pulsars (MSPs) in highly absorbed regions like the Galactic center and bulge (where soft X-ray sources are obscured). MSPs have hard power-law spectra ($\Gamma \sim 1$) from the intrabinary shock between the NS and its companion, or magnetospheric emission. Identifying the Fe-K line would allow us to distinguish MSPs from other accreting hard X-ray sources like cataclysmic variables.

The X-ray spectra of SNRs can also be modeled by thermal plasma components with high metal abundance. Young SNRs (where the reverse shock has not reached the core) usually have a cooler component with $kT < 2\text{keV}$, and strong Mg and Si lines. Thus ANN models that can pick out these lines will be able to differentiate the young SNRs in distant galaxies from low-luminosity AGN (see for instance Hebbar et al. (2019)).

3.4.3 Current and Future missions

In this work, we have focused on *Chandra* observations with similar response functions. Our results in §3.3.6 indicate that we need a more detailed modeling of the responses during our training before we can combine the spectra from multiple *Chandra* observations to increase the total number of counts. This will enable us to classify the faint sources more accurately.

XMM-Newton has ≈ 5 –10 times the effective area of *Chandra* at 1 keV (depending on the year), but also has a higher background for point-source extraction. Thus for individual observations we would only expect similar or a slight improvement in our performance. However, *XMM-Newton*’s soft energy response is more stable than *Chandra*’s, implying that spectra from multiple observations of the faint sources can be readily combined to increase the sensitivity in detecting emission lines. The *eRosita* probe also has similar characteristics as *XMM-Newton* but has observed a wider portion of the sky, and we can expect similar results.

Our results also point to the characteristics of future missions that could be ideal for our work. The Resolve instrument onboard *XRISM*, set to be launched later this year, will have an excellent spectral resolution allowing detection of abundant emission lines in the spectra of X-ray sources. However, its effective area in the soft energies is lower than *Chandra*, and it has a higher background (for point source spectroscopy). Thus, studying faint serendipitous sources will be difficult with *XRISM*. Proposed X-ray missions like the *AXIS*-probe with ~ 10 times the effective area of *Chandra* and excellent angular resolution throughout the field will allow for the detection of higher counts for the faint sources, and allow classifying them accurately. The higher effective area will also lead to the detection of more serendipitous sources making machine learning methods much more efficient than traditional methods. X-ray missions like the Line Emission Mapper and Athena X-IFU would be the most ideal for our work. Their large effective area in the soft-energy X-rays (< 2 keV) and spectral resolution of a

few electron-volts will allow for the detection of individual emission lines from elements with different ionization states, even in faint sources.

3.5 Conclusion

We discussed the application of a neural network model in distinguishing *Chandra* X-ray spectra of AGN in CDFS and stars in the Orion nebula cluster. We are able to achieve accuracy, recall and precision of $\sim 92\%$ on the simulated spectra and $\sim 91\%$ on the observed spectra. The algorithm is most efficient when the net counts in the 0.3–8.0 keV regime are $\gtrsim 200$, background contribution is $\lesssim 5\%$, the stars have absorption column densities $N_H < 10^{22} \text{ cm}^{-2}$, $kT \lesssim 2$ keV, and the AGN are hard with power-law index $\Gamma \leq 2$ and have a strong Fe-K emission line with equivalent width $\gtrsim 0.5$ keV.

We also tested the robustness of our method with the changing soft energy response of *Chandra* ACIS, and found that the performance of our model, trained on the COUP and CDFS responses, decreased on simulated spectra from responses in the Cycle 25 *Chandra* Call for proposals, especially in identifying AGN. Thus, combining multiple observations of faint sources would need more detailed modelling of the *Chandra* responses. **Applying these algorithms to *XMM-Newton* observations will allow us to utilize its larger effective area and near-constant response to increase the signal-to-noise ratio for faint sources.**

In this paper, we have only used the X-ray spectra for classifying the source. For sources with known optical/radio counterparts, we can use their position, variability, and multiwavelength properties in addition to the output of our machine learning methods (in place of commonly-used hardness ratios) to improve the accuracy of the classification.

X-ray catalogs have a unique property that while hundreds and thousands of sources have available spectra, only a few of them are labelled. Such datasets allow the use of semi-supervised learning such as auto-encoders. These al-

gorithms use clustering methods (i.e., use the distribution of features in the unlabelled data), representation of the data into smaller dimensions, etc., along with the information from the classified/labelled sources to efficiently train the ML model. Using these kinds of training algorithms would allow us to study the data without the need to simulate artificial spectra.

We plan to extend this method to distinguish and identify other kinds of X-ray spectra, like those of neutron stars and SNRs, from stars and AGN. We would also like to check the performance of our model on sources that have slightly different properties than the ones used in training. These methods will be especially useful with future higher-spectral-resolution X-ray data from upcoming microcalorimeter missions, such as Athena.

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Data Availability

The COUP source list and the membership catalogues used to extract the properties of the stars are available at the VizieR online databases Getman et al. (2005c,a). The catalogue of AGN in CDFS can also be found at the VizieR online catalogue Tozzi et al. (2007). The *Chandra* data themselves are available via the *Chandra* Source Catalog, <https://cxc.cfa.harvard.edu/csc/>, and/or the *Chandra* Data Archive <https://cxc.cfa.harvard.edu/cda/>. The spectra of simulated stars and their properties are available upon reasonable request to the first author.

Chapter 4

Identifying candidate X-ray binaries and millisecond pulsars in Galactic Bulge using Fe-K line diagnostics

4.1 Introduction

The inner kiloparsec of the Milky Way galaxy shows a high density and spherical distribution of stars known as the Galactic Bulge (GB, Dwek et al. [1995](#); Wegg and Gerhard [2013](#)). With a mass of $\approx 2 \times 10^{10} M_{\odot}$ (Valenti et al. [2016](#); Portail et al. [2017](#)) and an age of 7.5–11 Gyr (Surot et al. [2019](#)), the GB is an exciting site to study the structure and evolution of the galaxy, formation and evolution of compact objects (e.g. Muno et al. [2003](#), [2009](#); Genzel et al. [2010](#); Jonker et al. [2011](#); Zhu et al. [2018](#)), and probe possible dark matter annihilation at the galactic center (GC; e.g. Bergström et al. [1998](#); Regis and Ullio [2008](#)).

Fermi observations of the GC show an excess of γ -ray emission from the inner $\sim 1.25^{\circ}$ of our galaxy, peaking at 2 GeV (e.g. Goodenough and Hooper [2009](#); Abazajian and Kaplinghat [2012](#); Gordon and Macías [2013](#); Ajello et

al. 2016). Some studies have argued that this galactic center excess (GCE) is produced from dark matter annihilation, based on the γ -ray spectra (Hooper and Goodenough 2011) and the spatial distribution of the GCE (Daylan et al. 2016; Di Mauro 2021). However, other studies have maintained that the GCE can be explained, both in spectra and morphology, through known astrophysical sources such as unresolved millisecond pulsars (MSPs) formed in the GB (Abazajian and Kaplinghat 2012; Yuan and Zhang 2014; Brandt and Kocsis 2015; Macias et al. 2018). Identifying the required population of MSPs in the GB would resolve the issue, but this is difficult in the gamma-ray regime (due to the limited angular resolution of gamma-ray telescopes, Gonthier et al. 2018), and in the radio (due to distance and interstellar scattering: Deneva et al. 2009; Macquart and Kanekar 2015; Calore et al. 2016; Hyman et al. 2019; Frail et al. 2024).

Most MSPs produce principally blackbody-like X-ray emission (e.g. Zavlin et al. 2002; Bogdanov et al. 2006), that is too faint and soft to be detectable at the distance and extinction typical of the Galactic Center region, as X-ray photons with energies $\lesssim 2$ keV are heavily obscured by the large absorption column towards the GC. However, some MSPs produce relatively bright, magnetospheric, highly pulsed X-ray emission, such as PSR B1821-24 in M28 at $L_X \sim 10^{33}$ erg/s (Saito et al. 1997; Becker et al. 2003; Rutledge et al. 2004; Bogdanov et al. 2011), and the non-cluster pulsars PSR J0218+4232 and PSR B1937+21 at $L_X \sim 4 \times 10^{32}$ erg/s (Verbunt et al. 1996; Kuiper et al. 1998; Takahashi et al. 2001; Gotthelf and Bogdanov 2017). Another fraction of MSPs (the “spiders”; Roberts 2013) are in compact binary systems with a non-degenerate companion that produces a wind, generating an intra-binary shock (e.g. Wadiasingh et al. 2018; Kandel et al. 2019) which emits bright X-rays with energies $\gtrsim 2$ keV. A significant fraction of redback MSPs, in the field and in globular clusters, are now known to emit hard X-rays with $L_X > 10^{32}$ erg/s (Berteaud et al. 2021; Zhao and Heinke 2022; Koljonen and Linares 2023). Resolving bright MSPs

through X-ray telescopes like *Chandra* and *XMM-Newton* telescopes, which have a better angular resolution than Fermi, will allow us to constrain the MSP population in the GB and probe the source of the GCE.

However, other sources like cataclysmic variables (CVs, e.g., Revnivtsev et al. 2009; Perez et al. 2015; Hailey et al. 2016; Xu et al. 2019), X-ray binaries (XRBs, e.g., Revnivtsev et al. 2008; Jonker et al. 2011), and background active galactic nuclei (AGN) also emit X-rays with energy $\gtrsim 2$ keV and are detected in X-ray surveys. CVs, some active XRBs, and many nearby AGN show prominent Fe lines — 6.4 keV fluorescent Fe emission, 6.7 keV Fe XXV (He like) K line, and 7.0 keV Fe XXVI (H like) K line (e.g., Xu et al. 2016; Torrejón et al. 2010; Fabian et al. 2000; Ricci et al. 2014). The large amounts of interstellar dust in the Galaxy absorb all optical and ultraviolet information, and the high stellar density towards the GC makes it challenging to identify infrared counterparts of GB sources (DeWitt et al. 2010, 2013; Greiss et al. 2014; Wu et al. 2015). While the detailed spectroscopic analysis of all X-ray sources in the GB would be difficult and time-consuming, selecting a narrower subset of sources based on photometric methods (like broadband hardness ratios (HR) or narrow-band color diagnostics) and analyzing only these sources further would be more efficient.

Hailey et al. (2018) used the HR between 2-4 and 4-8 keV (we shall call this $HR2$) to select a group of soft non-thermal sources with $HR2 < 0.3$ within a parsec of the Galactic Center, which they argue are X-ray binaries with likely black hole accretors (Mori et al. 2021). They distinguish these sources from those with $HR2 > 0.3$, dominated by Fe emission lines, that they argue are dominated by CVs. However, such a broadband hardness ratio can be influenced by Galactic absorption and intrinsic hardness of the X-ray continuum, in addition to the strength of Fe lines.

In this paper, we use X-ray colors, defined narrowly around the Fe-line complex, to select a small set of possible continuum sources. We study these

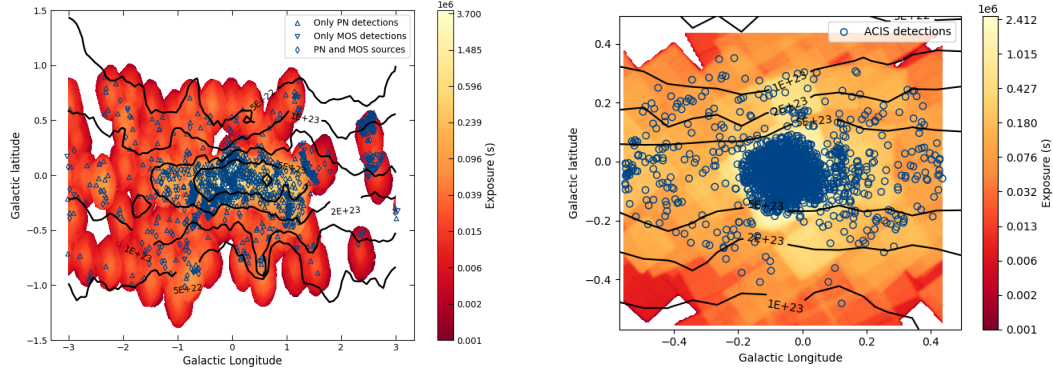


Figure 4.1: *XMM-Newton* EPIC (*left*) and *Chandra* ACIS (*right*) observations around the Galactic center. We combine the EPIC-PN and EPIC MOS exposures, using a weight of 0.4 for MOS data to account for the differing effective areas. We also overlay contours of N_H , and positions of all the sources we consider. Sources detected in EPIC-PN but not EPIC-MOS (blue upper triangle), detected in EPIC-MOS but not EPIC-PN (downward blue triangle), and detected in both EPIC-PN and MOS (blue diamond) are marked separately. *Chandra* ACIS sources are shown in blue circles. We only consider *XMM Newton* sources with $N_H \geq 5 \times 10^{22} \text{ cm}^{-2}$ and *Chandra* sources within $30'$ of the Galactic Center.

sources in detail as an efficient way to search for candidate MSPs in the GB, along with other continuum sources (X-ray binaries, slower pulsars, and distant AGN). We download the X-ray spectra of GB sources from *Chandra* and *XMM-Newton* data archives and check for any excess Fe emission compared to the continuum spectra of MSPs. We discuss our methodology in detail in §4.2. We report our results in §4.3 and discuss the candidate sources in §4.4. We summarize and discuss the future prospects of our work in §4.5.

4.2 Data Acquisition and Analysis

We use the *XMM-Newton* and *Chandra* observations around the GC to search for potential millisecond pulsar (MSP) candidates. The *XMM-Newton* source catalog (4XMM-DR13, Webb et al. 2020) and the *Chandra* source catalog (CSC 2.0, Evans et al. 2010; Evans et al. 2020) provide properties of the sources

detected in this region and allow us to access the associated data products directly, including the spectra. We use these spectra for our analysis.

4.2.1 XMM-Newton sources

The X-ray telescope *XMM-Newton* has extensively observed the region around the GC. Ponti et al. (2015) presents exposure maps reaching to 1.5 Ms equivalent of EPIC-pn observations around the central 1.5° from the GC. For our purpose, we select a $6^\circ \times 6^\circ$ field around the GC. This region is characterized by high interstellar extinction and substantial absorption column density. The absorption column density also varies significantly within this region, particularly affecting soft X-rays. We only consider regions with $N_H > 5 \times 10^{22} \text{ cm}^{-2}$, to identify only highly absorbed sources that are consistent with GC distances (e.g. Munro et al. 2003). We model the interstellar extinction from the Galactic Dust Reddening and Extinction Service (Schlegel et al. 1998; Schlafly and Finkbeiner 2011) and use the correlation reported in Bahramian et al. (2015) to calculate the N_H from extinction values (i.e. we use regions where $A_v \geq 17.8$).

We then look into the 4XMM-DR13 catalog (Webb et al. 2020) to search for sources in this region and the corresponding *XMM Newton* EPIC observations. We only select point sources (EP_EXTENT = 0), and require at least one spectrum (SPECTRA=True). The GC region has bright diffuse emission and is crowded with many bright sources. This leads to several moderately bright sources where the summary flag of the detection, SUM_FLAG, is flagged as 1 (i.e. source parameters may be affected). We consider detections flagged as 0 or 1 for our analysis, and ignore detections flagged > 1 . We use the hardness ratio, HR_3 , provided in the source catalog to remove likely foreground sources. This hardness ratio is calculated from count rates (CRs) in bands 3 (1–2 keV) and 4 (2–4.5 keV) i.e. $HR_3 = (CR_4 - CR_3)/(CR_4 + CR_3)$. Given the high N_H in our regions of interest, we expect very few photons with energy < 2 keV. We therefore only choose sources with 1σ upper limits on $HR_3 > 0.65$ for our

analysis (consistent with XMM observations of a source with $N_H = 5 \times 10^{22} \text{ cm}^{-2}$ and $\Gamma = 1.5$). This gives us 716 X-ray sources, with 2147 detections in total. We extract the EPIC-PN and EPIC-MOS spectra of these 716 X-ray sources from the HEASARC data archive. We combine the spectra for sources detected across multiple observations using the `epicspeccombine` method. We process EPIC-PN and EPIC-MOS spectra separately as their responses are very different. We show the exposure map, absorption column contours around the Galactic Center (Schlegel et al. 1998), and the selected XMM sources in Fig. 4.1.

For comparison, we also simulate 20000 spectra of highly absorbed MSPs, as they would be observed by XMM-Newton EPIC-PN and EPIC-MOS detectors. Since we do not know the exact distribution of the MSP properties in the GB, we use a log-uniform distribution for absorption column density, N_H , between $5 \times 10^{22} \text{ cm}^{-2}$ and $5 \times 10^{23} \text{ cm}^{-2}$; a uniform distribution of power-law indices, Γ , between 1.0 and 2.0; and a log-uniform distribution of 2–10 keV X-ray luminosity L_X between $10^{31} \text{ ergs s}^{-1}$ and $10^{34} \text{ ergs s}^{-1}$. The lower limit on N_H is consistent with our selection of sources and the higher limit is where $\gtrsim 50\%$ of the flux is absorbed in the 6–7 keV range. The limits on Γ are based on the distribution among the bright MSPs detected in GCs (Zhao and Heinke 2022). For each simulated MSP spectrum, we use the combined exposure, background, and response of a randomly selected XMM observation; values of N_H , Γ , and L_X chosen at random from their corresponding distributions; and then generate the fake spectrum through the `fakeit` method in XSPEC v12.9. If an observed source is not detected in one of the detectors, the MSP spectrum for that detector is not generated.

4.2.2 Chandra sources

The *Chandra* X-ray observatory has ≈ 2 Ms of total ACIS-I observations pointed towards Sgr A*. These are ideal for probing sources close to Sgr A*, due to the

depth of these observations and *Chandra*'s excellent spatial resolution in this crowded (and relatively high-background) environment. Farther from Sgr A*, XMM's angular resolution is generally sufficient, and as XMM collects more photons per second, XMM detections often deliver superior spectra. Thus, we only use the central 30' around Sgr A* for analysis of *Chandra* data. We only consider sources and regions with $N_H \geq 5 \times 10^{22}$, to be consistent with the Galactic Center distance (consistent with XMM constraints). We only use ACIS observations without a grating inserted (cf. Nowak et al. 2012; Zhu et al. 2018), to allow for combining spectra from multiple observation epochs.

We use the Chandra source catalog (CSC) v2.0 (Evans et al. 2010, 2024) to search for sources in this region. We select sources that have a `significance` greater than 3σ and `likelihood_class = TRUE` (`likelihood_class` is based on false detection rate, and not solely on the source significance or likelihood of detection). We require that the best estimate of the net source flux and photon flux in the "hard" (2–7 keV) band (`flux_aper_h`, `photflux_aper_h`) is positive, and that the source has at least one detection stack where the significance of the detection in the "hard" band (`s.detect_significance_h`, `s.flux_significance_h`) is higher than 3σ . We also set the upper limit of the hardness ratio between the "hard" and "medium" (1.2–2 keV) bands, $HR_{hm,upper} > 0.7$ to remove foreground sources. CSC v2.0 does not specify HR_{hm} or its error limits for sources if its aperture flux values and errors cannot be determined in either the "hard" or the "medium" bands. Since several highly absorbed sources may not be detected below 2 keV, i.e. in the "medium" band, we also allow for sources where HR_{hm} is not specified (we have already selected that `photflux_aper_h` > 0, so this only adds hard sources with non-detection in the "medium" band). Manually inspecting the CSC catalog shows that the confusion flag, `conf_flag`, and extended source flag, `extent_flag`, are not very accurate for sources close to the GC (due to the bright background diffuse emission). Therefore we do not constrain these properties. We also require that the "hard"

band flux significance in the selected observations (`o.flux_significance_h`) is greater than 3σ , and that the off-axis angle is less than $8.0'$. This gives us 2068 sources with 34761 observations. We extract spectra and detector responses of the selected sources in the selected observations from the Chandra Data Archive. While the *Chandra* ACIS soft energy response has degraded significantly over the years¹, its 2–10 keV response has remained nearly constant. Therefore, we combine the source spectra from all the observations using the `combine_spectra` tool in CIAO. We also generate 10000 simulated *Chandra* MSP spectra with the same N_H , Γ , and L_X used for XMM spectra, and the combined background and responses observed for the detected *Chandra* ACIS sources.

4.2.3 Simulated source selection

The GB has large amounts of diffuse X-ray emission, leading to a high background for the simulated and detected sources. Simulating sources by randomly choosing N_H , L_X , and background can lead to some low-luminosity sources being associated with high N_H and/or background; such sources would not be detected in observations. We only consider simulated spectra with more than 3σ (net counts by square root of background counts) significance of detection in the 2–10 keV range for analysis. Our simulations will also include bright sources simulated with low backgrounds. However, these do not significantly affect our results as their spectra are dominated by the Poisson noise from source emission rather than the background. Manually inspecting the source and background extraction regions in the 4XMM-DR13 shows that the selected background regions for some of the sources may be incorrect and lead to over-subtraction of the background spectra. Limiting ourselves to observed spectra with more than 3σ sources allows us to remove such sources.

¹<https://cxc.cfa.harvard.edu/proposer/POG/html/chap6.html>

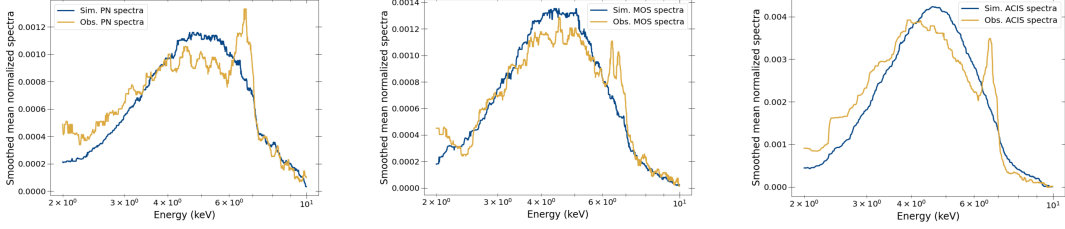


Figure 4.2: Mean normalized spectra of simulated MSPs and observed sources with EPIC-PN (*left*), EPIC-MOS (*center*), and ACIS (*right*) detectors. Each spectrum is divided by its net counts in the 2–10 keV energy range to get its normalized spectrum. The mean of the normalized spectra for observed sources shows an excess in 6–7 keV corresponding to the fluorescent Fe (6.4 keV) and Fe-K (6.7 keV & 7.0 keV) lines. This emission is from CVs embedded in our sample. The observed spectra also differ from simulated spectra in terms of hardness (possibly due to a more complex N_H & L_X distribution) and apparent absorption features (due to over-subtraction of the background).

Fig. 4.2 compares the mean normalized X-ray spectra of observed and simulated *XMM-Newton* and *Chandra* sources. The mean normalized spectrum is calculated by dividing the spectrum of each source by its net counts in the 2–10 keV energy range, and calculating the mean across all sources. Taking the mean spectra after such a normalization allows us to see the features in the faint spectra without being masked by those of bright sources. We see that the mean normalized spectra of the observed sources show excess emission in the 6–7 keV range, when compared with the normalized spectra of simulated MSPs. This excess emission is likely from CVs, X-ray binaries, and background AGN embedded among the observed sources that emit fluorescent Fe and Fe-K lines (6.4, 6.7 keV and 7.0 keV). Fig. 4.2 also shows that the mean normalized spectra of observed PN and ACIS sources have higher 2-4 keV emission, and the mean normalized spectrum of observed MOS sources shows a dip at ~ 2.5 keV. These features could be due to differences in the inherent distribution of N_H and L_X , or improper background subtraction. Limiting ourselves to the 5.5-7.5 keV energy interval allows a more robust comparison of simulated and observed sources.

We compare the distribution of net counts, background-to-net ratio, and

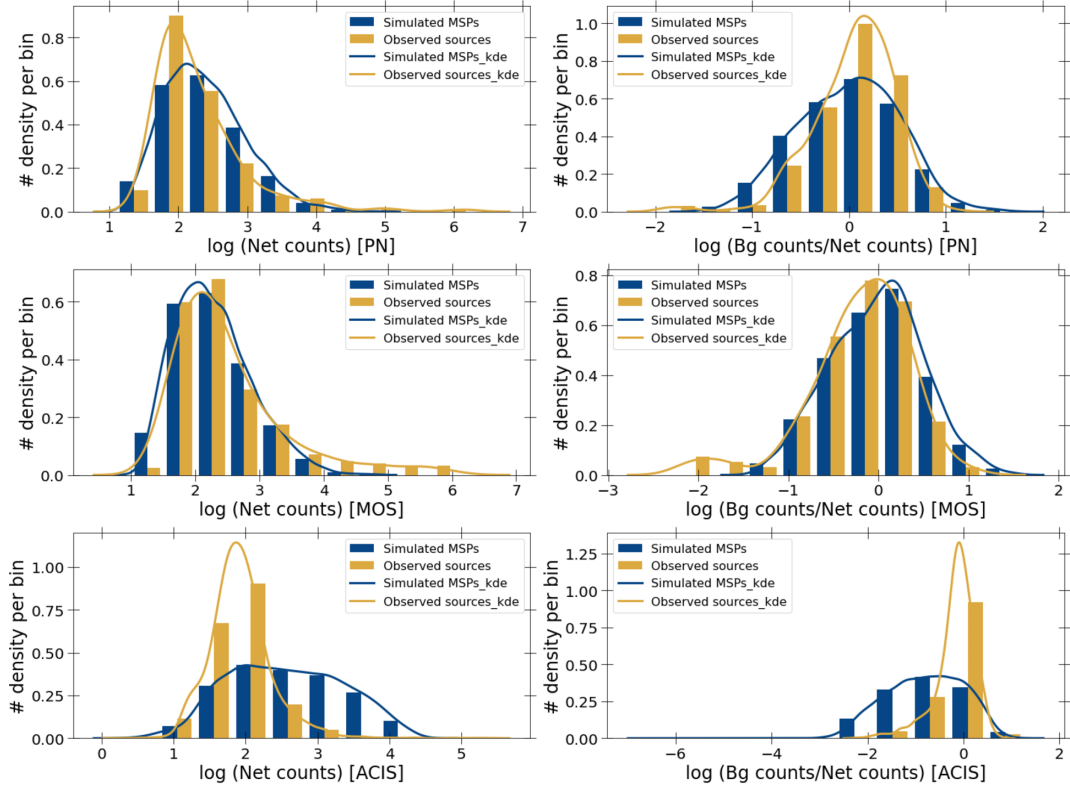


Figure 4.3: Distribution of the net and background counts in the EPIC-PN, EPIC-MOS, and ACIS sources (observed sources and simulated MSPs). We only show simulated sources whose net counts are greater than 3 times the square root of their background counts. In general, these sources show a high background due to the diffuse emission close to the Galactic Center. Our simulated sources cover the entire range of net counts and background contributions spanned by the observed sources.

signal-to-noise ratio of simulated and observed sources in EPIC-PN, EPIC-MOS, and ACIS detectors in Fig. 4.3. While the exact distributions of these properties for the simulated sources and observed sources are different, the simulated sources cover the entire range that the observed source and background counts cover.

4.2.4 X-ray color analysis.

We use the simulated spectra to analyze the distribution of X-ray colors for MSPs (and other quiescent X-ray sources) that do not show any excess Fe

emission. We define our X-ray colors as the ratio between the net counts in 6.2–7.2 keV and the sum of net counts in 5.8–6.2 and 7.2–7.6 keV. The 6.2–7.2 keV emission range includes the Fe XXV ($7 \rightarrow 1$) 6.7 keV line and the Fe XXVI ($3 \rightarrow 1$, $4 \rightarrow 1$) 6.95, 6.97 keV lines², as well as the 6.4 keV line from neutral Fe. The regions 5.8–6.2 keV and 7.2–7.6 keV do not show any lines for plasma with $kT > 5$ keV. So, we choose them as the continuum regions with which the 6.2–7.2 keV emission can be compared.

We plot the X-ray colors of the simulated sources against the net counts in the 2–10 keV energy range to constrain the X-ray color distribution of MSPs. We divide the simulated MSPs into 20 bins based on the logarithm of 2–10 keV net counts, and get the mean color, and the standard deviation in color, for each bin. We then use the mean X-ray colors as a function of net counts, and the 2σ upper and lower limits on the observed X-ray colors, to search for candidate sources lacking iron lines.

4.3 Results

Fig. 4.4 compares the distribution of X-ray color and net counts in the simulated MSPs and the observed sources, across EPIC-PN, EPIC-MOS, and ACIS detectors. As expected, the scatter in the X-ray color decreases as the net counts of the simulated MSPs increase (i.e. the signal-to-noise ratio increases). To define a 2-sigma range in X-ray color for each net counts bin of our simulated sources (left panels of Fig. 4.4), we assume a Gaussian distribution of the X-ray color. Note that our definition of an X-ray color is a ratio of two Poisson variables, resulting in a heavier tail than a strict Gaussian distribution. Therefore we calculate the standard deviation using 5σ -clipping; i.e., we exclude values $> 5\sigma$ away from the mean, and calculate σ iteratively, until convergence.

The scatter in the observed source colors (right panels of Fig. 4.4) is higher

²See, e.g., <http://app.atomdb.org>

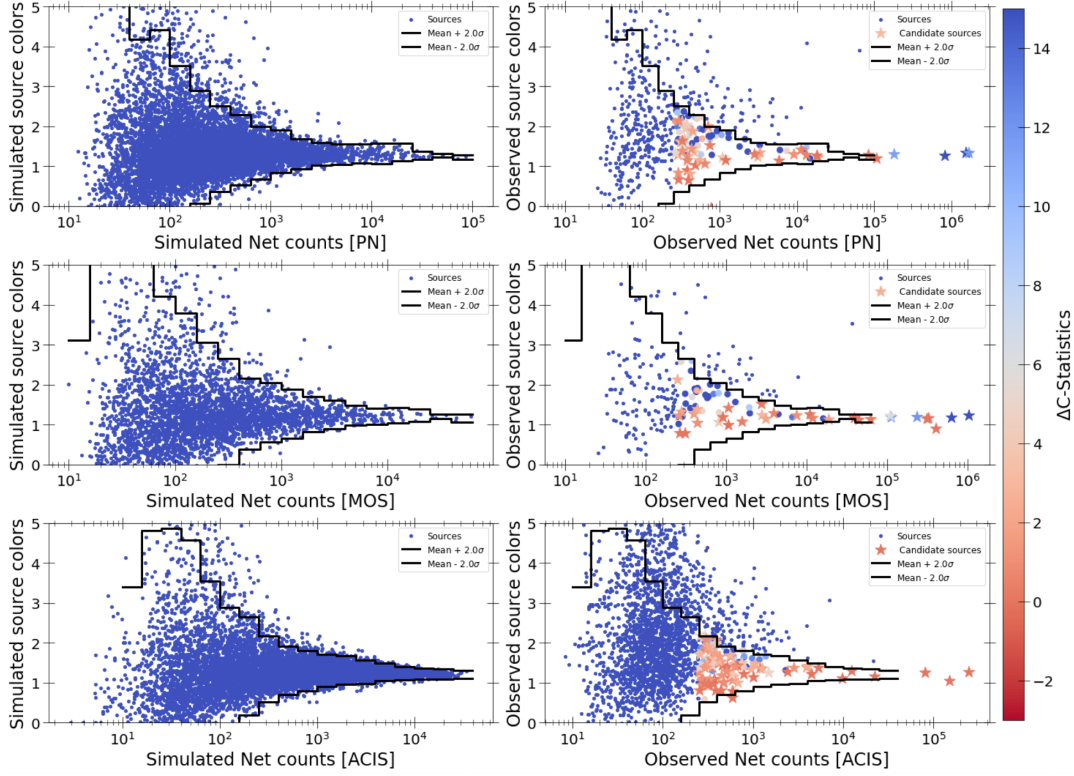


Figure 4.4: Distribution of simulated MSPs and observed sources in the X-ray color vs. net counts plane. We calculate the X-ray color as the ratio of the net counts in the 6.2–7.2 keV region, to the sum of the net counts in the 5.8–6.2 and 7.2–7.6 keV regions, so larger colors indicate Fe line emission. We also show the 2σ bounds on the color calculated empirically from our simulated sample of MSPs. The observed sources show a greater scatter in the X-ray colors, particularly to higher X-ray color, compared to the simulated MSPs, due to the presence of line-emitting CVs among the observed sources. Our final list of candidate continuum sources, that appear to lack lines both by X-ray color and in spectral fitting, are marked as stars.

than that of simulated MSPs even at high net counts, specifically towards higher X-ray colors. This is due to sources with fluorescent Fe and Fe-K lines (i.e. CVs, XRBs and AGN) in the observed sample. Fig. 4.4 shows the 2σ X-ray color limits for the simulated sources as a pair of black lines. We expect the observed MSPs to mostly lie within these bounds. We only consider EPIC-PN, EPIC-MOS, and ACIS sources with net counts ≥ 250 , as the 2σ bounds on the X-ray colors get very large for sources with low net counts. These two constraints give us 93 EPIC-PN, 65 EPIC-MOS (42 in common between PN

and MOS selections), and 130 Chandra ACIS candidate continuum sources. Differences in the selected set of PN and MOS sources could be due to our imposed counts cutoff, to sources falling on the corners of chips or in chip gaps, to scattering from source or background noise, etc.

We then perform spectroscopic analysis on this selected sample of sources, using XSPEC v12.11.1. We compare three models – an absorbed power-law (model I), an absorbed power-law with a Gaussian (Fe) line, with a fixed width of 0.1 keV and energy free to vary from 6.3 to 7.1 keV (model II), and an absorbed power-law with three Gaussian lines with fixed positions at 6.4, 6.7 and 7.0 keV, and 0 keV width (model III). We assume Wilms et al. (2000b) interstellar abundances and perform the spectral fitting in the 2–10 keV energy range using Cash statistics (C-statistics). Model II has 2 more free parameters than model I, while model III has 3 more free parameters than model I. We show the difference in the C-statistics between model I and model III fits for the selected sources in Fig. 4.4. We summarize the results of our spectroscopic analysis of these selected source in the Table 4.1. The full version of the table will be available online. We describe the columns of the table here.

For a source to be identified as a continuum source, i.e. that model I is more likely than model II or model III, we require that either a) ($cstat_I < cstat_{II} + 4$) & ($cstat_I < cstat_{III} + 6$); or b), that the normalization of all the Gaussian lines in models II and III are consistent with zero (i.e., the lower limit of the 90% confidence error bars reaches zero). Here, $cstat_I$, $cstat_{II}$, and $cstat_{III}$ are the best fitting C-statistic values for models I, II and III respectively. Criterion a) essentially applies the Akaike Information Criterion, by comparing the C-statistic, a log likelihood, of the models, applying a penalty that increases with the number of free parameters, and criterion b) directly checks for the log-likelihood of the Fe emission line(s). Using these constraints, we find 46 continuum EPIC-PN sources, 32 continuum EPIC-MOS sources (54 continuum XMM sources in total), and 82 ACIS continuum sources.

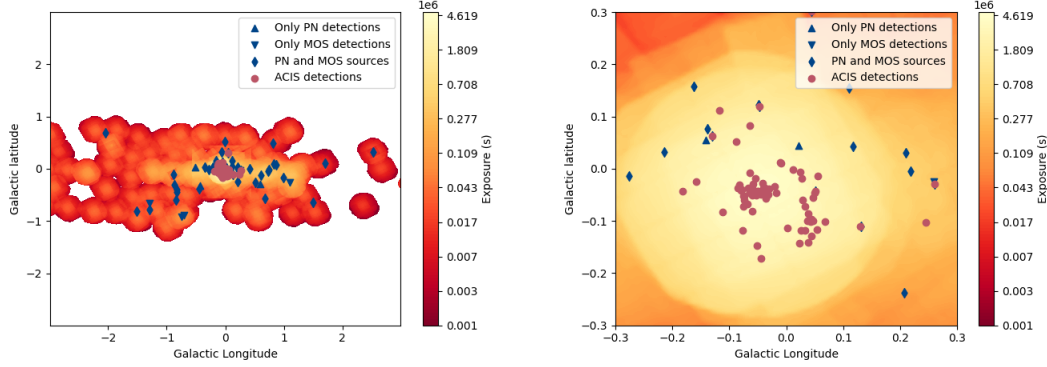


Figure 4.5: Sky positions of candidate continuum sources detected in *XMM-Newton* (blue circles) and *Chandra* (red circles) observations. Among the XMM detections, sources only identified through EPIC-pn (EPIC-MOS) analysis are shown as upward (downward) triangles, and sources detected in both EPIC-pn and EPIC-MOS are marked as diamonds. *Left*, a view of the full region. *Right*, a zoomed view of the Galactic center region. The background maps are the combined exposure maps of *XMM-Newton* EPIC-pn, EPIC-MOS, and *Chandra* ACIS observations (EPIC-MOS and ACIS exposure maps were weighted by 0.4 compared to EPIC-pn exposure maps). *Chandra* detections dominate the central few arcminutes, while *XMM-Newton* detections dominate further away.

The continuum emission of an X-ray source might be more complicated than a simple power-law. The X-ray spectra can be further complicated by sources falling on chip gaps in some observations, improper selection of the background, etc. Therefore, the quality (or "goodness") of the absorbed power-law fit alone is not a good indicator of a continuum source, and we do not use it as a criterion. These complications in the X-ray spectra are more evident when fitting a bright source; e.g., models II and III can sometimes fit the spectra better with spurious Gaussians that do not coincide with the expected positions and equivalent widths of Fe lines. Therefore we include c) all sources with $> 10^5$ counts with colors consistent with simulated MSPs in our list of line-less sources. These are mostly bright X-ray binaries. The positional distribution of the candidate sources is shown in Fig. 4.5.

4.4 Discussion

4.4.1 X-ray sources with counterparts

First, we check our candidate continuum source-list against known sources close to the Galactic center. We find 16 known sources in our list - 14 X-ray binaries/transients, one pulsar wind nebula (PWN), and one magnetar (SGR J1745-2900, Mori et al. 2013; Rea et al. 2013). Checking the spectra of the X-ray binaries shows that these do not show Fe fluorescent lines during their X-ray outbursts. The PWN, 4XMM J174722.8-280904, in the supernova remnant G0.9+0.1, was only detected in our *XMM-Newton* data (outside our *Chandra* source selection region), which does not have the resolution to distinguish the PWN from the likely pulsar CXOU J174722.8-280915 (Gaensler et al. 2001; Porquet et al. 2003; Camilo et al. 2009).

We also check for other multi-wavelength counterparts of our candidate X-ray sources. Due to the large number of stars and dust clouds in the Galactic plane, the probability of finding optical and infrared counterparts by coincidence is very high. Therefore, we limit ourselves to radio counterparts with $\nu \leq 5.5$ GHz. We start by searching for counterparts within $2''$ of the given source position (a conservatively large error radius for astrometric offsets between catalogs). The probability of random matches is high for radio surveys around the GC. We randomly shift the positions of the candidate X-ray sources by $10''$ and check for the number of chance counterparts. We perform 100 such realizations to estimate the false alarm rates (FAR), i.e. the expected number of XMM and Chandra sources that could have spurious matches given the separation distance between the X-ray and radio source position. We only select radio sources where the separation corresponds to an $\text{FAR} < 0.2$. We find counterparts to eight *Chandra* sources in the Jansky Very Large Array S-band observations of 2014 and 2019 (Zhao et al. 2020), one *Chandra* source in the 3 GHz Very Large Array Sky Survey quick-look images (Gordon et al. 2021), and

one *XMM-Newton* source in the VLA C-band (Lu et al. 2019) observations (in addition to known sources). Four of these sources have been suggested to be associated with PWN (Park et al. 2005; Munro et al. 2008; Zhao et al. 2013). We summarize these counterparts in Table 4.3 and Table 4.4. The full version of the tables are available online. Here we only show the first three candidates from the Chandra observation.

4.4.2 X-ray variability

We also check for X-ray variability of our candidate sources in the 4XMM-DR13 and CSC v2.0 catalogs. Although the 4XMM catalog has a variability flag, it only considers variability within the observation/detection. In some cases, the source variability flag `SC_VAR_FLAG` is set to True (T) due to observed variability in a detection with summary flag, `SUM_FLAG` > 1. In addition, we choose to use a looser threshold for variability than the strict $p < 10^{-5}$ used in the 4XMM-DR13 catalog (p is the probability that the flux is constant). The CSC v2.0 catalog uses a much looser threshold for variability, $p < 0.1$, and could mark a source variable in the master catalog even when some of the source detections in the stack might be confusing (e.g., high off-axis angle, low significance of detection, etc.). Therefore, we use other source and detection properties in these catalogs to determine intra- and inter-observation variability.

The 4XMM-DR13 catalog also gives the chi-square probability that the observed detection is constant, `EP_CHI2PROB`. We check for these values among the “good” detections that we have analysed here (check §4.2.1 for the selection criteria). For sources with multiple detections we use the minimum value of `EP_CHI2PROB`. We set a threshold probability, $p < 3.3 \times 10^{-4}$ to check if the source is variable on short time scales (i.e., within an observation). Given that we have 54 candidate sources with 302 detections (in total), the false alarm rate for our XMM candidate continuum sources is ≈ 0.1 .

The CSC v2.0 catalog provides three measures for the probability that the

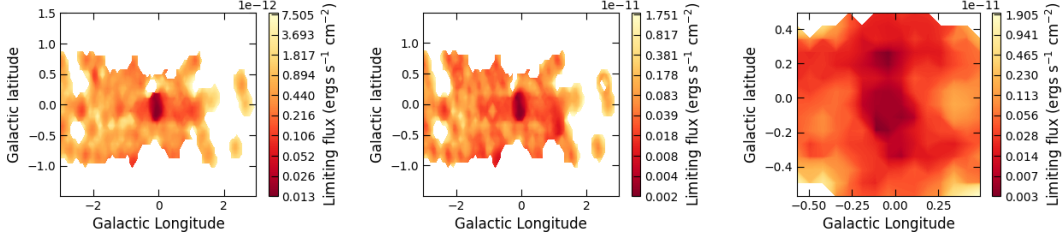


Figure 4.6: Limiting flux for AGN in our analysis for EPIC-PN (*left*), EPIC-MOS (*center*) & ACIS-I (*right*) detectors. The limiting flux indicates the minimum flux required for the AGN to be detected with ≥ 250 counts, after suffering Galactic absorption, given the total exposure time towards that region. The flux is calculated using N_H values and combined exposures of all observations used in our analysis. We use a grid size of $4.5'$.

source is variable within an observation—the Gregory-Loredo estimate of variability (`var_prob_h`), the Kolmogorov–Smirnov test for variability (`ks_prob_h`), and Kuiper’s test for variability (`kp_prob_h`). We check these values for each “good” observation (check §4.2.2 for selection criteria) of our candidate sources. For a source to be variable within an observation (short-term variability), we require that at least two of these three values be ≥ 0.999 . Given that we have 82 candidate continuum *Chandra* sources with 838 observations (in total), the false alarm probability for our list of *Chandra* sources is ≈ 0.1 .

To calculate whether different observations/detections of a given source show inter-observation variability, we compare the flux calculated in each observation in the hard band (2–7 keV for *Chandra*, and 2.0–10.0 keV for *XMM-Newton*) and check the χ^2 probability that the flux is constant. We use a threshold $p < 0.001$ to consider a source to be variable between observations (long-term variability).

4.4.3 Background AGN

AGN, often at high redshift, make up the majority of X-ray sources in deep X-ray observations at high Galactic latitude. While many AGN show fluorescent Fe lines, their redshift can shift the 6.4 keV line to outside the 6.2–7.2 keV

range that we consider here. We use the exposure maps of *XMM Newton* and *Chandra*, the absorption column density, N_H from extinction values of Schlafly and Finkbeiner (2011), and the Gilli et al. (2007) X-ray background model to estimate an upper limit to the number of AGN among our candidate continuum sources. This is an upper limit estimate, since we consider the extreme case that all AGN lack Fe lines within our selected energy range.

4XMM-DR13 provides separate exposure maps for several energy bands. We notice that the band 4 (2.0–4.5 keV) and band 5 (4.5–12 keV) exposure maps have similar exposure values (exposure values are provided in the units of seconds). Therefore we only combine band 5 exposure maps of all the *XMM Newton* observations we use in our analysis. The exposure maps of different detectors and filters are combined separately (i.e., three combined exposure maps of PN for “thin”, “medium”, and “thick” filters, and three for the different MOS filters). CSC 2.0 provides ACIS exposure maps for three different energy bands. We use the hard band (2–7 keV) exposure maps to calculate total exposures. Since these exposure maps are provided in the units of $\text{cm}^2 \text{ s}$, we divide the exposure map by 358.13 cm^2 — the effective area of ACIS-I at 3.8 keV, the effective energy of the hard band — to get the value of exposures in seconds (we consider the ACIS-I and ACIS-S exposures together as they have similar responses in the hard band).

We use a grid size of $4.5' \times 4.5'$ for exposure and N_H maps. Since we only consider sources with ≥ 250 counts in the 2–10 keV energy range for identifying candidate continuum sources, we calculate the minimum count rate to reach this threshold for different sky positions based on the exposure values. We then use the corresponding Galactic N_H values and PIMMS v4.13 to calculate the lower limit for the flux of AGN with ≥ 250 counts at that location. We show our resulting limiting flux maps in Fig. 4.6. We notice that the calculated limiting flux of EPIC-pn observations reaches $\sim 10^{-14} \text{ ergs/s}$, but none of the observed XMM sources with counts ≥ 250 had such a low flux. This could be

due to the effect of the bright diffuse emission background. Therefore, we place an empirical floor on the limiting flux of 4.0×10^{-14} ergs cm $^{-2}$ /s. We then calculate the number of AGN at each grid point using these values with the Gilli et al. (2007) X-ray background model, and estimate the predicted total number of AGN with ≥ 250 counts in our observations. We predict an upper limit of ~ 14 AGN in EPIC-pn observations, ~ 12 AGN in EPIC-MOS observations and ~ 5 AGN in ACIS observations. Comparing the estimated number of pn and MOS observations with the number of candidate sources, we can argue that there could be ~ 25 AGNs in total among the XMM sources.

The N_H values generated from Schlaflly and Finkbeiner (2011) maps are significantly higher than the best-fit absorption values for X-ray spectra of our sources. We infer that this is likely due to the majority of our X-ray sources lying near the Galactic Center, while the Schlaflly and Finkbeiner (2011) maps include dust and gas on the far side of the Galaxy, which we expect would indeed affect distant AGN. Repeating our AGN estimates using N_H values from the best-fitting spectra of our selected X-ray sources (check appendix), we get ~ 50 background AGN among *XMM* detections and ~ 26 AGN among the *Chandra* sources. These numbers would suggest that most of our candidate continuum sources from XMM are background AGN, but we reiterate that we think the true N_H values for background AGN are higher.

4.4.4 Probability of CVs among candidate quiescent sources

While most bright CVs (dominated by intermediate polars [IPs]; Suleimanov et al. 2022) show strong Fe lines, a few show weaker Fe lines (see e.g. Ezuka and Ishida 1999). This, along with Poisson noise from low counts in the 5.8-7.6 keV range, and high background diffuse X-ray emission, might make it difficult to detect Fe lines from some CVs, even using spectral analysis. We use the equivalent widths (EWs) of IPs in the solar neighborhood as reported in Xu et al. (2016) and N_H , L_X values from our MSP simulations to estimate the

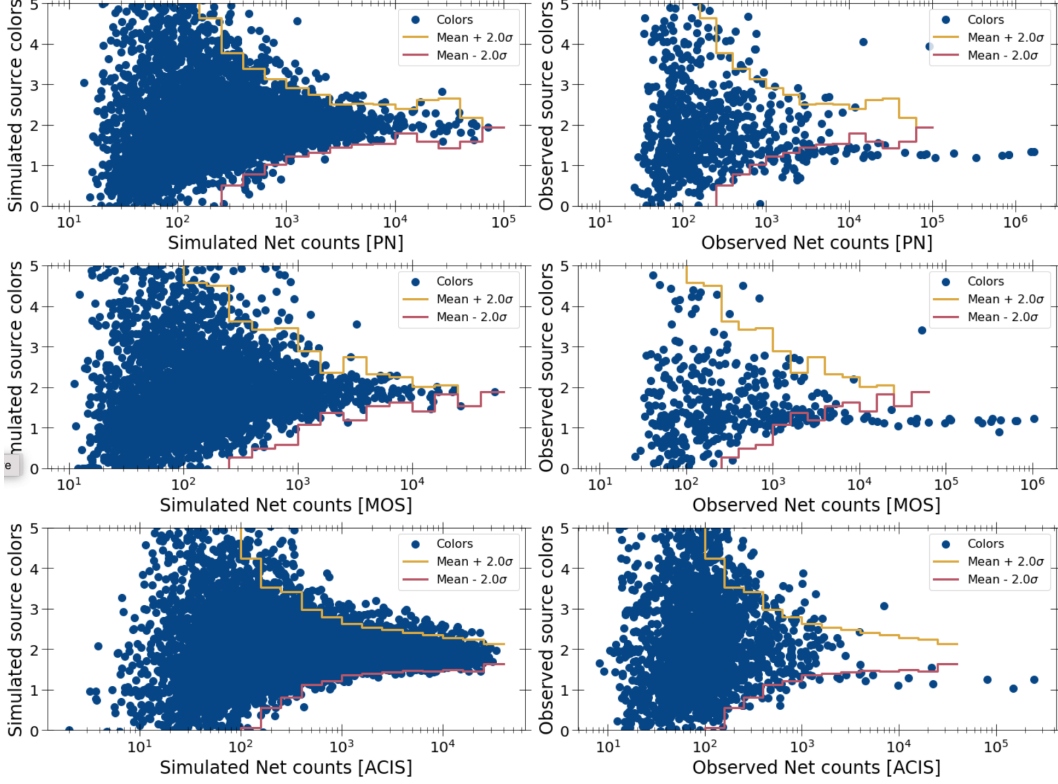


Figure 4.7: Comparing the Fe colors of simulated CVs with that of observed sources. Using twice the EWs of Fe lines of IPs in Xu et al. (2016) fits our observed sources better. We use these simulated CVs to estimate the probability of CVs among our candidate continuum sources.

number of CVs among our candidate continuum sources. (Other kinds of CVs will likely be too faint and/or soft to be seen in the Bulge through the high Galactic absorption.) CVs simulated using EW values of nearby IPs from Xu et al. (2016) do not reproduce the distribution of Fe colors vs net counts among the observed sources. Therefore, we try using twice the EWs of lines when simulating each source reported in Xu et al. (2016). This agrees better with the observed color of sources (compare the simulations in the left panels of Fig. 4.7 with the observed population of sources with lines in the right panels). This suggests that the metallicity of IPs in the GB is super-solar. This is consistent with expectations of a higher metallicity in the Bulge, e.g. Henry and Worthey 1999. It is also consistent with prior discussions of the Fe line strength of sources

in the GB, which found that roughly twice supersolar metallicity appears to be needed (Xu et al. 2019; Yamamoto et al. 2023; Anastasopoulou et al. 2023). The right panels of Fig. 4.7 show two populations of observed X-ray sources, one with a distribution of Fe line strength reasonably represented by our CV simulations, and another with an X-ray color consistent with no Fe line flux (cf. Fig. 4.4).

We then select simulated CVs that fall within the bounds we set for continuum-source Fe colors, and perform a spectral analysis of these sources. We check for simulated CVs that do not show evidence of Fe lines by using the same criteria that we use for selecting continuum sources: they do not show significant improvement in C-statistic when fitting to a model with Fe lines, and/or the photon flux [norm] in these lines is consistent with zero. Using calculations detailed in the Appendix, we estimate that there could be ~ 12 CVs among the EPIC-pn detections, ~ 4 in EPIC-MOS detections, and ~ 16 in ACIS detections, that cannot be distinguished from sources lacking Fe lines.

It is possible that there are additional CVs that for some reason (e.g. very low metallicity) do not show significant Fe lines, contaminating our sample of apparently line-less X-ray sources.

4.4.5 Comparison with Hailey et al. (2018)

Hailey et al. (2018) identified 12 Chandra X-ray sources near Sgr A* as likely quiescent black hole X-ray binaries, based on a lack of strong Fe lines inferred using broad hardness ratios (2-4 keV vs. 4-8 keV). We check whether these 12 soft sources are detected through our method. Only five of these sources (2CXO J174540.4-290036, 2CXO J174539.4-290041, 2CXO J174539.4-290046, 2CXO J174540.3-290049, 2CXO J174539.2-290049) are listed in CSC v2.0. Among these, 2CXO J174539.4-290041 and 2CXO J174540.3-290049 are detected among our candidate continuum sources. We see that the remaining three sources lie outside our Fe color limits, indicating the presence of Fe-lines in the X-ray

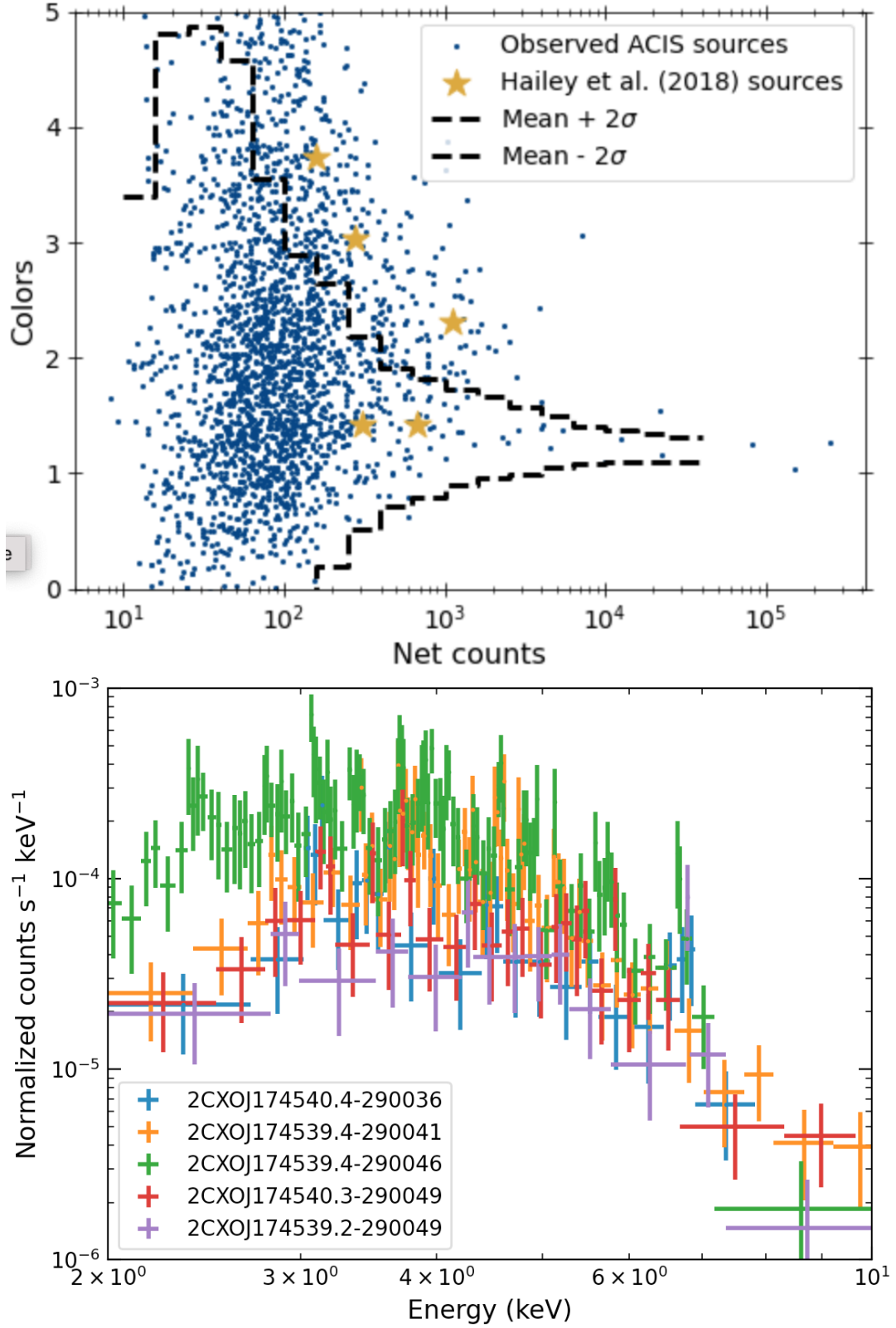


Figure 4.8: Fe excess (*left*) and spectra (*right*) of Hailey et al. (2018) sources detected in CSC v2.0. Only 2CXO J174539.4-290041 and 2CXO J174540.3-290049 lie within the 2σ bounds of MSP-like Fe colors. We also show the spectra of these sources, which reveal Fe lines⁸⁵ in the other three; 2CXO J174540.4-290036, 2CXO J174539.4-290046, 2CXO J174539.2-290049.

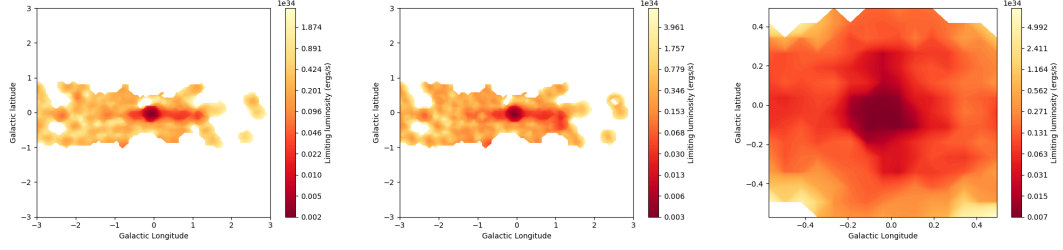


Figure 4.9: Limiting unabsorbed luminosity for GB sources to have ≥ 250 counts in our analysis for EPIC-PN (*left*), EPIC-MOS (*center*) & ACIS-I (*right*) detectors. The flux is calculated using N_H values and combined exposures of all observations used in our analysis. We use a grid size of $4.5'$.

spectra. The position of these five sources in the color vs. net counts plot, and their spectra, are shown in Fig. 4.8, which illustrate that the remaining three of these sources appear to show significant 6.7 keV Fe lines, suggesting that they may be CVs.

4.4.6 Implications for the γ -ray GCE

We use the Di Mauro (2021) estimates of γ -ray flux around the galactic center, the γ -ray luminosity function of MSPs (Gonthier et al. 2018; Hooper and Mohlabeng 2016), and the X-ray luminosity function of MSPs in globular clusters (Zhao and Heinke 2022) to compare the numbers of candidate continuum sources we detect to the predicted number of detectable MSPs, if the γ -ray GCE is indeed produced by MSPs. First, we compare the number of *Chandra* candidates in the central 0.2° around Sgr A*. We fit the $dS/d\Omega$ vs θ values in Fig. 10 of Di Mauro (2021) (where $dS/d\Omega$ is the γ -ray surface brightness calculated with annuli of size 1° , and θ is the angular distance from the Galactic Center) with a power-law (i.e, $dS/d\Omega = A(\theta/1^\circ)^{-\Gamma}$, where Γ is the power-law index). The best-fit power-law index is 1.55 ± 0.05 and power-law norm is 0.047 ± 0.004 . Using this power-law fit, we estimate the γ -ray flux within 0.1° and 0.2° to be $(1.2 \pm 0.4) \times 10^{-10}$ ergs/cm²/s and $(1.6 \pm 0.5) \times 10^{-10}$ ergs/cm²/s, respectively, corresponding to a GCE luminosity of $(9 \pm 3) \times 10^{35}$ ergs/s and $(1.3 \pm 0.4) \times 10^{36}$

ergs/s, for a Galactic Center distance of 8.0 kpc.

We use the various spectral energy distributions of MSPs presented in Gonthier et al. (2018) to convert these 1–10 GeV luminosities to 0.1–1000 GeV luminosity, L_γ . Depending on the SED used, we get L_γ values of $(1.8 \pm 0.7) \times 10^{36}$ ergs/s within 0.1° and $(2.5 \pm 0.9) \times 10^{36}$ ergs/s within 0.2° . For an average MSP γ -ray luminosity of $\langle L_{\gamma, MSP} \rangle = 1.1 \times 10^{33}$ ergs/s (GCE MSPs from Gonthier et al. 2018), this gives ≈ 1700 MSPs within 0.1° and ≈ 2300 MSPs within 0.2° . Alternatively, Hooper and Mohlabeng (2016) gives $\langle L_{\gamma, MSP} \rangle = 3.0 \times 10^{33}$ ergs/s for MSPs in globular clusters. If we use this value for MSPs in the GB, we get ≈ 600 and ≈ 800 MSPs within 0.1° and 0.2° , respectively. Another way of estimating the number of MSPs is to consider the fraction of 1-10 GeV flux within 0.1° and 0.2° compared to the 7° (region of interest in Gonthier et al. 2018). Our fit to the Di Mauro (2021) GCE surface brightness gives a flux of $(7.8 \pm 0.8) \times 10^{-10}$ ergs/cm²/s within 7° . For 11,500 MSPs within 7° (the estimate of Gonthier et al. 2018), this gives ≈ 1800 and ≈ 2400 MSPs within 0.1 and 0.2° of the galactic center, which is similar to our first estimate.

We correct the N_H values for GB sources (see Appendix B.1) and calculate the limiting luminosity for GB X-ray sources that can be detected with ≥ 250 counts in the 2–10 keV energy range by the EPIC-pn, EPIC-MOS and ACIS detectors, given the exposure times. Our 2–10 keV limiting luminosity reaches $\approx 10^{32}$ ergs/s around Sgr A* in the ACIS observations. From the Zhao and Heinke (2022) luminosity function, we notice that only 6 of the 173 MSPs have $L_{2-10\text{keV}} > 10^{32}$ ergs/s. Thus, we should only be able to detect 20–62 and 28–83 MSPs within 0.1° and 0.2° . Our analysis of *Chandra* sources has revealed 80 candidate continuum sources (mostly within 0.2deg), of which we have estimated ~ 5 possible AGN and ~ 16 possible CVs. If most of the remaining line-less X-ray sources are MSPs, they could be enough to produce the observed GCE.

At an angular distance of 1.5° away from the galactic center, the γ -ray GCE surface-brightness reduces to 3×10^{-8} ergs/cm²/s/sr (which translates to ~ 100

MSPs per square degree) and falls off exponentially. However, our limiting X-ray luminosity in these regions is $\gtrsim 10^{33}$ ergs/s, i.e. approximately 1 out of every 173 MSPs will be detected. Thus we expect very few MSPs outside the central region; with about 30 XMM fields around this radius, each of ~ 0.2 square degrees, this suggests of order 4 detectable MSPs. This is consistent with our previous observation that the number of candidate line-less X-ray sources in the XMM observations is very similar to the sum of CVs with faint Fe emission, plus background AGN.

4.5 Summary & Conclusion

In this paper, we attempted to identify MSPs in the GB to constrain the gamma-ray excess observed in Fermi observations of the GC. We searched for *Chandra* and *XMM-Newton* sources in the GB that exhibit continuum spectra and do not show fluorescent Fe or Fe-K lines typical of CVs. We find a total of 54 *XMM-Newton* sources and 83 *Chandra* sources, among which 16 X-ray sources are known X-ray sources including known X-ray binaries, one PWN and one magnetar. We estimate that ~ 35 of the *XMM-Newton* sources and ~ 5 *Chandra* sources could be background AGN. Our candidate sources are also likely to include 10–20 CVs with weak Fe lines. Comparison of the number of lineless XMM candidates with the number of possible CVs and background AGN show that only a small number of XMM candidates are likely MSPs. Calculating the number of MSPs needed to explain the GCE surface brightness distribution shows that most of the detectable MSPs will be close to the Galactic Center, and could be detected in our ACIS list of continuum X-ray sources. Indeed, our list of 83 *Chandra* candidate continuum sources is consistent with the expected number of 20–80 MSPs detectable in the X-ray, if the γ -ray GCE is indeed produced by MSPs.

The high absorption from dust and gas towards the Galactic center and

the large density of stars in the Galactic plane give us limited information on the counterparts for our candidate sources. Nevertheless, 16 of our suggested candidates have been independently identified as X-ray binaries or pulsars/magnetars/pulsar wind nebulae. We suggest that further follow-up of unclassified radio sources that have potential matches with these X-ray sources may be particularly rewarding.

The bright diffuse emission close to the Galactic center increases the background noise in the detected X-ray sources. Future X-ray telescopes with high effective area and a narrow point-spread-function (e.g. the proposed AXIS telescope; Reynolds et al. 2023; Safi-Harb et al. 2023) would allow us to reach lower limiting luminosities, and constrain the population of MSPs.

Acknowledgments

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Table 4.1: Descriptive version of the “Spectral fitting of *XMM-Newton* sources” table

Number	Column name	Units	Explanation
1	Name	—	IAU name of the XMM source
2	Source_ID	—	Unique source identifier in the 4XMM-DR13 catalogue
3	Counts_PN	photons	2–10 keV net counts in the combined PN spectra
4	Color_PN	—	Fe color in the combined PN spectra
5	Counts_MOS	photons	2–10 keV net counts in the combined MOS1 and MOS2 spectra
6	Color_MOS	—	Fe color in the combined MOS1 and MOS2 spectra
7	NH ¹	cm ⁻²	Best fitting absorption column density for the source
8	NH_low ¹	cm ⁻²	Lower limit of absorption column density
9	NH_high ¹	cm ⁻²	Upper limit of the absorption column density
10	PL_index ¹	—	Best fitting power law index for the source
11	PL_index_low ¹	—	Lower limit of the power-law index
12	PL_index_high ¹	—	Upper limit of the power-law index
13	Flux ¹	ergs cm ⁻² s ⁻¹	2–10 keV flux
14	Flux_low ¹	ergs cm ⁻² s ⁻¹	Lower limit of 2–10 keV flux

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Table 4.1: (continued)

Number	Column name	Units	Explanation
15	Flux_high ¹	ergs cm ⁻² s ⁻¹	Upper limit of 2–10 keV flux
16	C-statistics	—	C-statistics for the best-fitting absorbed power-law model
17	DOF	—	Degrees of freedom for the absorbed power-law model
18	Line_pos ²	keV	Position of the Gaussian line in model II
19	Line_pos_low ²	keV	Lower limit of the Gaussian line position
20	Line_pos_high ²	keV	Upper limit of the Gaussian line position
21	Line_photflux ²	photons cm ⁻² s ⁻¹	Photon flux from the Gaussian line
22	Line_photflux_low ²	photons cm ⁻² s ⁻¹	Lower limit of the photon flux from Gaussian line
23	Line_photflux_high ²	photons cm ⁻² s ⁻¹	Upper limit of the photon flux from Gaussian line
24	Delta_cstat_II-I	—	Difference in C-statistics between model I and II
25	Photflux_6.4 ³	photons cm ⁻² s ⁻¹	6.4 keV line photon flux
26	PhotFlux_6.4_low ³	photons cm ⁻² s ⁻¹	Lower limit of 6.4 keV line photon flux
27	Photflux_6.4_high ³	photons cm ⁻² s ⁻¹	Upper limit of 6.4 keV line photon flux
28	Photflux_6.7 ³	photons cm ⁻² s ⁻¹	6.7 keV line photon flux

Continued on next page

Table 4.1: (continued)

Number	Column name	Units	Explanation
29	Photflux_6.7_low ³	photons cm ⁻² s ⁻¹	Lower limit of 6.7 keV line photon flux
30	Photflux_6.7_high ³	photons cm ⁻² s ⁻¹	Upper limit of 6.7 keV line photon flux
31	Photflux_7.0 ³	photons cm ⁻² s ⁻¹	7.0 keV line photon flux
32	Photflux_7.0_low ³	photons cm ⁻² s ⁻¹	Lower limit of 7.0 keV line photon flux
33	Photflux_7.0_high ³	photons cm ⁻² s ⁻¹	Upper limit of 7.0 keV line photon flux
34	Delta_cstat_III-I ³	—	Difference in C-stat between models I and III

Notes: This table describes the columns of the machine-readable table accompanying the thesis.

¹Values correspond to model I (`tbabs*pegpw`)

²Values correspond to model II (`tbabs*(pegpw+gaussian)`)

³Values correspond to model III (`tbabs*(pegpw+gaussian+gaussian+gaussian)`)

Table 4.2: Descriptive version of the “Spectral fitting of *Chandra* sources” table

Number	Column name	Units	Explanation
1	Name	—	IAU name of the XMM source

Continued on next page

Table 4.2: (continued)

Number	Column name	Units	Explanation
2	Counts	photons	2–10 keV net counts in the combined ACIS spectra
3	Colors	—	Fe color in the combined ACIS spectra
4	NH ¹	cm ⁻²	Best fitting absorption column density for the source
5	NH_low ¹	cm ⁻²	Lower limit of absorption column density
6	NH_high ¹	cm ⁻²	Upper limit of the absorption column density
7	PL_index ¹	—	Best fitting power law index for the source
8	PL_index_low ¹	—	Lower limit of the power-law index
9	Pl_index_high ¹	—	Upper limit of the power-law index
10	Flux ¹	ergs cm ⁻² s ⁻¹	2–10 keV flux
11	Flux_low ¹	ergs cm ⁻² s ⁻¹	Lower limit of 2–10 keV flux
12	Flux_high ¹	ergs cm ⁻² s ⁻¹	Upper limit of 2–10 keV flux
13	C-statistics	—	C-statistics for the best fitting absorbed power-law model
14	DOF	—	Degrees of freedom for the absorbed power-law model
15	Line_pos ²	keV	Position of the Gaussian line in model II

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Table 4.2: (continued)

Number	Column name	Units	Explanation
16	Line_pos_low ²	keV	Lower limit of the Gaussian line position
17	Line_pos_high ²	keV	Upper limit of the Gaussian line position
18	Line_photflux ²	photons cm ⁻² s ⁻¹	Photon flux from the Gaussian line
19	Line_photflux_low ²	photons cm ⁻² s ⁻¹	Lower limit of the photon flux from Gaussian line
20	Line_photflux_high ²	photons cm ⁻² s ⁻¹	Upper limit of the photon flux from Gaussian line
21	Delta_cstat_II-I	—	Difference in C-statistics between model I and II
22	Photflux_6.4 ³	photons cm ⁻² s ⁻¹	6.4 keV line photon flux
23	PhotFlux_6.4_low ³	photons cm ⁻² s ⁻¹	Lower limit of 6.4 keV line photon flux
24	Photflux_6.4_high ³	photons cm ⁻² s ⁻¹	Upper limit of the 6.4 keV line photon flux
25	Photflux_6.7 ³	photons cm ⁻² s ⁻¹	6.7 keV line photon flux
26	Photflux_6.7_low ³	photons cm ⁻² s ⁻¹	Lower limit of 6.7 keV line photon flux
27	Photflux_6.7_high ³	photons cm ⁻² s ⁻¹	Upper limit of 6.7 keV line photon flux
28	Photflux_7.0 ³	photons cm ⁻² s ⁻¹	7.0 keV line photon flux
29	Photflux_7.0_low ³	photons cm ⁻² s ⁻¹	Lower limit of 7.0 keV line photon flux

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Table 4.2: (continued)

Number	Column name	Units	Explanation
30	Photflux_7.0_high ³	photons cm ⁻² s ⁻¹	Upper limit of 7.0 keV line photon flux
31	Delta_cstat_III-I ³	—	Difference in C-stat between models I and III

Notes: This table describes the columns of the machine-readable table accompanying the thesis.

¹Values correspond to model I (tbabs*pegpw)

²Values correspond to model II (tbabs*(pegpw+gaussian))

³Values correspond to model III (tbabs*(pegpw+gaussian+gaussian+gaussian))

95

Table 4.3: Descriptive version of the “*XMM-Newton* candidates without lines” table

Number	Column name	Units	Explanation
1	Name	—	IAU name of the XMM source
2	Source ID	—	4XMM-DR13 unique source ID
3	RA	deg	Mean Right ascension in the 4XMM-DR13 SC_RA column
4	Dec	deg	Mean Declination in the 4XMM-DR13 SC_DEC column
5	Position error	arcsec	Source position error in the 4XMM-DR13 SC_Poserr column

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Table 4.3: (continued)

Number	Column name	Units	Explanation
6	Variability	—	Flags for short-term (S) and long-term (L) variability
7	Known Source	—	Known counterpart for the source
8	Counterpart	—	Marked as “Y” if radio counterpart exists, else left blank
9	Ref.	—	Reference for the known source or radio survey with the counterpart

Notes:This Table is published in its entirety as a machine-readable table accompanying this thesis. Here, we only describe the columns for guidance regarding its form and content.

Table 4.4: Descriptive version of the “*Chandra* candidates without lines” table

Number	Column name	Units	Explanation
1	Name	—	IAU name of the Chandra source
2	RA	deg	Right ascension in the CSC v2.0 master catalog
3	Dec	deg	Declination in the CSC v2.0 master catalog
4	Position error	arcsec	Source position error in the CSC v2.0 master catalog
5	Variability	—	Flags for short-term (S) and long-term (L) variability

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Table 4.4: (continued)

Number	Column name	Units	Explanation
6	Known Source	—	Known counterpart for the source
7	Counterpart	—	Marked as “Y” if radio counterpart exists, else left blank
8	Ref./Comments	—	Reference for the known source or radio survey with the counterpart/other comments.

Notes: This Table is published in its entirety as a machine-readable table accompanying this thesis. Here, we only describe the columns for guidance regarding its form and content.

Chapter 5

Semi-supervised classification of XMM-Newton spectra

5.1 Introduction

The X-ray universe consists of hundreds of thousands of hot objects that emit radiation through various thermal processes like blackbody and bremsstrahlung emission and non-thermal processes like synchrotron radiation and inverse Compton scattering. These X-ray sources include compact high-energy sources like isolated neutron stars (NSs, Kaspi et al. 2006; Pavlov et al. 2002), X-ray binaries (XRBs, Remillard and McClintock 2006; Campana et al. 1998) and active galactic nuclei (AGN, Netzer 2015; Padovani et al. 2017) that host exotic environments with extreme densities (up to 10^{14} – 10^{15} g cm $^{-3}$), high temperatures ($\gtrsim 10^6$ K) with energetic particles, and immensely strong gravitational and magnetic fields (10^{14} – 10^{15} G) (see Remillard and McClintock 2006; Lattimer and Prakash 2007; Turner and Miller 2009, etc. for detailed reviews). Thus, studying the X-ray universe could allow us to understand nuclear forces, the interaction of matter with strong gravitational and magnetic fields, etc. However, our knowledge of these compact objects is limited by the small number of identified objects.

The X-ray sky is continuously monitored by several X-ray instruments like *Chandra* (Weisskopf 2003) and *XMM-Newton* (Jansen et al. 2001) that not only record photons from the specific target but also observe other sources in the field of view. Catalogues that combine the detections of X-ray sources across multiple observations could be incredibly useful in revealing new kinds of X-ray emitters, studying the variability of sources and understanding the properties of various X-ray source populations. Catalogs that summarize the photometric, spectral and variability properties and provide standardized data products like spectra and light curves can allow us to study the properties of sources in detail. While early X-ray missions detected very few objects, modern X-ray catalogs have detected hundreds of thousands of X-ray sources in the sky — e.g, the *Chandra* source catalog, CSC v2.1 has 407,806 unique objects (Evans et al. 2010; Evans et al. 2020), the *XMM-Newton* serendipitous sky survey, 4XMM-DR14 (Webb et al. 2020), has identified 692,109 X-ray sources, and the final eROSITA survey is expected to detect millions of X-ray sources (Predehl et al. 2021).

However, most of these sources have very few counts and have not been studied in detail to understand the X-ray emission and classify the source. X-ray spectroscopy on a source-by-source basis allows us to study each source in detail, but is time-consuming for the thousands of sources in X-ray catalogs. Automated spectral fitting of XMM spectra (e.g. Corral et al. 2015) could speed up the process. However, without manual oversight, automated fitting can lead to improper model selection, due to presence of local minima and/or complicated source spectra. Hardness ratios that compare the numbers of photons in soft vs hard bands can reveal some properties of the source (Yokogawa et al. 2000; Prestwich et al. 2003; Brassington et al. 2008), but they can miss information about the nature of sources that is contained in spectral lines (Hebbar et al. 2019). Thus, we need efficient ways to capture the important details in the X-ray spectra and classify X-ray sources.

Our earlier results have shown that supervised training of *Chandra* X-ray

spectra can separate AGN and stars with an accuracy of $\sim 92\%$ (Hebbar and Heinke 2023). However, in most catalogs, we only have a limited amount of labelled data — e.g. the CSC training set by Yang et al. (2022) and Chen et al. (2023) have ~ 3000 sources, the dataset by Lin et al. (2024) have $\sim 10,000$ known X-ray sources; much smaller than the number of unique sources detected in CSC v2.1 and 4XMM-DR14. Thus, semi-supervised training, where we use the unlabelled dataset to select important features before supervised classification, or extend supervised classification to an unlabelled dataset to stabilize the performance, is needed to classify the data. Below, we introduce important machine learning concepts that will be used in this paper.

5.1.1 Convolutional Neural Networks

Convolutional neural networks (CNNs) are a type of artificial neural network (ANN) that applies the convolution operation on the input with a kernel of a fixed size to get the feature map (refer O’Shea and Nash 2015; Goodfellow et al. 2016). While the traditional fully connected (FC) ANN applies the weight matrix to the entire input, the CNN applies weights to only a part of the input (with a pre-determined kernel size) at a given time, and then translates through the input to generate the second layer. We can then apply a pooling operation (such as taking the average of neighboring values in the second layer) to reduce the number of nodes in that layer. CNNs can also have multiple set of weights that are applied together to the same input to explore different kinds of input relationships at the same time. Like the ANNs, we can also have multiple layers of CNNs for deeper learning.

The use of CNNs reduces the number of weights used in a model by focusing on the connectivity of neighbouring values in the input, and ignoring units that are far apart. Given that the same weights are applied to different parts of the inputs, the CNNs can detect the same feature in different parts of the input, i.e., CNNs are invariant to translation. This is very useful in our case given

that we are interested in detecting X-ray emission lines in different parts of the spectra. However, we also note the non-linearity of the CNNs allows them to overfit the data if proper regularization is not used.

5.1.2 Autoencoders

Autoencoders are a type of ANN that learn to non-linearly represent the data into a lower dimension (latent space) and reconstruct the data given the latent space values (Bank et al. 2021). Thus, the auto-encoder learns two functions — an encoder that compresses the input, and a decoder that recreates the input data from the compressed values. The autoencoder can then be trained based on how well the input values are reconstructed by using losses such as the mean squared error (MSE). These loss functions can be tuned for specific user needs, such as using Poisson loss instead of MSE, adding L_1 & L_2 loss on the latent values to constrain their spread, etc.

Unlike principal component analysis that uses linear dimensionality reduction to capture dimensions where the data has the most variance, the non-linearity of autoencoders allows us to explore a wider range of relationships to parameterize the input. Being able to manually tune the encoder, decoder, and loss functions used for training allows us to extract interesting features that may not show the largest variance in the data.

5.1.3 Semi-supervised learning

When the number of unlabelled datapoints is much higher than the number of labelled datapoints, using the unlabelled data to learn features in the input space and regularize the learning algorithm could be useful to improve the performance as compared to just training on the limited labelled set only. While numerous semi-supervised learning techniques exist in this chapter, we only focus on two methods — pseudo-labelling and training auto-encoders with

classification loss.

Pseudo-labelling



In this method, we first train the model on the labeled data and then use the trained model to assign labels to the unlabeled set. We then use the pseudo-labels that are assigned with confidence above a set threshold to further train the model on the labeled and pseudo-labeled set simultaneously. The pseudo-labels are re-calculated after each weight update. The loss from the pseudo-labeled data can be scaled using a balancing term to optimize the contribution from the unlabeled dataset while not disturbing the training for labeled data.

Pseudo labeling works on the assumption that data that lie close to each other in a high density region belong to the same class and that the classes are separated by a low density region. Given this, pseudo-labeling minimizes the spread in probabilities that the unlabeled data belongs to a particular class. Proper scheduling of the confidence threshold and the balancing term can overcome confirmation bias (i.e. the model getting better on data it already performs well on). Using a low threshold and a low balancing term during the early epochs of training allows the model to see unlabeled inputs where the model is performing marginally, while still giving priority to the labeled set. Higher thresholds and balancing values can be useful for the fine-tuning during the final epochs.

Autoencoder with classification loss

Standard autoencoders are very efficient in compressing the input data to a latent space without losing much information. However, this might not ensure that the latent space still includes information useful for separating the classes. To overcome this, we train a classifier simultaneously with the autoencoder that checks if the labeled dataset can be separated even in the latent space. This is achieved by using an additional classification loss for the labeled data and

minimizing the classification loss and reconstruction loss simultaneously. The classification loss and reconstruction losses can be scaled appropriately to meet the specific user needs.

5.2 Data Acquisition and analysis

In earlier chapters, we saw that the neural networks performed with an accuracy of $\geq 90\%$ on sources with $\gtrsim 100$ counts, and depend on the response of the detector to the changing photon energy. Given its higher effective area and stable detector response, we choose to use data from *XMM-Newton* for our analysis.

5.2.1 Extraction of *XMM-Newton* Spectra

The 4XMM-DR13 catalog summarizes the properties of 1,035,832 detections of 692,109 unique sources drawn from 13,864 *XMM-Newton* EPIC observations made between February 3, 2000 and December 31, 2023 (Webb et al. 2020). The 4XMM-DR14 pipeline also produces spectra for 372,603 source detections that have ≥ 100 EPIC counts (flagged by `SPECTRA = True`).

The detection of a source by the *XMM-Newton* pipeline can be influenced by the detector artifacts, cosmic X-ray background, presence of hot pixels, a source falling on chip gaps or corners, a source being close to other bright/extended sources, etc. These can lead to detection of spurious sources, erroneous calculation of source properties, and improper source & background extraction regions for the data products generated. Therefore, we limit ourselves to the cleanest detections that have `SUM_FLAG = 0`. Extracting spectra from extended *XMM-Newton* sources will need a more careful analysis of source shape and background. Therefore, we only use point sources with `EP_EXTENT = 0`. We further limit the detections to not being marked as `CONFUSED` (i.e. detection could be associated with multiple sources) or having a `HIGH_BACKGROUND` (man-

ually set when instruments have a consistently high background), and having `OBS_CLASS` ≥ 2 (i.e. less than 10% of the detector area has been manually flagged as “bad”). These constraints give us 158,657 unique sources with 190,459 PN spectra, 75,921 MOS1 spectra, and 87,922 MOS2 spectra.

Most X-ray CCDs are also sensitive to optical photons. Therefore, X-ray telescopes use optical blocking filters to minimize the contamination of X-ray signals by these photons. *XMM-Newton* observations can use one of the three THIN, MEDIUM or THICK optical blocking filters. Among our selected spectra, 183,611 (95,170 PN and 88,441 MOS) are observed with the THIN Filter, 149,022 (83,774 PN and 65,248) with the MEDIUM filter, and 18,335 (10,896 PN and 7,439 MOS) with the THICK filter. The effective areas of the EPIC cameras with the THIN and MEDIUM filters is very similar for energies ≥ 0.5 keV. Given this, and the stable response of the *XMM-Newton* EPIC cameras over time, we combined the THIN and MEDIUM spectra of all sources detected across multiple observations and epochs using the `epicspeccombine` method (we do not use the THICK filter spectra). We process the EPIC-PN and EPIC-MOS spectra separately, as their effective areas and responses are very different (but we combine the spectra from MOS1 and MOS2). Finally, we have 131,945 EPIC-PN spectra and 62,521 MOS spectra, corresponding to 147,157 unique sources.

Given the effective area of the EPIC-PN and EPIC-MOS cameras and the difference in the effective areas with the THIN and MEDIUM filters for low energies, we only use energy ranges 0.5–10.0 keV for our analysis. The energy resolution of the CCD detectors is ~ 100 –200 eV, but the spectra are binned to 5 eV. This can lead to very noisy spectra. Therefore, we rebin the spectra to bin sizes of 25 eV, i.e. 380 bins between 0.5–10.0 keV. This allows us to smooth the spectra slightly without losing the ability to extract ~ 100 eV emission lines from the spectra. For our analysis, we only use spectra with background-subtracted net counts ≥ 100 . This gives us 60,606 PN spectra and 28,026 MOS spectra.

5.2.2 Generating a set of labeled spectra

We use the 4XMM-DR13 training dataset generated by Lin et al. (2024) for our analysis, using the visualization tool by Yang et al. (2021). The dataset classifies $\sim 10,000$ known X-ray sources in the 4XMM-DR13 catalog into AGN (4758), CV (253), HM-STAR (153), HMXB (103), LM-STAR (4678), LMXB (101), NS (135), NS_BIN (41), and YSO (690) (some of these sources may not have X-ray spectra generated by the 4XMM-DR14 catalog). Given the limited number of X-ray sources in some of the classes, and the similarity of the X-ray spectra, we regroup HM-STAR, LM-STAR, and YSO into one group ‘STARS’, and LMXB, HMXB, NS and NS_BIN into group ‘NS/BH’. This gives us 1339 PN and 720 MOS spectra for AGN, 1193 PN and 621 MOS spectra for Stars, 99 PN and 81 MOS spectra for NS/BH, and 99 PN and 81 MOS spectra for CVs.

The net counts among the selected spectra (labeled and unlabeled) show a huge range from 100 counts (our set threshold) to 10^6 counts. In order to ensure that our model trains using the shape of the spectra rather than the brightness of the source, we use normalized spectra for our analysis. I.e. we divide the net spectrum of each source by the total number of net X-ray photon counts in that spectrum between 0.5–10.0 keV (in other words, the sum of each normalized spectrum across 380 bins is 1).

We show the distribution of net counts (0.5–10.0 keV) for our selected PN and MOS sources in Fig. 5.1. We see that most of the spectra, both PN and MOS, have net counts ≤ 1000 . Fig. 5.2 shows the distribution of the background-to-net count ratio in the PN and MOS spectra used in our analysis. Most PN and MOS spectra have background counts similar to the net counts in the 0.5–10.0 keV range

Fig. 5.3 shows the mean normalized EPIC-PN spectra of different groups of sources. The Fe-L and Ne-K lines result in a strong peak at ~ 0.8 keV for stars, while the mean spectra of AGN, NS/BHs and CVs do not seem to show peaks from X-ray emission lines. Checking individual spectra of CVs

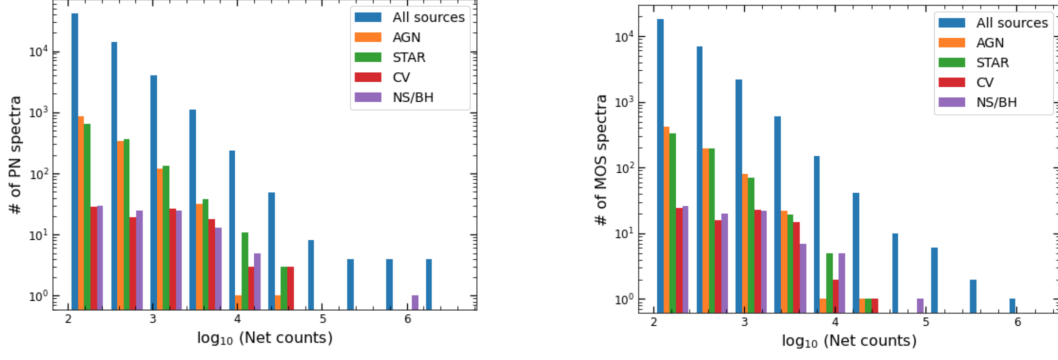


Figure 5.1: Distribution of net counts between 0.5–10.0 keV in our selected PN (*left*) and MOS (*right*) spectra. We also show the distribution of net counts for different kinds of X-ray sources in our labelled set. We only use spectra with ≥ 100 counts for our analysis.

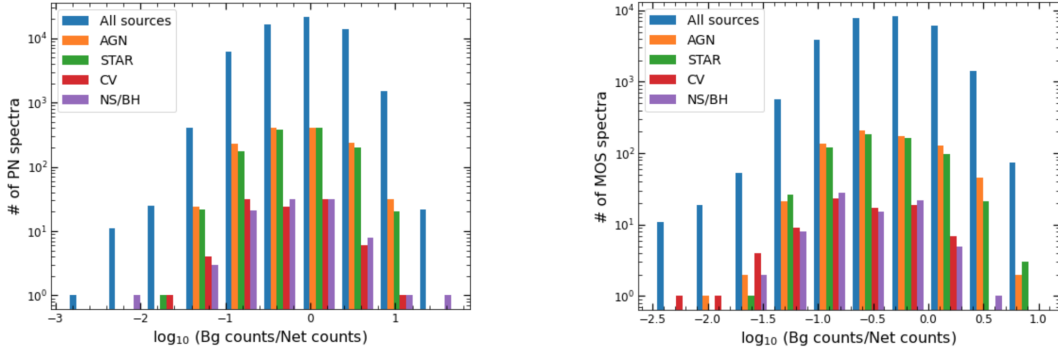


Figure 5.2: Distribution of background-to-net count ratio for our selected PN (*left*) and MOS (*right*) spectra (in the 0.5–10.0 keV range). We also show these distributions for different kinds of X-ray sources in our labelled set.

does show the presence of Fe-K lines in some of them (especially intermediate polars). The slight hard X-ray excess in the mean normalized spectrum of NS/BH is due to inclusion of normalized spectra from HMXBs, that usually show strong absorption in soft X-rays. Fig. 5.4 shows the mean normalized EPIC-MOS spectra for the different classes of X-rays in our labeled set. Given the smaller effective area of EPIC-MOS at lower energies, these spectra show a lower fraction of counts $\lesssim 1$ keV. Other than this difference, they show similar features as in Fig. 5.3.

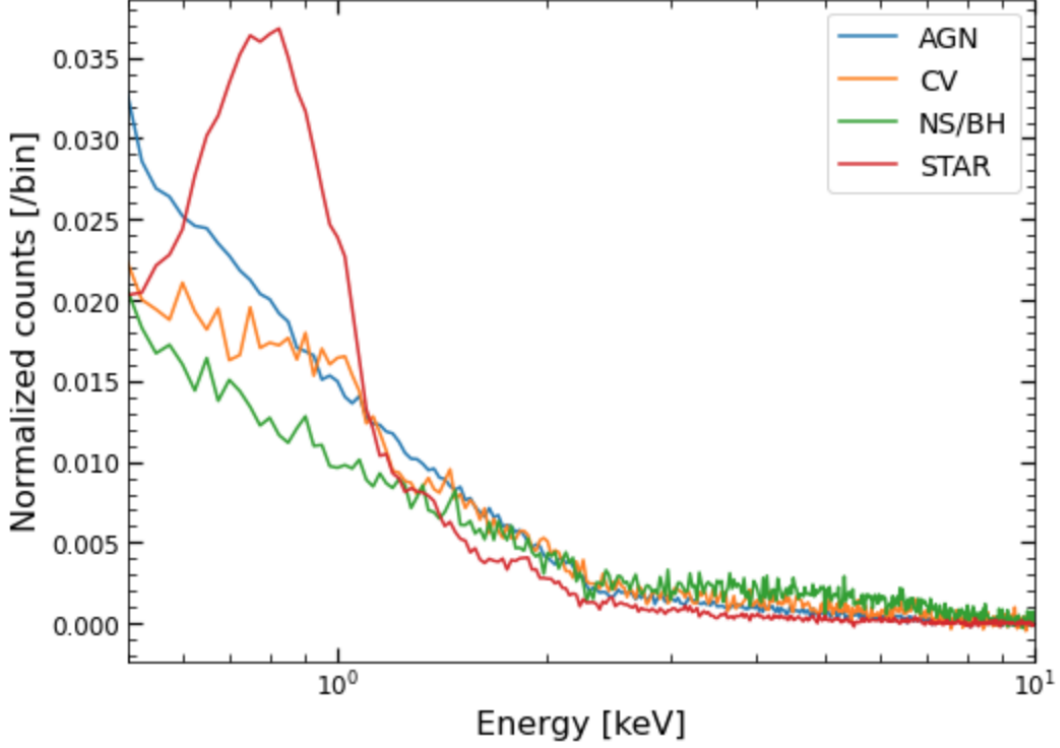


Figure 5.3: Mean normalized PN spectra of the different classes of X-ray sources in our labelled set. The spectrum of stars shows a peak around ~ 0.8 keV. This is due principally to the combined effects of Fe-L/Ne-K emission lines. The mean spectra of AGN, CVs and NS/BH are largely featureless.

5.2.3 Semi-supervised learning of the X-ray spectra

We start with classification of the PN spectra alone. Given that we have a large number of labeled AGN and star spectra (in comparison to NSs/BHs and CVs), and we expect most of the unlabeled spectra to be distant background AGN or chromospherically active stars, we first consider two-class classification methods to identify the AGN and stars in the unlabeled set. From the set of 2532 labelled AGN and star PN spectra, we separate out 506 spectra as our hold-out test set. The training algorithms do not see the test set during training and the test set is only used to evaluate the performance of the classification method. We use the Pytorch module in Python for modelling, training and visualizing the machine learning algorithms.

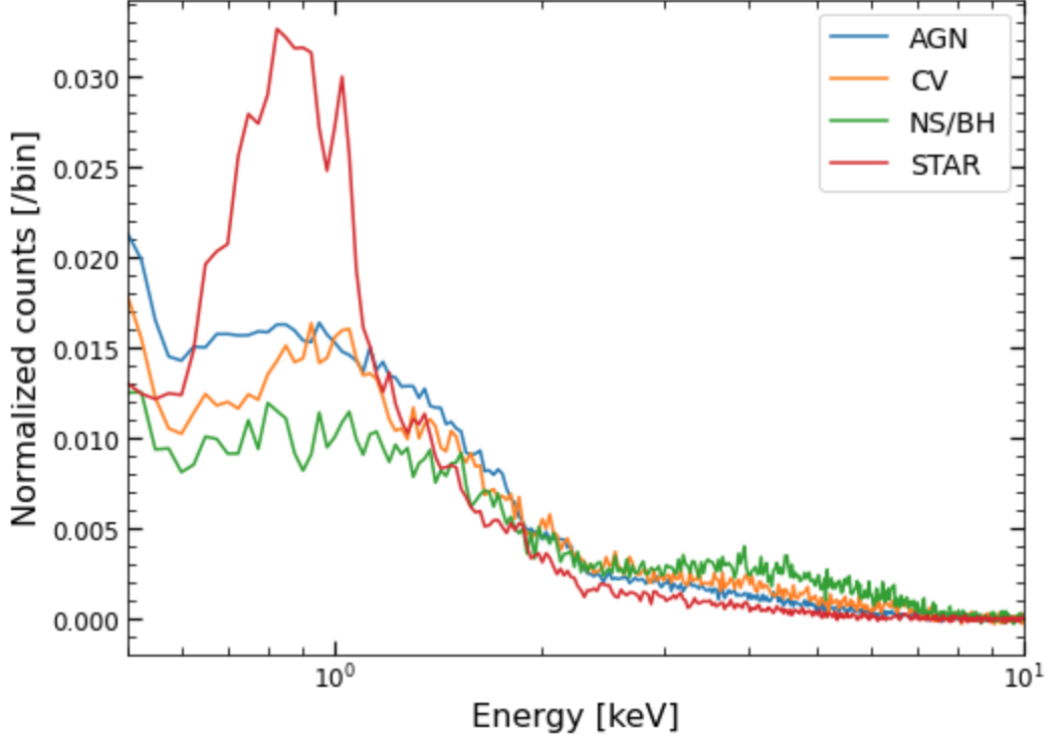


Figure 5.4: Mean normalized MOS spectra of different classes of X-ray sources in our labelled set. These spectra show similar features as the mean normalized PN spectra.

Pseudo-labeling

First, we use a pseudo-labeling method to classify the unlabeled sources. Our model consists of 2 CNN layers and two FC layers. The first CNN layer has 8 output channels and uses a kernel size of 7 (i.e. 175 eV). We apply a max-pooling function over two nodes to condense the output to 190 bins. We then apply a second CNN layer with 16 output channels and a kernel size of 5 (i.e. 250 eV). We use another max pooling over two nodes to condense the outputs to 95 bins. We also use batch-normalization after each CNN layer for faster training. These outputs are then flattened and fed into an FC layer with 64 nodes. The last layer has 2 nodes giving the probability of the source being a star or an AGN. We use the ReLU activation function on all intermediate layers (the two CNNs and one FC layer). No activation function is used on the final output

node, since most loss functions in Pytorch assume that the outputs are in logits (i.e., natural logarithm of the probabilities). We use cross-entropy loss for calculating the classification loss and an Adam optimizer with a learning rate (which determines the size of steps in the gradient descent algorithm) of 10^{-4} for the training. We use an L1 regularizer with $\lambda_1 = 10^{-5}$ and L_2 loss with $\lambda_2 = 10^{-4}$, and a dropout rate of 0.2 during the training to prevent overfitting of the training data. We use a 5-fold cross-validation to understand the spread in the performance of the ML model and select the best classification. We shall use the accuracy, recall, precision, true negative rate (TNR), and negative predictive value (NPV) for evaluating the performance on the test set.

Autoencoder

Our alternative method is to use an autoencoder model with classification loss to identify AGN and stars. This model is useful to understand the properties of the latent space that can best distinguish these two classes while being able to reconstruct the output spectra. For the encoder, we use two CNN layers with ReLU activation – first with 8 channels and a kernel size of 7, and second with 16 channels and a kernel size of 5 (both CNN layers use max-pooling with size 2, and batch normalization, like the previous model), followed by a fully connected layer to convert the 1520 ($16 \times 380/4$) values generated to 32 nodes using Softplus activation. These 32 nodes define the latent space. For the decoder, we first use a fully connected layer to convert 32 latent nodes to 1520 values, and then two transpose CNN layers with (16, 8) channels and kernel sizes (7, 5). We use ReLU activation on the outputs of the first transpose CNN layer, and Softplus activation for the final outputs, to ensure that the reconstructed spectra have positive values. We use the logarithm of the Poisson loss to check the difference between the input and the reconstructed spectra. Since Poisson loss is defined over counts rather than normalized spectra, we multiply the input normalized and reconstructed spectra by the net counts in

the input spectra (before it was normalized), calculate the Poisson loss and then divide the log-Poisson loss by the total counts (so that we give equal weights to bright and faint sources) to get the reconstruction loss.

The non-linearity of the autoencoder can allow it to explore a wide range of latent dimensions and decoder functions to reconstruct input. Our main constraints are that the stars and AGN remain separable in the latent space and that the reconstructed spectra be smooth and not capture the noise in the input spectra. We calculate a smoothness/total variation(TV) loss to ensure that the reconstructed spectra are smooth. To ensure separation in the latent space, we train a one-layer neural network function with two output nodes that correspond to log-probabilities of the sources being a star or an AGN. For the labeled data, we also compute the cross-entropy loss from the classifier. Thus the total loss for unlabeled data is the sum of the reconstruction loss and the smoothness loss. The total loss for the labeled data is the sum of the reconstruction loss, smoothness loss and cross-entropy loss.

To prevent over-training of the training data, we add artificial noise to the latent variables before reconstruction and classification (only during training). The artificial noise added is scaled by the standard deviation of the latent variables across a given batch, i.e., for each vector of latent variables $z \in \mathbb{R}^{m \times 32}$, we first calculate $\sigma_z \in \mathbb{R}^{32}$. Then we add a random noise with $\sigma_{recon} = 0.05\sigma_z$ before passing z to the decoder, and a random noise of $\sigma_{class} = 0.2\sigma_z$ before applying the classifier. We also apply an L_2 regularization with $\lambda_{L2} = 0.001$ and L_1 regularization with $\lambda_{L1} = 0.0001$.

5.3 Results

We start with analysis of the PN spectra only, for a two-class classification to identify AGN and stars. We analyze the performance of our models using accuracy, recall, true negativity rate (TNR), precision and negative predictive

value (NPV). Our pseudo-label gives an accuracy of $(93.6 \pm 0.6)\%$ on the validation sets (during cross-validation) and an accuracy of 93.3% on the hold-out test set. We notice that this model has higher TNR (98%) and precision (97%) as compared to recall (88%) and NPV (90%). We consider stars as class 1 and AGN as class 0, implying that recall signifies the fraction of stars correctly identified, TNR is the fraction of AGN identified properly, precision is the confidence we have on sources predicted to be stars, and the NPV is the proportion of predicted AGN that are truly AGN. This indicates that the model is placing a higher threshold for identifying stars as compared to AGN. This could happen if the unlabeled data contains many more AGN than stars. Training the models on the MOS spectra alone gives an accuracy of $(93.8 \pm 0.5)\%$ on the validation sets and 92.2% on the hold out test set. The recall, TNR, precision, and NPV values for the test set in MOS spectra are 87%, 97%, 96%, and 90%, respectively, indicating a higher threshold for the detection of stars.

Using the autoencoder with the classification loss gives a validation accuracy of $(93 \pm 2)\%$ with a hold-out test accuracy of 92%. The recall, TNR, precision, NPV values for the test data are 91%, 95%, 94%, and 92%, respectively. Thus, the autoencoder model also places a higher threshold for the identification of stars. For the autoencoder model trained on MOS, the overall accuracy is $(94 \pm 2)\%$ on the validation sets and 93.7% on the hold-out test sets. The recall is 93%, TNR is 94%, precision is 94%, and NPV is 94% on the test set.

We first analyze how the performance changes with the number of net counts in the spectra (plotted in Fig. 5.5). All the performance metrics for both models — accuracy, recall, precision, TNR and NPV — increase with increasing net counts in the spectra. This is expected, since the Poisson noise reduces with increasing net counts, and features like emission lines in the X-ray spectra become clearer. The limited number of sources with counts $\gtrsim 10^5$ can result in fluctuations in performance calculated for these bins. The limited number of sources might also be the reason for a slight decrease in the autoencoder recall

Table 5.1: Performance comparison between pseudo-label and autoencoder models for *XMM-Newton* data under different cases.

Case	Model	Detector	Accuracy	Recall	TNR	Precision	NPV
Counts > 100	Pseudo-label	PN	93	88	98	97	90
		MOS	92	87	97	96	90
	Autoencoder	PN	93	91	95	94	92
		MOS	94	93	94	94	94
Confidence > 90%	Pseudo-label	PN	95	87	96	98	92
		MOS	94	85	94	95	93
	Autoencoder	PN	98	83	86	99	98
		MOS	96	87	87	97	95
Counts < 100	Pseudo-label	PN	80	80	81	81	80
		MOS	78	71	85	82	76
	Autoencoder	PN	78	86	70	74	84
		MOS	79	80	78	77	81

and NPV in some of the higher-count bins. In general, we get accuracy $\gtrsim 90\%$ for spectra with net counts $\gtrsim 300$.

Fig. 5.6 shows the variation in the performance metrics with the ratio of background to net counts in the spectra. In general, the performance tends to decrease with increasing background. This is to be expected, since the difficulties in achieving perfect background subtraction, and the noise in the background, can make it difficult to separate out the emission line features in the source spectra. In general, we get accuracy (and other metrics) $\geq 90\%$ for a background-to-net-count ratio $\lesssim 2$ in case of PN spectra. For MOS spectra performance values $\geq 90\%$ occur for background-to-net-count ratio $\lesssim 0.3$.

We also examine the difference between correctly and incorrectly classified test sources. We plot the mean normalized spectra of correctly and incorrectly identified sources in Fig. 5.7. As expected, the stars that are mis-classified as AGN do not show the strong peak at 0.8–1.0 keV from the Fe-L/Ne-K lines. The wrongly classified sources also show higher noise. This is likely due to the lower photon counts and higher background contribution in these spectra, and the small number of incorrect sources in our test set (given a test set of ~ 500 sources and 90% accuracy, only 10–30 stars and AGN are assigned in the wrong group).

Since EPIC-PN and EPIC-MOS are independent cameras and we have

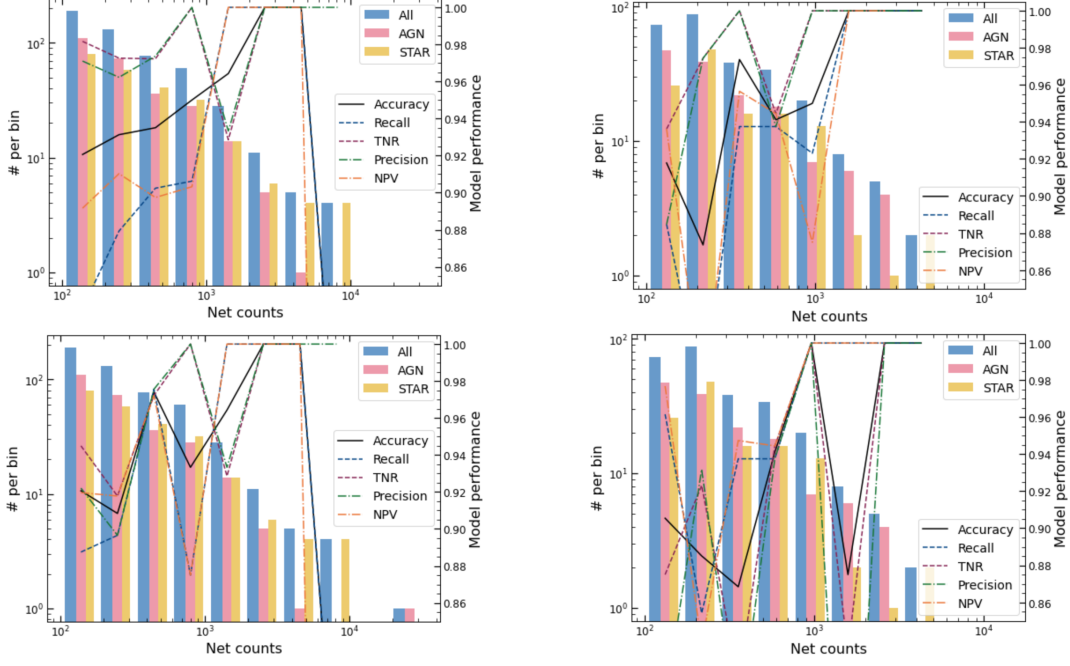


Figure 5.5: Performance of our models vs. net counts in the spectra. The top-left sub-figure shows the performance of PN spectra using the pseudolabeling algorithms, the top-right sub-figure shows the performance of MOS spectra with the pseudolabeling algorithm, the bottom-left sub-figure shows the performance of the autoencoder model for the PN spectra, and the bottom-right sub-figure shows the performance of the autoencoder model on the MOS spectra. In general, we see that the model classifies the spectra with an accuracy, recall and precision $\geq 90\%$ on spectra with net counts $\gtrsim 300$

processed the two spectra separately, we check to see if combining the PN and MOS predictions would improve our results. For this, we first multiply the ‘STAR’/‘AGN’ probabilities as output from the models trained on PN with those trained on MOS spectra, re-normalize the results so that the sum of the probabilities of star and AGN is 1, and choose the class with the higher combined probability. However, we do not notice any significant change in the performance. The change in performance was within the error observed for validation sets. Thus, for cases where the PN and MOS predictions are different, we will need more detailed analysis to accurately classify the source.

We also check the performance of our trained models on the sources with low

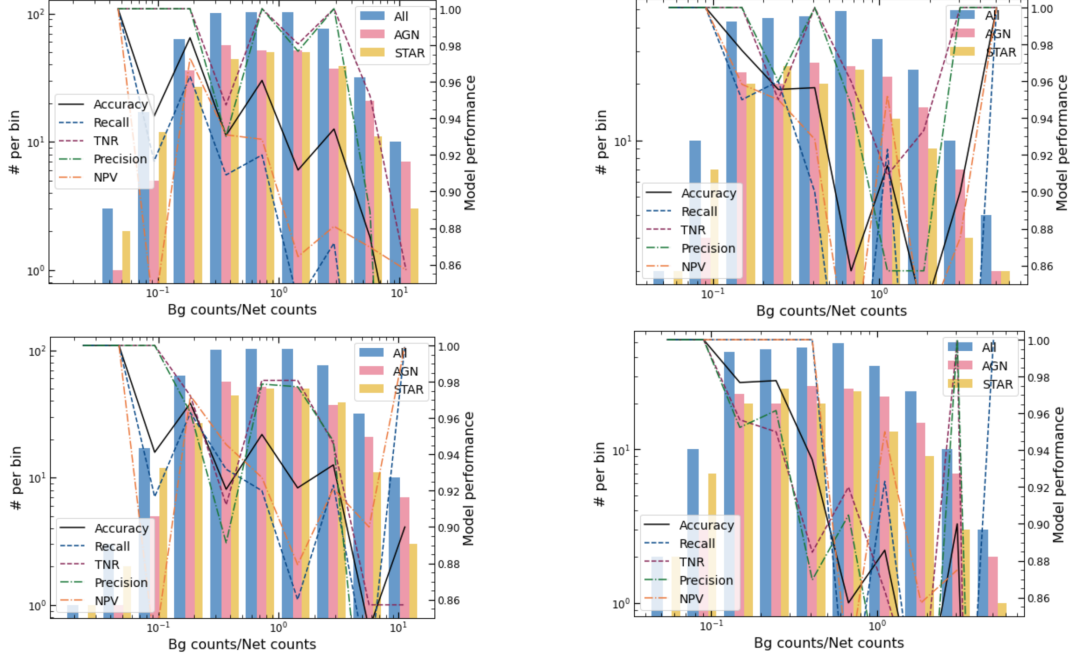


Figure 5.6: Performance of our models with background-to-net-count ratio in the spectra. The top-left sub-figure shows the performance of PN spectra using the pseudolabeling algorithm, the top-right sub-figure shows the performance of MOS spectra with the pseudolabeling algorithm, the bottom-left sub-figure shows the performance of the autoencoder model for the PN spectra, and the bottom-right sub-figure shows the performance of the autoencoder model on the MOS spectra. The performance of the models decreases with increasing background.

photon counts in Fig. 5.8. We notice that the performance decreases steeply for net counts < 100 , with accuracy of only 75% for spectra with ~ 50 counts. This is to be expected due to the high Poisson noise in these spectra. The fluctuations in accuracy seen in the plots are likely due to the limited number of sources in each bin. However, the models trained on PN spectra tend to have a relatively higher recall and lower NPV; i.e., most of the stars are identified correctly, and the sources that are identified as AGN are mostly AGN. In other words, most sources are being identified as stars, and we have a strong confidence in sources classified as AGN. This could be because the Poisson noise detected in the soft energy ranges could be confused as the soft X-ray emission lines in the spectra.

Given that the unlabeled spectra include other classes of X-ray sources in

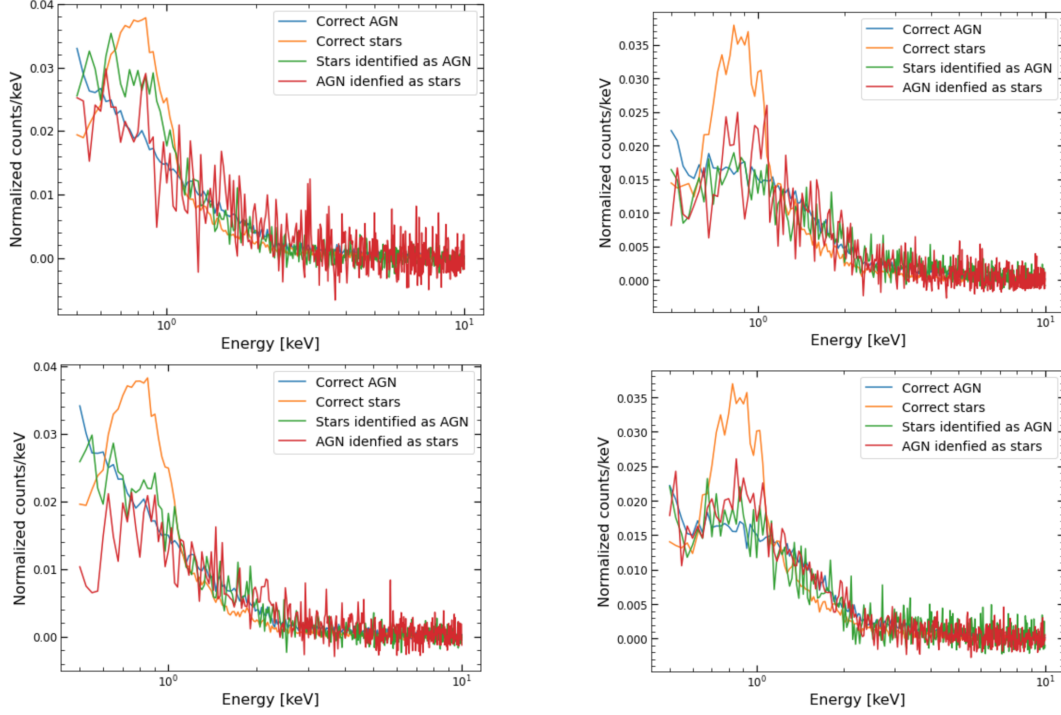


Figure 5.7: Mean normalized spectra of correctly vs. incorrectly classified sources. The top-left sub-figure shows the mean PN spectra using the pseudolabeling algorithm, the top-right sub-figure shows the mean MOS spectra with the pseudolabeling algorithm, the bottom-left sub-figure shows the mean PN spectra for the autoencoder model, and the bottom-right sub-figure shows the mean MOS spectra using the autoencoder model. Stars mis-identified as AGN do not tend to show the strong peak in the 0.8–1.0 keV range. The increased noise in the incorrectly classified spectra is from the lower photon counts in these spectra, and the small number of incorrectly classified sources in the test set ($\mathcal{O}(10)$).

addition to AGN and stars, we only want to accept predictions that have a very high confidence of $\geq 90\%$. Applying this confidence cutoff for our labeled set gives us a recall of 87%, TNR of 96%, precision of 98%, and NPV of 92% for PN spectra with the model using pseudolabels. The autoencoder model gives recall, TNR, precision, and NPV of 83%, 86%, 99% and 98%, respectively, if we apply a 90% confidence cutoff. For the MOS spectra, the pseudolabel model gives values of 85%, 94%, 95%, 93% for recall, TNR, precision, and NPV, respectively, and the autoencoder model gives values of 87%, 87%, 97%, 95% for these performance metrics.

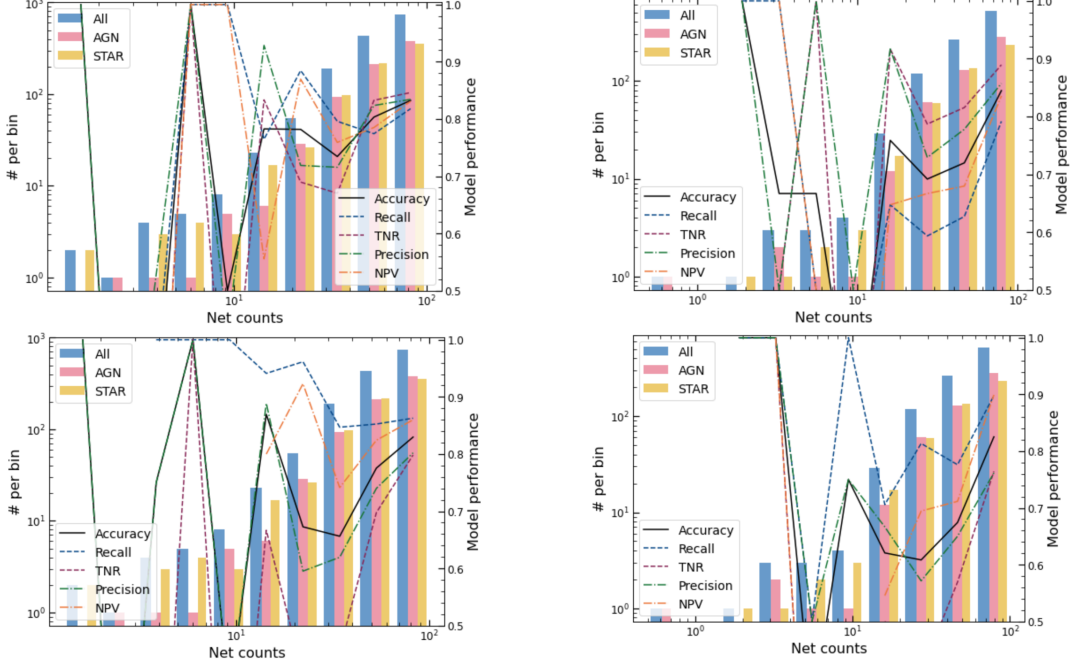


Figure 5.8: Performance of our trained ML models on PN and MOS spectra with < 100 counts. The top-left sub-figure plots the PN spectra classified using the pseudolabelling algorithm, the top-right sub-figure plots the MOS spectra classified with the pseudolabelling algorithm, the bottom-left sub-figure shows the classifications on PN spectra using the autoencoder model, and the bottom-right sub-figure shows classifications on the MOS spectra using the autoencoder model. The performance of our models declines steeply for the spectra with low photon counts.

We then apply these models to the unlabeled spectra. We keep the confidence threshold of 90% for our predictions. Applying the pseudolabel model to the 57876 unlabeled PN spectra gives us 15,605 stars and 40,338 AGN. The autoencoder model predicts 11652 stars and 31650 AGN among the PN spectra. For the 26523 unlabeled MOS spectra, the pseudolabel model predicts 6696 stars and 18611 AGN and the autoencoder model predicts 6126 stars and 16372 AGN. We show the mean spectra of predicted AGN and stars in the unlabeled set in Fig. 5.9. The spectra of stars are similar to those in our labeled set and show a peak in the 0.8–1.0 keV region, while the AGN spectra are largely featureless.

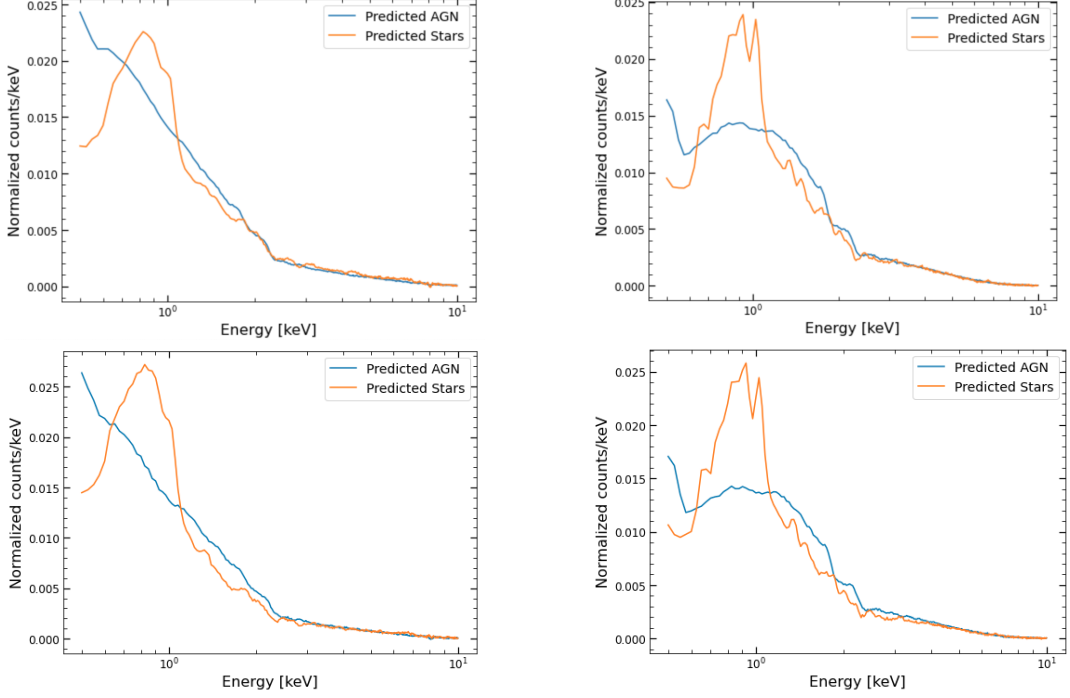


Figure 5.9: Mean normalized spectra of correctly vs. incorrectly classified sources. The top-left sub-figure shows the mean PN spectra using the pseudolabeling algorithm, the top-right sub-figure shows the mean MOS spectra with the pseudolabeling algorithm, the bottom-left sub-figure shows the mean PN spectra for the autoencoder model, and the bottom-right sub-figure shows the mean MOS spectra using the autoencoder model. The spectra show similar features as the mean normalized spectra of stars and AGN in the labeled set.

5.4 Discussion

5.4.1 Comparison with hardness ratios

Hardness ratios are often used directly to differentiate between non-thermal emission of AGN and the soft thermal emission of stars. However, young stellar objects (YSOs) and high-mass stars (HM-STAR) can have strong absorption from the surrounding medium, which can make the spectra look hard. Therefore, we check if our methods identify stars across a range of absorption column densities, N_H , and intrinsic hardness of the spectra.

Moderate absorption from the interstellar medium and surrounding gas ($N_H \lesssim 10^{22} \text{ cm}^{-2}$) reduces the amount of soft X-rays below $\lesssim 1 \text{ keV}$. Therefore,

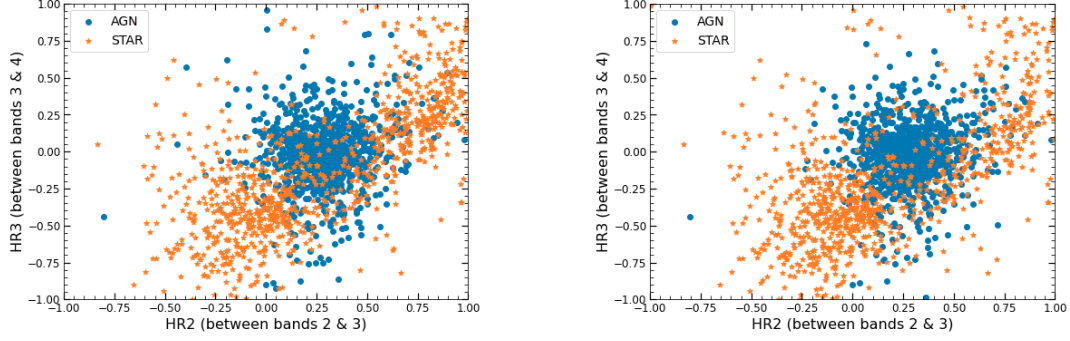


Figure 5.10: Hardness ratios of 1000 predicted AGN and stars, calculated from the PN spectra. We show the predictions of the pseudo-label model on the left, and those of the autoencoder on the right. We see that both models are able to predict stars over a wide range of intrinsic hardnesses and absorption column densities.

comparing the hardness ratios (HR) between band 2 (0.5–1.0 keV) and band 3 (1.0 – 2.0) is usually a good indication of N_H . The HR between the bands 3 and 4 (2.0–4.5) is generally a good measure of the intrinsic hardness of the X-ray spectrum. Fig. 5.10 shows the joint distribution of these two hardness ratios for a thousand predicted AGN and stars, calculated from the PN spectra. From the figure, we see that both the autoencoder model and the pseudo-label model have identified stars with a range of HR2 (signifying the absorption) and HR3 (signifying the intrinsic hardness of the spectra). Comparing this to HRs of the labelled set presented in Lin et al. (2024), and using the visualization tool of Yang et al. (2021), we notice that the observed HRs for stars match the combined distribution of low-mass stars (LM-STAR), HM-STARs and YSOs. From this comparison, we also conclude that the stars with low HR2 and HR3 (bottom left in the plots) are probably LM-STARs, and those with high HR2 and HR3 are mostly HM-STARs and YSOs.

5.4.2 Visualizing the CNN activations

While CNNs enable superior performance, they lack interpretability for understanding why the model made a particular decision. Since CNN layers

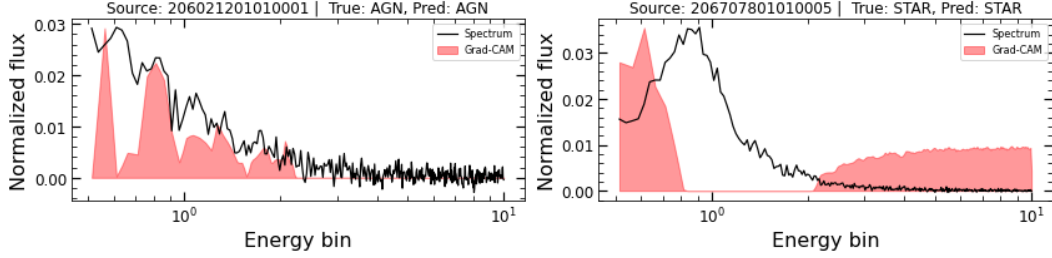


Figure 5.11: Grad-CAM heatmap for an example AGN and a star. Depending on the spectra, different features can drive the classification of the source. We note that the left plot shows specific searches by the algorithm for line features at 0.6 (O), 0.8 (Fe), 1.0 (Ne), 1.3 (Mg) and 1.8 (Si) keV, typical of stars.

use shared weights that slide through the model, revealing the regions in the input that contribute the most gradient towards the decision would allow us to interpret the CNN results. We use the Grad-CAM algorithm (Selvaraju et al. 2019) to understand the CNN model trained by the pseudo-label algorithm.

Grad-CAM is applied on the final CNN layer of the trained model. First, we pass the selected inputs through the model, activating the 16 feature maps, A^k (corresponding to the 16 channels), in the final CNN layer. Each of these A^k has 95 bins (because max-pooling has been applied two times). We then calculate the gradient on the output logits (log-probability) of ‘AGN’/‘STAR’ from each bin of a given feature map, i.e. $\partial y_c / \partial A_i^k$, where y_c is the output logit of the predicted class, and A_i^k is the i^{th} bin of feature map A^k . We then take the mean of these values across all bins to get the importance weight for feature map A^k . We then calculate the Grad-CAM weight $L_{\text{GradCAM}}^c = \text{ReLU}(\sum_k A^k)$. Thus, Grad-CAM gives a heatmap across the bins in the conv2 feature maps that drive the classification.

Fig. 5.11 shows the Grad-CAM heatmap for example AGN and stars. Depending on the spectra, the model looks at different parts of the input to predict the output class. Even different spectra of ‘AGN’/‘STAR’ show different activations. For the AGN spectrum shown, the algorithm seems to search for emission lines, especially ~ 1 keV, and the lack of such lines drives the classification. For

the shown AGN, the model might also be considering the lack of any absorption ~ 0.5 keV. For the bright star the model seems to focus more on the sharp rise below the 1 keV peak and the lack of hard X-rays, rather than the peak itself.

5.4.3 Separation between AGN and stars in latent space

The autoencoder model allows us to condense the 380 bins of the X-ray spectra to 32 latent variables. We use the Uniform Manifold Approximation and Projection (UMAP) algorithm to visualize this latent space in two dimensions. UMAP is a non-linear dimensionality reduction technique that constructs a graph representation of the relationships between data points and then optimizes a low-dimensional embedding that preserves both local and global structure—that is, points that are close in the original high-dimensional space remain close in the reduced representation. Unsupervised fitting of a UMAP on all the data in the latent space gives us Fig. 5.12. From the projections we see that the labeled AGN and stars are well-separated. Based on the density plots of the entire set of data, we can see that there are likely many more AGN than stars in the unlabeled data.

5.4.4 Extending to 4-class classification

Our labelled set also includes ~ 100 ‘NS/BH’ and ‘CV’ spectra. Extending our models to include and identify these spectra accurately faces two challenges. The dataset is heavily imbalanced, in that we have ten times more AGN and stars (the majority classes) than ‘NS/BH’ and ‘CV’ (the minority classes) spectra. Given so few spectra, the model could overfit these spectra. The unlabelled set might help us to prevent overfitting. First, we try extending our pseudo-label model to 4 classes. In order to ensure that prospective ‘NS/BH’ and ‘CV’ sources are not all classified as ‘AGN’ or ‘STAR’ during pseudo-labeling, we keep the threshold of the majority classes as 0.9, but fix the threshold of minority

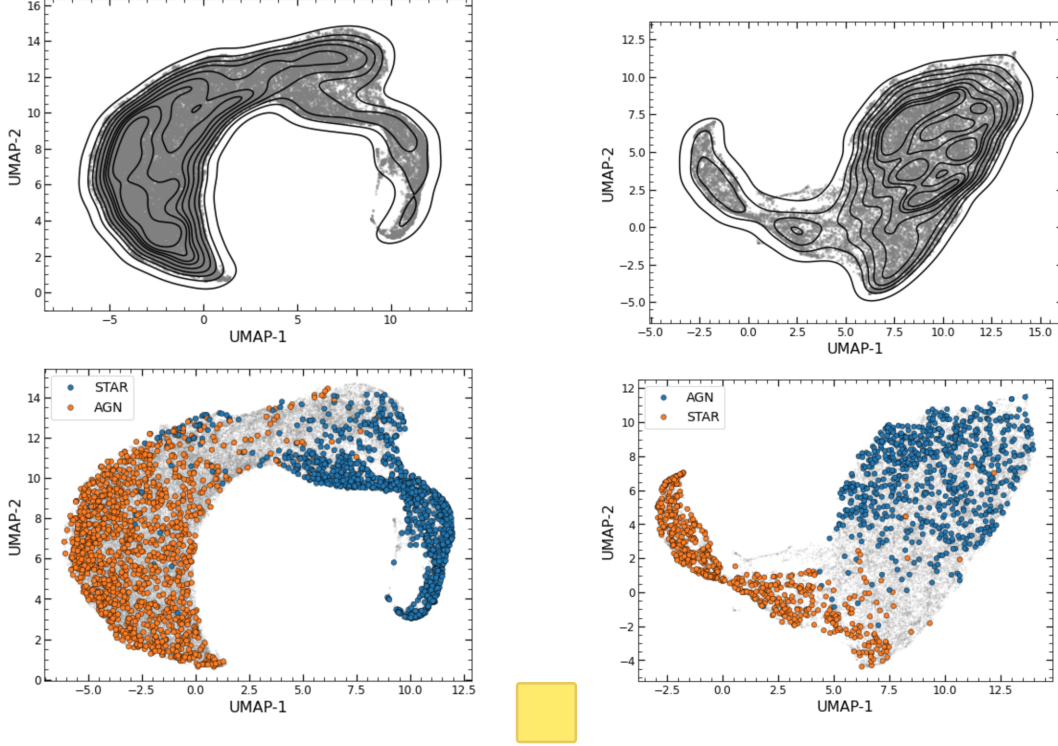


Figure 5.12: Unsupervised UMAP projection of the latent variables encoded using the trained autoencoder model. We see that the labeled AGN and stars are separated as distinct clusters in this manifold. The figures on the left show the distributions of the latent variables for the PN spectra, and those on the right for the MOS spectra. We see that the labeled AGN and stars are mostly separated in these projections.

classes to 0.75. Training this model gives an overall validation accuracy of $(87 \pm 1)\%$, but the accuracy of the ‘NS/BH’ and ‘CV’ classes is only $\sim 30\%$ (the accuracy on ‘NS/BH’ and ‘CV’ classes fluctuates a lot depending on which sources are in the training, validation and test sets).

We add class weights to address the imbalance between the majority and the minority classes. We first use weights inversely proportional to the square root of the number of spectra in each class, i.e. the ‘NS/BH’ and ‘CV’ classes are weighted ~ 3 times more than the ‘AGN’ and ‘STAR’ classes. This gives an overall accuracy of $\sim 75\%$, while the accuracy for NS/BH/CV classes is $\sim 45\%$. Extending the auto encoder model to 4 classes gives us an overall accuracy of $\sim 85\%$, and NS/BH/CV accuracy of $30 - 40\%$. We show two instances of the

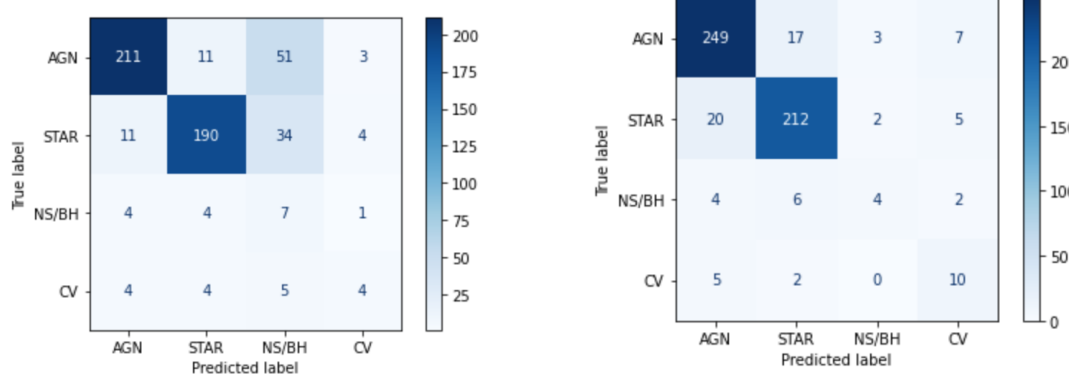


Figure 5.13: Example confusion matrices from 4-class pseudo-labeling (left) and autoencoder (right) methods on the PN spectra. While the algorithms are able to identify AGN and stars, the ‘NS/BH’ and ‘CV’ sources are not classified accurately.

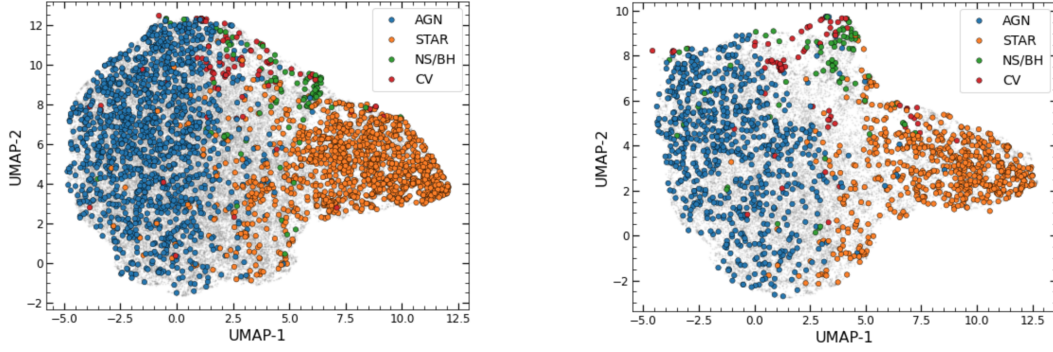


Figure 5.14: UMAP projection of the latent space in the 4-class autoencoder. AGN and stars show good separation. While we see grouping of some of the ‘NS/BH’ sources, the ‘NS/BH’ and the ‘CV’ classes do not show clear clustering of X-ray spectra as seen for the AGN and stars.

confusion matrix in Fig. 5.13. From the figure, we see that the difference in the performance of the pseudo-label and autoencoder methods mainly comes from the pseudo-label method classifying several AGN and stars as ‘NS/BH’. We see that this happens because of the relaxed threshold on ‘NS/BH’ and ‘CV’ during the pseudo-labeling process. Keeping the same pseudolabeling threshold of 0.75 for all classes gives an overall accuracy of 87% but ‘NS/BH/CV’ accuracy of $\sim 35\%$.

We also check the distribution of the ‘NS/BH’ and ‘CV’ sources in the latent

space using the autoencoder setup. We repeat the previous process of getting the UMAP projection of the latent space for the entire data and plotting our labelled data on top to check for any separation between classes. Fig 5.14 shows the UMAP projection of this latent space. We see that AGN and stars still show clear separation. There is some grouping among ‘NS/BH’ sources but we also see many of them scattered among the stars. The ‘CV’ sources are even less localized, and are largely scattered among the AGN. While the ‘NS/BH’ and ‘CV’ class are mostly at the top region in Fig. 5.14, they do not appear to form clusters where they make up the clear majority of the sources in a region.

5.5 Conclusion

We have discussed the application of machine learning methods, especially semi-supervised training using CNN architecture, in classifying X-ray spectra in the 4XMM-DR14 catalog. We trained two models — one that applies pseudo-labels to the unlabelled data to check for more variety in the classes and ensure a smooth decision boundary, and the other using auto-encoders to optimize a lower-dimensional latent space that can retain the information necessary both to retain the ability to separate the classes, and to reconstruct the original source spectra. When the algorithms are trained to identify only AGN and stars, both algorithms give an accuracy of $\sim 93\%$. The algorithm is most efficient when the net counts in the spectra are $\gtrsim 300$, and the ratio of background to net counts is $\lesssim 2$ for PN spectra and $\lesssim 0.3$ for MOS spectra. We also notice that the latent space of the PN and MOS spectra show clear separation between the labelled AGN and stars.

We also consider the mean normalized spectra of correctly classified vs. misclassified sources and see that the correctly identified spectra of stars have a strong peak ~ 1 keV, while stars identified as AGN seem to have a weaker peak, or no peak, in this region. Visualizing the CNN activations with Grad-CAM

shows that depending on the individual spectra, different parts of the spectra can influence the decisions made by the model. E.g., it might check for the absence of emission lines to conclude that one source is an AGN, while in a spectrum of a bright star showing strong peaks, the model might focus on the sharp rise before the peak and lack of hard X-rays to classify the source correctly.

Extending the two-class classification model to four classes to include the ‘NS/BH’ and ‘CV’ classes presented significant challenges due to their small number (minority classes). In general, we notice that the sources in the minority classes are being overfit by the model, so that anything that is even slightly different from the training data will be identified as an AGN or star. Visualizing the latent space of the 4-class autoencoder reveals only a very slight localization in the minority classes. Getting a larger set of NS/BH/CV type sources would allow us better selection of these models.

In this paper, we only use the X-ray spectra for our classification. Other properties of the sources like X-ray variability, position, multi-wavelength colors, etc, would allow more accurate classification. In such cases, we argue that using the latent variables generated from our autoencoder would be a better substitute to capture the properties of the spectra, as compared to the the broadband hardness ratios.

We also see that higher net counts and stronger signal-to-noise ratio (less background) allow better classification of sources. Thus, X-ray instruments with high sensitivity across a wide field of view can promise a multifold increase in performance –the stronger signal directly impacts the learning process, the larger number of observed sources increases the unlabelled dataset allowing it to reveal hidden structure in the data, and would also allow a more confident identification of X-ray sources, increasing the labelled set. Higher spectral resolution (without sacrificing effective area) would also be immensely helpful in picking out emission features in the spectra. Thus, future X-ray instruments like

the Athena mission (Barcons et al. [2017](#)), the proposed AXIS probe (Reynolds et al. [2023](#)), and/or the proposed Lynx telescope (Gaskin et al. [2019](#)) would be a huge step forward in improving such techniques.

Chapter 6

Summary and conclusions

In this thesis, we examined data-driven techniques to classify X-ray sources in large catalogs through their X-ray spectra. We argued that identifying X-ray emission lines in spectra, and differentiating between line-dominated and continuum emission, would allow us to classify X-ray spectra efficiently. Through our work, we explored different machine learning techniques and tailored X-ray colors to distinguish different classes of X-ray sources. We show that such techniques can better capture important information about the X-ray spectra as compared to standard broad-band hardness ratios.

We first test the performance of artificial neural networks (ANNs) on separating AGN from the CDFS catalog and young stars in the Orion Nebula in Chapter 3. Since most of these sources had a good spectral fit, we simulate 100,000 AGN and 10,000 stars. We use these simulated data to train a supervised classification model and test the results. Given that the CDFS AGN are particularly faint and do not represent typical *Chandra* sources in their average background levels, we reduce the background by a factor of 10 for our final simulations. Our model can classify the simulated *Chandra* X-ray spectra of these AGN and stars with an accuracy of $\sim 92\%$. When this trained model is used on the observed spectra of AGN and stars, we achieve $\sim 91\%$ accuracy of classification. Given that we simulate the spectra from known

properties, we also check for variation in the performance of the algorithms with different properties of X sources. Our algorithm performs best on *Chandra* sources where the net counts in the 0.3–8.0 keV energy range $\gtrsim 200$, and the ratio of background to net counts in this energy range is $\lesssim 5\%$. Stars with lower absorption, $N_H \lesssim 10^{22} \text{ cm}^{-2}$, and $kT < 2 \text{ keV}$, i.e., showing strong X-ray emission lines, are better classified. Similarly, AGN with a hard spectrum, i.e. $\Gamma \lesssim 2$, and a strong Fe-K emission line, are identified more easily.

Since most X-ray sources in catalogs are not well-studied, we need to implement semi-supervised and/or unsupervised training to classify X-ray spectra from large catalogs. Given the large variation in the X-ray spectra of sources, classifying X-ray spectra in an unsupervised fashion might lead to redundant groupings, for example, hard vs. soft sources that can already be distinguished by hardness ratios in the medium (1–2 keV)/hard bands (2–7 keV), interstellar absorption that can be characterized by hardness ratios in the soft (0.5–1 keV)/medium bands, etc. One way to overcome this would be to use manually tailored X-ray colors to distinguish line-dominated vs. continuum sources. We use this idea in Chapter 4 to distinguish the population of millisecond pulsars (MSPs) in the galactic bulge (GB) from the spectrally hard, X-ray luminous cataclysmic variables, likely dominated by intermediate polars (IPs), found in that region. Using simulations of MSPs, and Fe colors (defined as the ratio of X-ray photons in the 6.2–7.2 keV range to those in the ranges 5.8–6.2 keV and 7.2–7.6 keV), we constrain the typical ranges for Fe colors for different net count ranges. We use these Fe color limits to select potential MSPs among the observed sources. We also perform spectral analysis on these sources to ensure that there are indeed no lines in the X-ray spectra. Our analysis reveals 54 *XMM-Newton* and 83 *Chandra* sources that are consistent with the Fe-colors of MSPs and do not show any Fe lines in their X-ray spectra. These include 16 known X-ray sources including known X-ray binaries, one pulsar wind nebula, and one magnetar. From the exposure maps of *XMM-Newton* and *Chandra* and

the background AGN distribution, we estimate that ~ 35 of the *XMM-Newton* sources and ~ 5 *Chandra* sources could be background AGN. Our candidate sources may also include 10–20 CVs with weak Fe lines.

Tailoring X-ray colors are more challenging for identifying the soft emission lines, where several factors including the absorption column density, elemental abundances, temperature of the plasma, etc., influence the continuum and emission line strength. Therefore, semi-supervised learning techniques that use the labelled as well as the unlabelled data to extract the underlying structure and classify the sources would be more useful for large X-ray catalogs. We tested

two semi-supervised training algorithms on the 4XMM-DR14 spectra — one using pseudo-labelling of the unlabelled data, and the other using autoencoders to capture important features in the spectra that can reconstruct the original spectra while ensuring that the classes remain separable in the latent space.

We start with a two-class classification model to separate AGN and stars. We achieve $\sim 93\%$ accuracy in classifying these sources using both pseudo-labelling and autoencoder algorithms. In general, we observe that the accuracy is highest for $\gtrsim 300$ counts, and when the ratio of background to source counts $\lesssim 2$ for the PN spectra, and $\lesssim 0.5$ for the MOS spectra. We also see that the AGN and stars are well-separated in the latent space defined by the auto-encoder. Extending these models for 4-class classification and including classes for ‘NS/BH’ and ‘CV’ classes gives us poorer results. The overall accuracy reduces to (70–85)%, but the accuracy for just the ‘NS/BH’ and ‘CV’ classes is (30–50)%. This poor performance is due to the limited number of spectra in these classes, leading to overfitting, to these sources not contributing enough toward the loss gradients, and to their spectra being very similar to the spectra of the other classes. Expanding the labelled set to include more neutron stars, X-ray binaries, and CVs could help in better identification of these X-ray sources.

My thesis shows that ML methods are capable of distinguishing line-dominated spectra from continuum spectra. However, we see that X-ray spectra

alone might be insufficient to distinguish spectra of some X-ray source classes. In these cases, ML methods still allow us to capture the important features in the spectra through autoencoders and characterize the spectra using only a few latent variables. These latent variables allow us to move beyond simple hardness ratios and can be used along with other properties like the positions of X-ray sources, their X-ray variability, their optical colors and magnitude, their detection in radio/infrared wavelengths, etc., to determine the type of X-ray source. Future telescopes, such as Athena, AXIS and LynX, with a higher effective area, higher spectral resolution, ideally good angular resolution to reduce background, and a wider field of view, will allow us to study more sources in detail and increase our labelled set, as well as detect more serendipitous sources and add to our unlabelled dataset. The higher counts and signal-to-noise ratio from these instruments will also improve the performance of the ML methods.

Appendix A

Separating COUP and CDFS spectra

A.1 Typical spectra of simulated stars and AGN

We show the spectra of stars simulated by us with respect to the variation in the absorption column density, N_H , and plasma temperature, kT in Fig A.1. The values of N_H and kT shown correspond to the 90-percentile intervals. We use the model `tbabs*apec` to generate the spectra. We notice that while the Fe-L/Ne-K lines are clearly detected for stars with low- N_H , they are absorbed in the stars with very high N_H , leading to more pronounced Si-K and S-K lines (this is because we normalize the spectra by dividing each channel by the net counts). Similarly, as kT increases the spectra become less line-dominated.

We also show the spectra of our simulated AGN with changing N_H and Γ in Fig A.2. The values of N_H and Γ correspond to the 5th-percentile and 95th-percentile values. We use the model `tbabs*ztbabs*pegpwlw` to generate the spectra. The spectra of AGN remain continuum dominated throughout the range of N_H and Γ . Most of the features that we see in the AGN spectra are due to a combination of the absorption edges and effective area of the Chandra ACIS detector.

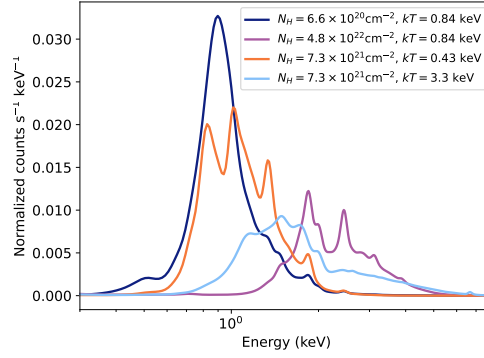


Figure A.1: Normalized model spectra of stars for 90-percentile intervals in N_H and kT . The spectra have been normalized such that the sum of counts in all energy channels is one. These spectra also do not account for the Poisson error due to low counts. For low N_H , we see a dominant Fe-L/Ne-K emission line, which disappears for higher N_H (we only see Si and S lines for $N_H = 4.8 \times 10^{22}$). Similarly at lower kT , we clearly see Fe-L, Ne-K, Mg-K and Si-K lines, but these aren't dominant at higher temperatures.

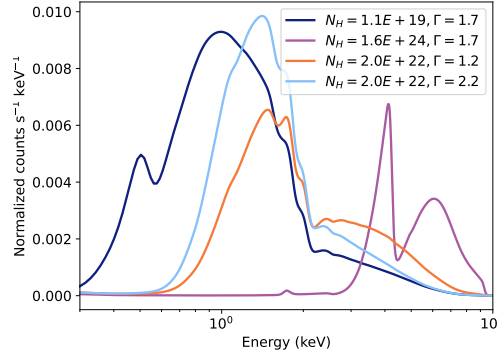


Figure A.2: Normalized spectra of AGN for 90-percentile intervals in N_H , and Γ . We fix the redshift to 0.7 (most probable value in the distribution of z). The features in the spectra are mainly due to the effective area of *Chandra* ACIS and absorption edges. In this figure, we only show AGN of type `tbabs*ztbabs*pegpwlw`. Some AGN also have an additional soft component and an Fe-K line. 4% of the AGN are better represented by a `pexrav` model.

A.2 TensorFlow model of the ANN

We use Keras and TensorFlow modules in python to model the ANN.

```
ann_model = tf.keras.Sequential([
```

```

layers.Dense(
    8, activation='relu',
    kernel_regularizer=regularizers.l1(
        0.001)),
layers.Dense(2, activation='softmax'))])
# Using 2 output nodes allows changing the
# threshold for the two classes separately
# and extend this to multi-class
# classification, if needed.

loss_fn = tf.keras.losses.BinaryCrossentropy(
    from_logits=False)
# We notice setting 'from_logits=False'
# more compatible with the calculation of
# keras metrics and does not affect our
# performance.

ann_model.compile(optimizer='adam',
                  loss=loss_fn)

```

A.3 Choosing number of nodes, L_1 value, and the size of training set

We also hyper-parameterize our NN model for the number of nodes and the value of the L_1 regularization. For this purpose, we train and test the ANN model on the simulated data with reduced background, and also test the performance on the observed spectra (thus ensuring that our model is robust for higher noise and background). We use a 5-fold cross-validation in our training

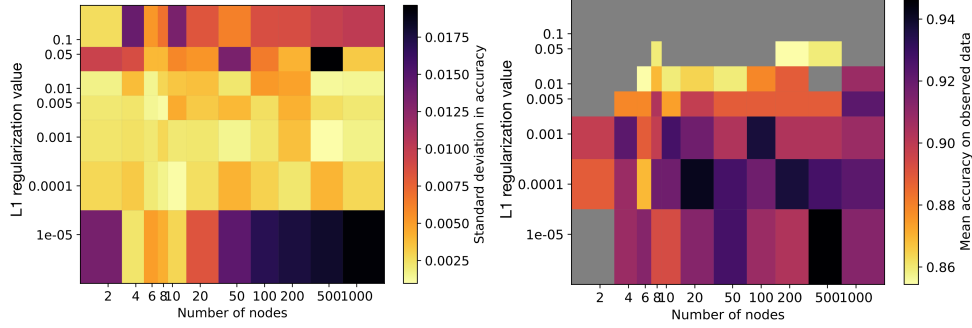


Figure A.3: (*Left*) Mean accuracy on the observed spectra and the (*Right*) standard deviation in accuracy on the simulated spectra for different numbers of nodes and the L_1 regularization values, λ s. We notice that while the accuracy seems to increase with the number of nodes and decreasing λ , the standard deviation also increases, making these models less robust when the data has low signal-to-noise ratios.

process. The standard deviation in the accuracy is calculated using the values of accuracy in the cross-validation process. In the top plot of Fig. A.3 we show the accuracy of the ANN algorithm on the observed spectra for a range of L_1 values and numbers of nodes. We also show the standard deviation in the accuracy values during the cross-validation in the bottom plot of Fig. A.3. We see that the accuracy of classification changes only slightly, given the standard deviations, for any number of nodes greater than 10 and any L_1 value over 10^{-3} . However, from Fig. A.4, we notice that for an L_1 value of $\lambda \lesssim 10^{-4}$, the distribution of weights becomes highly variable, i.e. while the higher values of weights still correspond to emission lines, they seem to pick out lines smaller than the resolving power of the CCD detectors. This can happen as a result of over-fitting, and might not yield good results for data with higher noise. Therefore, we choose $\lambda = 0.001$ for our model. For this value of λ , the accuracy does not change for number of nodes $\gtrsim 10$. Therefore, we choose 10 nodes and $\lambda = 0.001$ for our ANN model.

Fig. A.5 plots our the performance with the size of the training data. For small number of data sets, the model over-fits the data during training and we get poor performance on the test set. The accuracy seems to saturate at $\sim 90\%$

(within error-bars) for 10^5 data points in the training spectra.

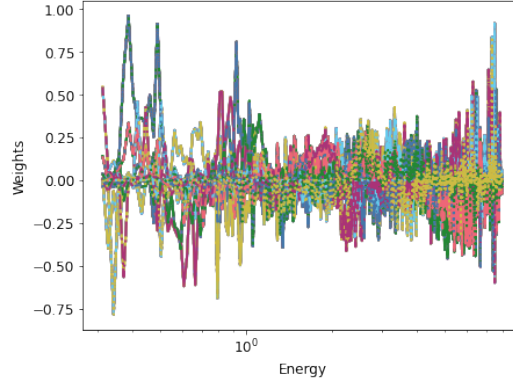


Figure A.4: Best fit weights for the ANN model with 20 nodes in the hidden layer and L_1 regularization value, $\lambda = 0.0001$. We see that these weights are more variable than those in Fig. 3.15.

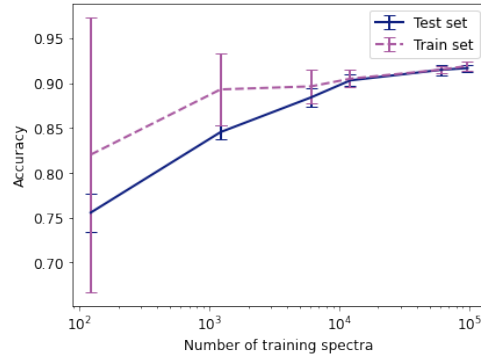


Figure A.5: Change in classification accuracy with the number of spectra used for training. The accuracy seems to saturate at $\sim 90\%$ at $\sim 10^5$ training points.

Appendix B

Identifying candidate MSPs and X-ray binaries in the Galactic Bulge

B.1 Calculating absorption column for Galactic Bulge sources.

The N_H values calculated from the dust extinction maps of Schlafly and Finkbeiner (2011) will include the Galactic absorption throughout the Galaxy. Therefore, they overestimate the absorption column for sources detected within the Galactic Bulge (GB). We need to calculate the absorption column for GB sources to determine the limiting luminosity of MSPs detectable in our observations and estimate the number of detectable MSPs required to explain the Fermi GCE.

We utilized our spectral analysis of the 116 *XMM-Newton* sources and 130 *Chandra* sources to map the N_H in the GB. We compare the best-fitting N_H value from spectra of sources ($N_{H,GB}$) with that calculated from dust extinction

($N_{H,dust}$) around the sources (Fig. B.1). From this comparison, we approximate:

$$N_{H,GB} = N_{H,dust}, \text{ if } N_{H,dust} < 10^{23} \text{ cm}^2, \quad (\text{B.1})$$

$$N_{H,GB} = 0.5N_{H,dust}, \text{ if } N_{H,dust} \in (10^{23}, 5 \times 10^{23}) \text{ cm}^2, \quad (\text{B.2})$$

$$N_{H,GB} = 0.2N_{H,dust}, \text{ if } N_{H,dust} > 5 \times 10^{23} \text{ cm}^2 \quad (\text{B.3})$$

The smaller values of $N_{H,GB}$ w.r.t $N_{H,dust}$ are consistent with our line of sight passing through larger amounts of gas beyond the GB as we approach the Galactic Center.

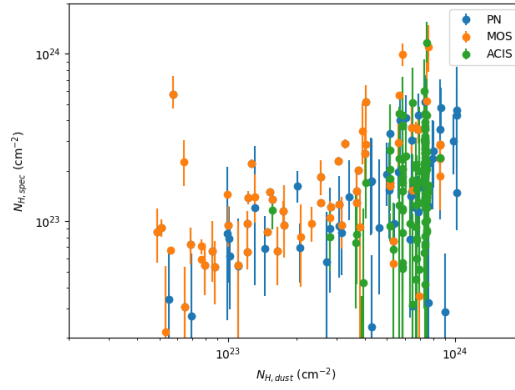


Figure B.1: Comparing the absorption column density from spectral fitting ($N_{H,spec}$) to that calculated from dust extinction maps ($N_{H,dust}$).

[h]

B.2 Estimating the number of cataclysmic variables in our sample.

For our purposes, we assume that all sources considered are either continuum sources (*CS*) or cataclysmic variables (*CV*). For calculations below, we only consider sources whose Fe colors are below the 2σ upper limit of continuum colors. We denote sources that do not reveal Fe lines upon spectroscopic analysis

by NL (no lines). To estimate the number of CVs among our candidate sources, we first calculate the probability that an observed source with no detected Fe lines is a CV, i.e., $p(CV|NL)$.

From Bayes' theorem, $p(CV|NL) = p(NL|CV) * p(CV) / p(NL)$; where $p(CV)$ is the probability that the observed source is a CV, and $p(NL)$ is the probability that the observed source has no Fe lines detected in its spectra. Since we assume all our sources are either continuum or cataclysmic variables, we can write,

$$p(NL) = p(NL|CS)p(CS) + p(NL|CV)p(CV) = p(NL|CS)(1 - p(CV)) + p(NL|CV)p(CV), \quad (B.4)$$

$$\Rightarrow p(CV) = \frac{p(NL|CS) - p(NL)}{p(NL|CS) - p(NL|CV)}, \quad (B.5)$$

where $p(CV)$, $p(CS)$ are the probabilities that the observed source is a CV, or a continuum source; $p(NL|CS)$, $p(NL|CV)$ are the probabilities that no Fe lines are detected in the X-ray spectra of a continuum source, or a CV.

We can estimate $p(NL|CV)$ and $p(NL|CS)$ from our simulated samples of Galactic CVs and MSPs, and approximate $p(NL)$ from the observed sample of sources. We divide sources with net counts ≥ 250 into 5 bins and calculate $p(CV|NL)$ for each bin. Since the number of observed candidate continuum sources (NL) with ≥ 1500 counts is limited, the error-bars on $p(NL)$ in these bins can be large. Therefore, in bins where $p(NL) > p(NL|CS)$, we use $p(CV) = 0$. We then multiply this by the number of candidate sources in each bin to estimate the number of CVs in our sample.

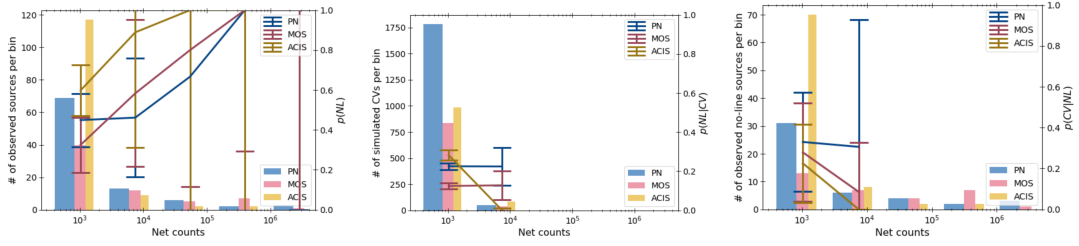


Figure B.2: *Left:* Probabilities that observed sources have no detected Fe lines $p(NL)$; *Center:* probabilities that simulated CVs do not show sufficient evidence of Fe lines i.e. $p(NL|CV)$; *Right:* probability that sources with no detected Fe lines could be CVs. All results shown here are for sources that have Fe colors consistent with that of MSPs. We also show the histograms of the observed sources, simulated CVs, and candidate continuum sources (with MSP-like colors). Given that none of our simulated CVs with ≥ 3000 counts had colors consistent with simulated MSPs, we assume that all sources with ≥ 3000 counts and showing no lines on spectral analysis are continuum sources.

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